# 3D position RMSE [m] per sequence

| dataset | sequence | KIMS | KIMS 10 seeds | LM | SW-PGO | SFUISE | filter |
|---|---|---|---|---|---|---|---|
| MILUV | default_1_remoteControlHighPace_1 | 0.205 | 0.205 | 0.235 | 0.712 | div. | 1.319 |
| MILUV | default_1_random3_1 | 0.232 | 0.232 | 0.265 | 0.541 | 0.488 | 0.255 |
| MILUV | default_1_random3_0 | 0.166 | 0.165 | 0.682 | 0.627 | 0.704 | 0.217 |
| MILUV | default_1_circular3D_0 | 0.093 | 0.093 | 0.091 | 0.144 | 0.123 | 0.303 |
| MILUV | default_1_remoteControlLowPace_0_v2 | 0.086 | 0.086 | 0.088 | 0.099 | 0.114 | 0.219 |
| MILUV | default_1_remoteControlLowPace_0 | 0.201 | 0.201 | 0.216 | 0.328 | 0.955 | 0.247 |
| MILUV | default_1_circular2D_0 | 0.133 | 0.134 | 0.131 | 0.275 | 0.192 | 0.335 |
| MILUV | default_1_remoteControlHighPace_0 | 0.100 | 0.100 | 0.096 | 0.100 | div. | 1.358 |
| MILUV | obstacles_1_random3_0b | 0.131 | 0.132 | 0.595 | 0.185 | 0.639 | 0.252 |
| MILUV | default_1_random3_2 | 0.135 | 0.135 | 0.140 | 0.209 | div. | 0.265 |
| MILUV | default_1_remoteControlHighPace_2 | 0.245 | 0.244 | 0.241 | 0.227 | 0.763 | 1.414 |
| MILUV | noAprilTags_1_random3_2b | 0.130 | 0.130 | 0.133 | 0.175 | 0.368 | 0.241 |
| MILUV | noAprilTags_1_remoteControlHighPace_2b | 0.191 | 0.191 | 0.197 | 1.338 | 1.144 | 0.273 |
| NTU VIRAL | eee_01 | 0.203 | 0.203 | 0.200 | 0.199 | 0.126 | - |
| NTU VIRAL | eee_02 | 0.370 | 0.365 | 0.363 | 0.357 | 2.537 | - |
| NTU VIRAL | eee_03 | 1.845 | 1.845 | 1.861 | 1.867 | 1.144 | - |
| NTU VIRAL | nya_01 | 0.793 | 0.789 | 0.786 | 0.786 | div. | - |
| NTU VIRAL | nya_02 | 3.085 | 2.590 | 3.083 | 0.445 | 0.395 | - |
| NTU VIRAL | nya_03 | 0.917 | 0.891 | 1.034 | 0.993 | div. | - |
| NTU VIRAL | sbs_01 | 0.191 | 0.190 | 0.183 | 0.174 | 0.207 | - |
| NTU VIRAL | sbs_02 | 0.411 | 0.575 | 0.404 | 0.891 | 3.109 | - |
| NTU VIRAL | sbs_03 | 0.435 | 0.436 | 0.433 | 0.433 | 3.146 | - |
| IULUI | CoplanarSquareTurn | 0.120 | 0.120 | 0.548 | 0.912 | 0.106 | - |
| IULUI | CoplanarCircleSpin | 0.117 | 0.117 | 0.418 | 0.406 | 0.235 | - |
| IULUI | NoncoplanarCircleSpin | 0.128 | 0.128 | 0.175 | 0.133 | 0.300 | - |
| UTIL | const1-trial1-tdoa2 | 0.096 | - | 0.096 | 0.094 | 0.095 | 0.108 |
| UTIL | const1-trial1-tdoa3 | 0.100 | - | 0.100 | 0.096 | 0.094 | 0.131 |
| UTIL | const1-trial2-tdoa2 | 0.112 | - | 0.112 | - | 0.115 | 0.122 |
| UTIL | const1-trial2-tdoa3 | 0.110 | - | 0.110 | - | 0.104 | 0.143 |
| UTIL | const1-trial3-tdoa2 | 0.098 | - | 0.098 | - | 0.094 | 0.108 |
| UTIL | const1-trial3-tdoa3 | 0.097 | - | 0.097 | - | 0.090 | 0.120 |
| UTIL | const1-trial4-tdoa2 | 0.101 | - | 0.102 | - | 0.099 | 0.111 |
| UTIL | const1-trial4-tdoa3 | 0.096 | - | 0.097 | - | 0.090 | 0.124 |
| UTIL | const1-trial5-tdoa2 | 0.107 | - | 0.107 | - | 0.103 | 0.114 |
| UTIL | const1-trial5-tdoa3 | 0.112 | - | 0.112 | - | 0.108 | 0.139 |
| UTIL | const1-trial6-tdoa2 | 0.111 | - | 0.111 | - | 0.104 | 0.123 |
| UTIL | const1-trial6-tdoa3 | 0.108 | - | 0.108 | - | 0.103 | 0.134 |
| UTIL | const2-trial1-tdoa2 | 0.095 | - | 0.095 | 0.092 | 0.094 | 0.103 |
| UTIL | const2-trial1-tdoa3 | 0.095 | - | 0.095 | 0.090 | 0.096 | 0.113 |
| UTIL | const2-trial2-tdoa2 | 0.106 | - | 0.106 | - | 0.103 | 0.114 |

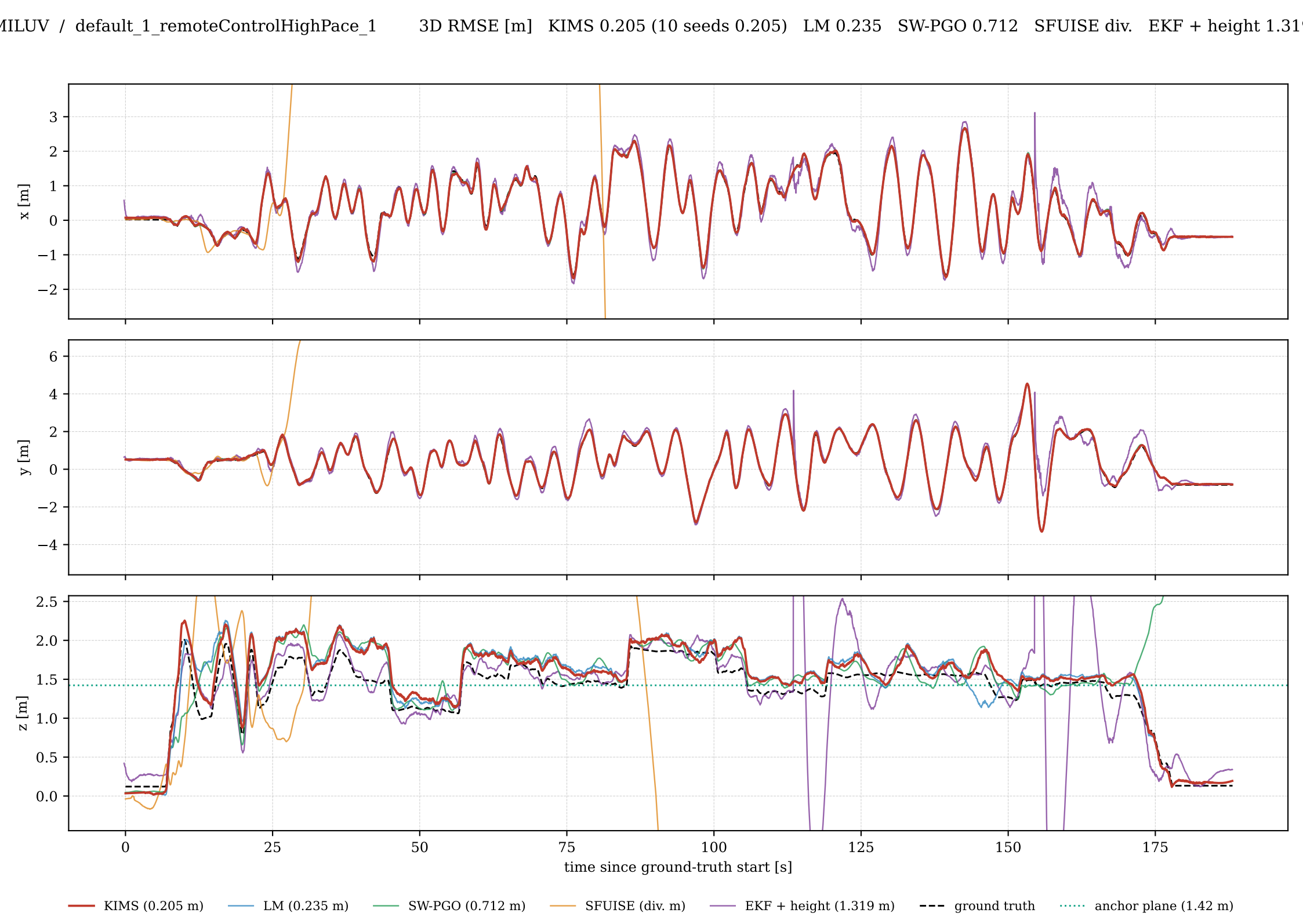
KIMS: seed 1234, N = 2048. KIMS 10 seeds: mean over seeds 1-10, the value of Table I on MILUV, NTU VIRAL and IULUI.
LM: the same window objective solved by Levenberg-Marquardt. SW-PGO: sliding-window pose graph with the Tukey loss. SFUISE: published configuration from the KIMS start pose.
filter: our runs of the code released with MILUV (EKF + laser height) and UTIL (ESKF).
Whole flight, no alignment. NTU VIRAL: Leica prism, yaw + translation fit per run, 10 s cut at both ends. div.: RMSE above 100 m. -: no output.

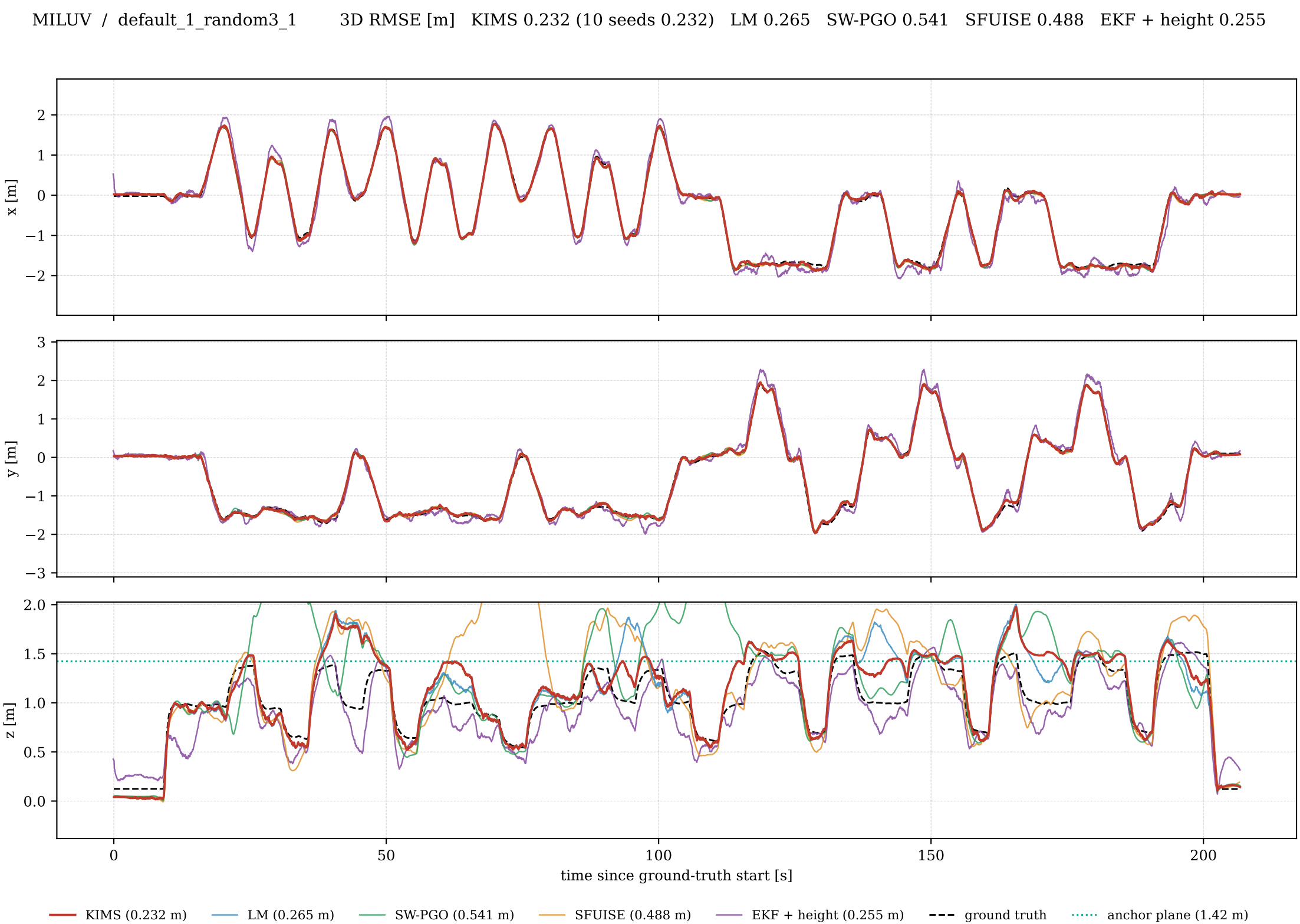
## 3D position RMSE [m] per sequence

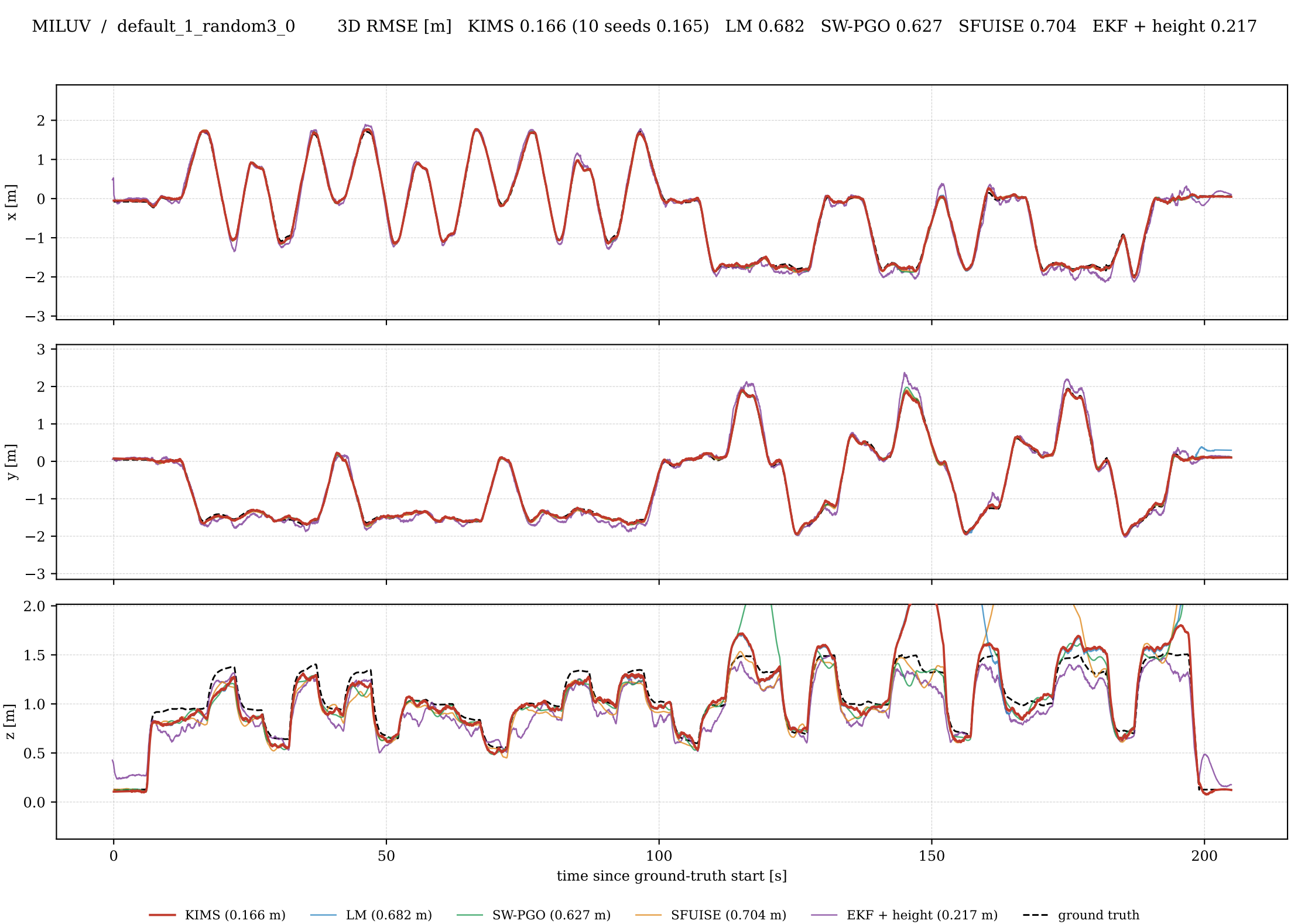
| dataset | sequence | KIMS | KIMS 10 seeds | LM | SW-PGO | SFUISE | filter |
|---|---|---|---|---|---|---|---|
| UTIL | const2-trial2-tdoa3 | 0.088 | - | 0.088 | - | 0.084 | 0.113 |
| UTIL | const2-trial3-tdoa2 | 0.107 | - | 0.107 | - | 0.110 | 0.108 |
| UTIL | const2-trial3-tdoa3 | 0.081 | - | 0.078 | - | 0.083 | 0.112 |
| UTIL | const2-trial4-tdoa2 | 0.088 | - | 0.088 | - | 0.096 | 0.093 |
| UTIL | const2-trial4-tdoa3 | 0.088 | - | 0.088 | - | 0.086 | 0.117 |
| UTIL | const2-trial5-tdoa2 | 0.111 | - | 0.111 | - | 0.111 | 0.121 |
| UTIL | const2-trial5-tdoa3 | 0.096 | - | 0.096 | - | 0.089 | 0.118 |
| UTIL | const2-trial6-tdoa2 | 0.101 | - | 0.101 | - | 0.101 | 0.106 |
| UTIL | const2-trial6-tdoa3 | 0.100 | - | 0.100 | - | 0.094 | 0.117 |
| UTIL | const3-trial1-tdoa2 | 0.230 | - | 0.234 | 0.190 | 0.330 | 0.245 |
| UTIL | const3-trial1-tdoa3 | 0.198 | - | 0.200 | 0.181 | 0.207 | 0.226 |
| UTIL | const3-trial2-tdoa2 | 0.187 | - | 0.193 | - | 0.207 | 0.197 |
| UTIL | const3-trial2-tdoa3 | 0.169 | - | 0.170 | - | 0.182 | 0.219 |
| UTIL | const3-trial3-tdoa2 | 0.187 | - | 0.191 | - | 0.175 | 0.189 |
| UTIL | const3-trial3-tdoa3 | 0.175 | - | 0.176 | - | 0.184 | 0.208 |
| UTIL | const3-trial4-tdoa2 | 0.197 | - | 0.201 | - | 0.183 | 0.217 |
| UTIL | const3-trial4-tdoa3 | 0.160 | - | 0.160 | - | 0.174 | 0.199 |
| UTIL | const3-trial5-tdoa2 | 0.202 | - | 0.206 | - | 0.186 | 0.221 |
| UTIL | const3-trial5-tdoa3 | 0.182 | - | 0.184 | - | 0.177 | 0.230 |
| UTIL | const3-trial6-tdoa2 | 0.169 | - | 0.174 | - | 0.171 | 0.195 |
| UTIL | const3-trial6-tdoa3 | 0.157 | - | 0.158 | - | 0.165 | 0.202 |
| UTIL | const3-trial7-tdoa2-manual1 | 0.234 | - | 0.238 | - | 0.138 | 0.258 |
| UTIL | const3-trial7-tdoa2-manual2 | 0.301 | - | 0.305 | - | 0.162 | 0.320 |
| UTIL | const3-trial7-tdoa3-manual3 | 0.219 | - | 0.221 | - | 0.224 | 0.301 |
| UTIL | const3-trial7-tdoa3-manual4 | 0.216 | - | 0.218 | - | 0.187 | 0.323 |
| UTIL | const4-trial1-tdoa2-traj1 | 0.198 | - | 0.198 | 0.195 | 0.202 | 0.201 |
| UTIL | const4-trial1-tdoa2-traj2 | 0.156 | - | 0.156 | - | 0.159 | 0.168 |
| UTIL | const4-trial1-tdoa2-traj3 | 0.173 | - | 0.174 | - | 0.184 | 0.194 |
| UTIL | const4-trial1-tdoa3-traj1 | 0.159 | - | 0.159 | 0.151 | 0.155 | 0.195 |
| UTIL | const4-trial1-tdoa3-traj2 | 0.139 | - | 0.139 | - | 0.131 | 0.160 |
| UTIL | const4-trial1-tdoa3-traj3 | 0.156 | - | 0.157 | - | 0.153 | 0.187 |
| UTIL | const4-trial2-tdoa2-traj1 | 0.235 | - | 0.235 | - | 0.240 | 0.231 |
| UTIL | const4-trial2-tdoa2-traj2 | 0.181 | - | 0.185 | - | 0.195 | 0.199 |
| UTIL | const4-trial2-tdoa2-traj3 | 0.214 | - | 0.215 | - | 0.230 | 0.221 |
| UTIL | const4-trial2-tdoa3-traj1 | 0.183 | - | 0.183 | - | 0.177 | 0.217 |
| UTIL | const4-trial2-tdoa3-traj2 | 0.179 | - | 0.181 | - | 0.189 | 0.212 |
| UTIL | const4-trial2-tdoa3-traj3 | 0.191 | - | 0.191 | - | 0.199 | 0.227 |
| UTIL | const4-trial3-tdoa2-traj1 | 0.287 | - | 0.296 | - | 0.504 | 0.615 |
| UTIL | const4-trial3-tdoa2-traj2 | 0.299 | - | 0.299 | - | div. | 0.592 |
| UTIL | const4-trial3-tdoa2-traj3 | 0.307 | - | 0.308 | - | 0.449 | 0.514 |

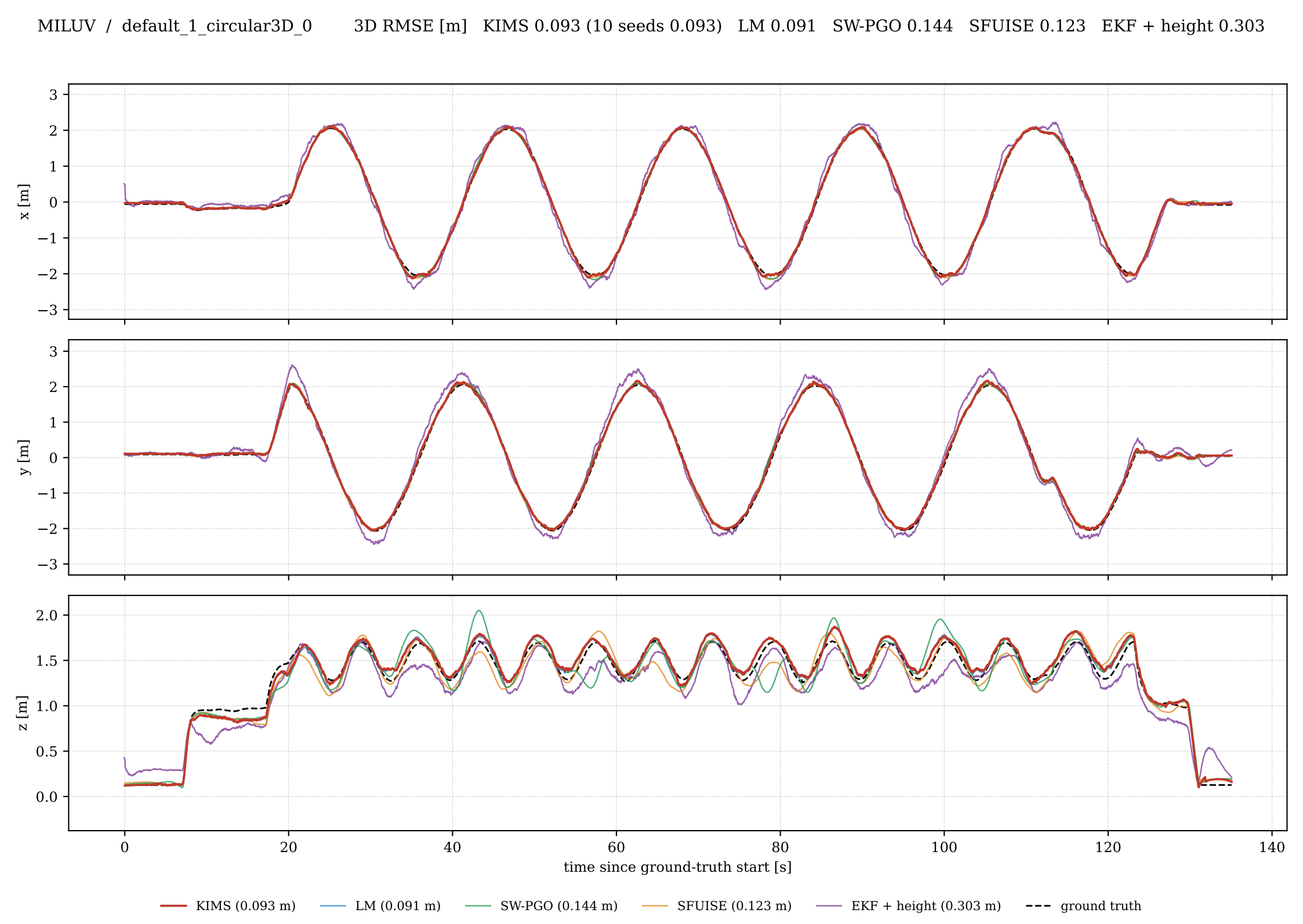
## 3D position RMSE [m] per sequence

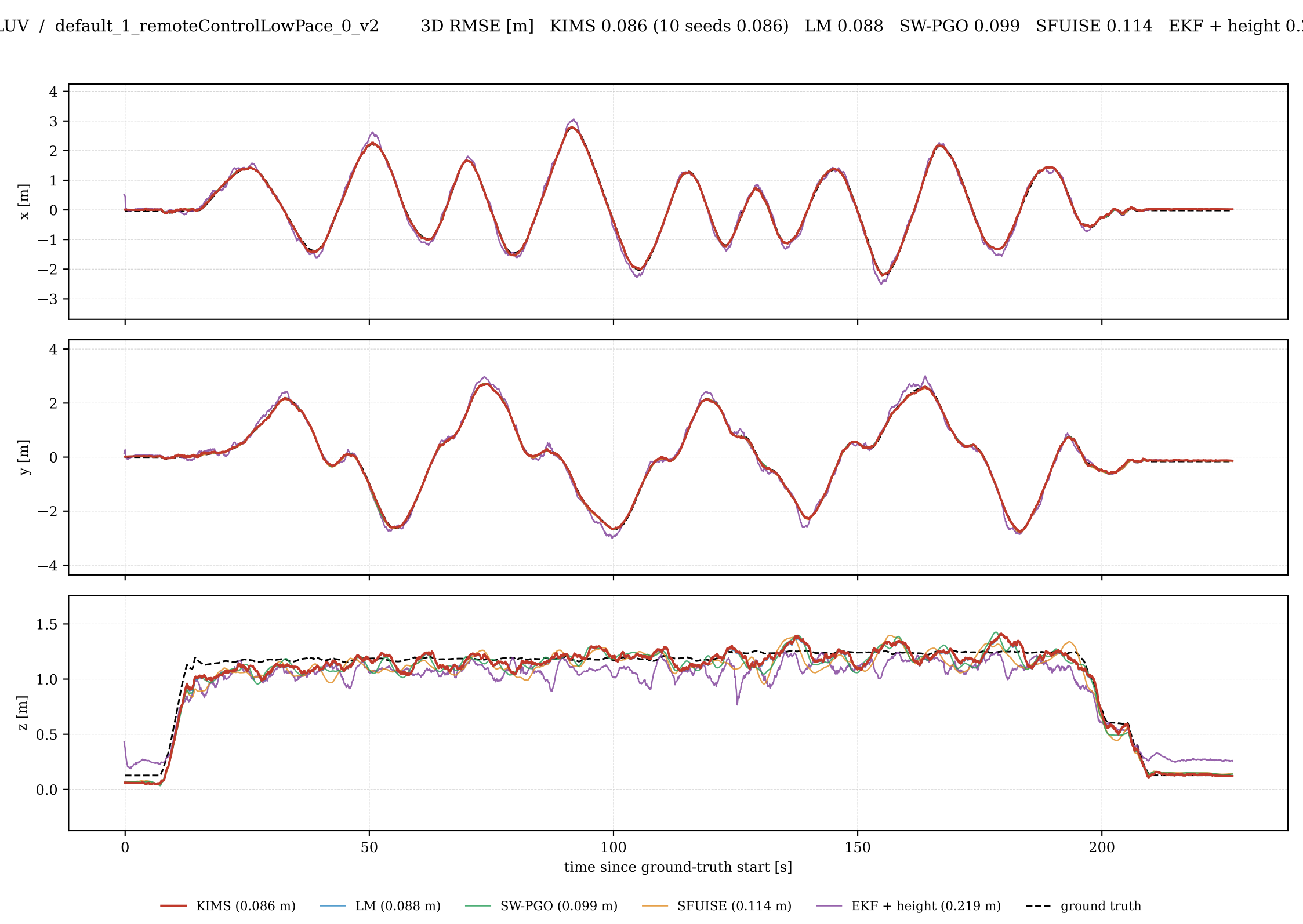
| dataset | sequence | KIMS | KIMS 10 seeds | LM | SW-PGO | SFUISE | filter |
|---|---|---|---|---|---|---|---|
| UTIL | const4-trial3-tdoa3-traj1 | 0.242 | - | 0.242 | - | 0.216 | 0.431 |
| UTIL | const4-trial3-tdoa3-traj2 | 0.233 | - | 0.233 | - | 0.722 | 0.405 |
| UTIL | const4-trial3-tdoa3-traj3 | 0.244 | - | 0.245 | - | 0.221 | 0.555 |
| UTIL | const4-trial4-tdoa2-traj1 | 0.595 | - | 0.596 | - | div. | 0.693 |
| UTIL | const4-trial4-tdoa2-traj2 | 0.505 | - | 0.508 | - | div. | 0.627 |
| UTIL | const4-trial4-tdoa2-traj3 | 0.489 | - | 0.491 | - | 1.296 | 0.562 |
| UTIL | const4-trial4-tdoa3-traj1 | 0.433 | - | 0.433 | - | 0.399 | 0.617 |
| UTIL | const4-trial4-tdoa3-traj2 | 0.396 | - | 0.396 | - | 0.349 | 0.631 |
| UTIL | const4-trial4-tdoa3-traj3 | 0.403 | - | 0.403 | - | 0.447 | 0.650 |
| UTIL | const4-trial5-tdoa2-traj1 | 0.380 | - | 0.392 | - | 0.272 | 0.411 |
| UTIL | const4-trial5-tdoa2-traj2 | 0.262 | - | 0.264 | - | 0.218 | 0.426 |
| UTIL | const4-trial5-tdoa2-traj3 | 0.286 | - | 0.287 | - | 0.227 | 0.363 |
| UTIL | const4-trial5-tdoa3-traj1 | 0.274 | - | 0.275 | - | div. | 0.387 |
| UTIL | const4-trial5-tdoa3-traj2 | 0.258 | - | 0.259 | - | 0.206 | 0.366 |
| UTIL | const4-trial5-tdoa3-traj3 | 0.255 | - | 0.256 | - | 0.210 | 0.384 |
| UTIL | const4-trial6-tdoa2-traj1 | 0.412 | - | 0.413 | - | div. | 0.454 |
| UTIL | const4-trial6-tdoa2-traj2 | 0.352 | - | 0.353 | - | 0.314 | 0.384 |
| UTIL | const4-trial6-tdoa2-traj3 | 0.394 | - | 0.395 | - | 0.351 | 0.458 |
| UTIL | const4-trial6-tdoa3-traj1 | 0.380 | - | 0.381 | - | 0.320 | 0.556 |
| UTIL | const4-trial6-tdoa3-traj2 | 0.401 | - | 0.402 | - | 0.320 | 0.540 |
| UTIL | const4-trial6-tdoa3-traj3 | 0.391 | - | 0.392 | - | 0.347 | 0.523 |
| UTIL | const4-trial7-tdoa2-manual1 | 0.204 | - | 0.235 | - | 0.184 | 0.281 |
| UTIL | const4-trial7-tdoa2-manual2 | 0.205 | - | 0.281 | - | 0.159 | 0.302 |
| UTIL | const4-trial7-tdoa2-manual3 | 0.211 | - | 0.212 | - | 0.190 | 0.270 |

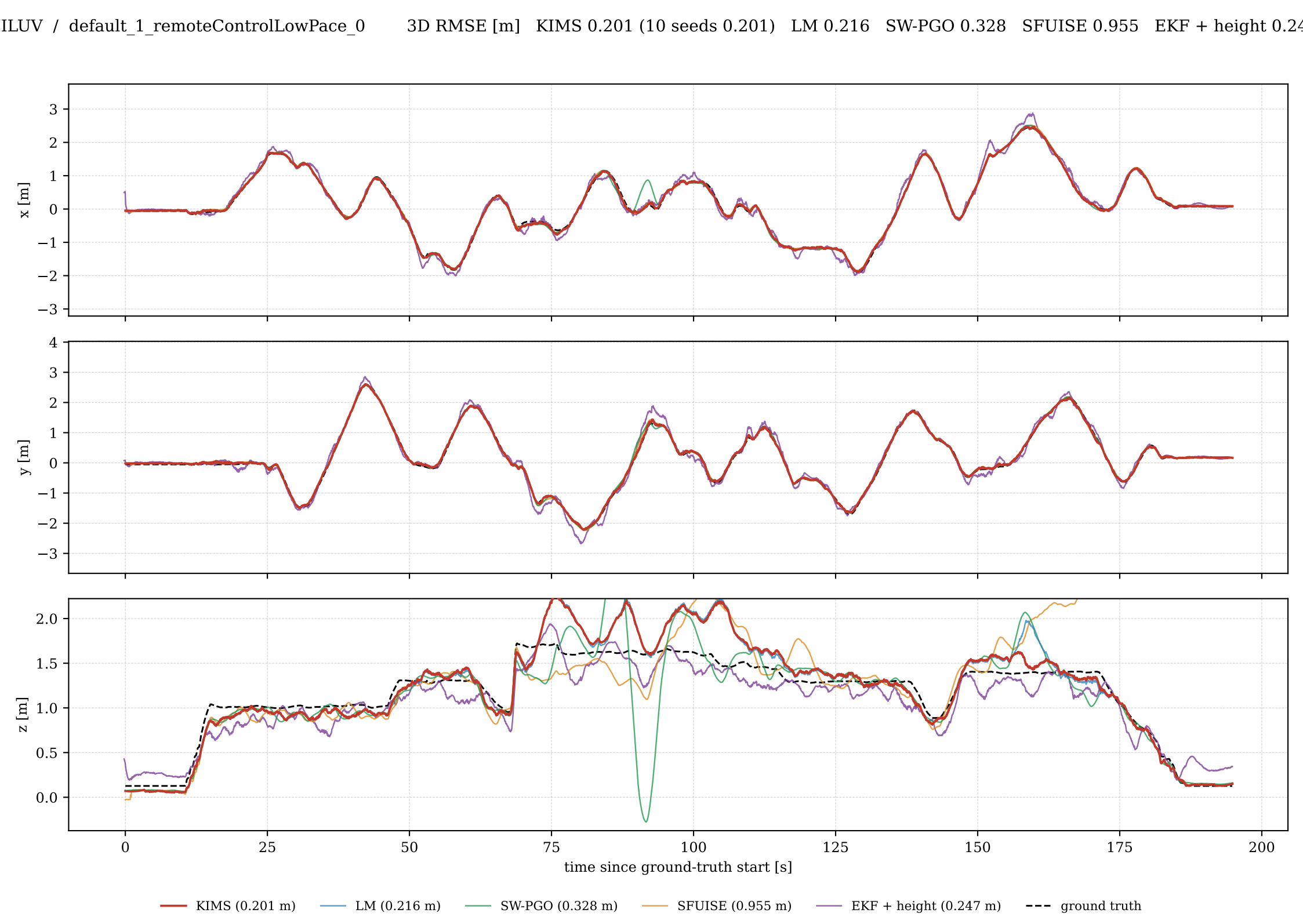

MILUV / default_1_remoteControlHighPace_1 3D RMSE [m] KIMS 0.205 (10 seeds 0.205) LM 0.235 SW-PGO 0.712 SFUISE div. EKF + height 1.31
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.205 m)
LM (0.235 m)
SW-PGO (0.712 m)
SFUISE (div. m)
EKF + height (1.319 m)
ground truth
anchor plane (1.42 m)

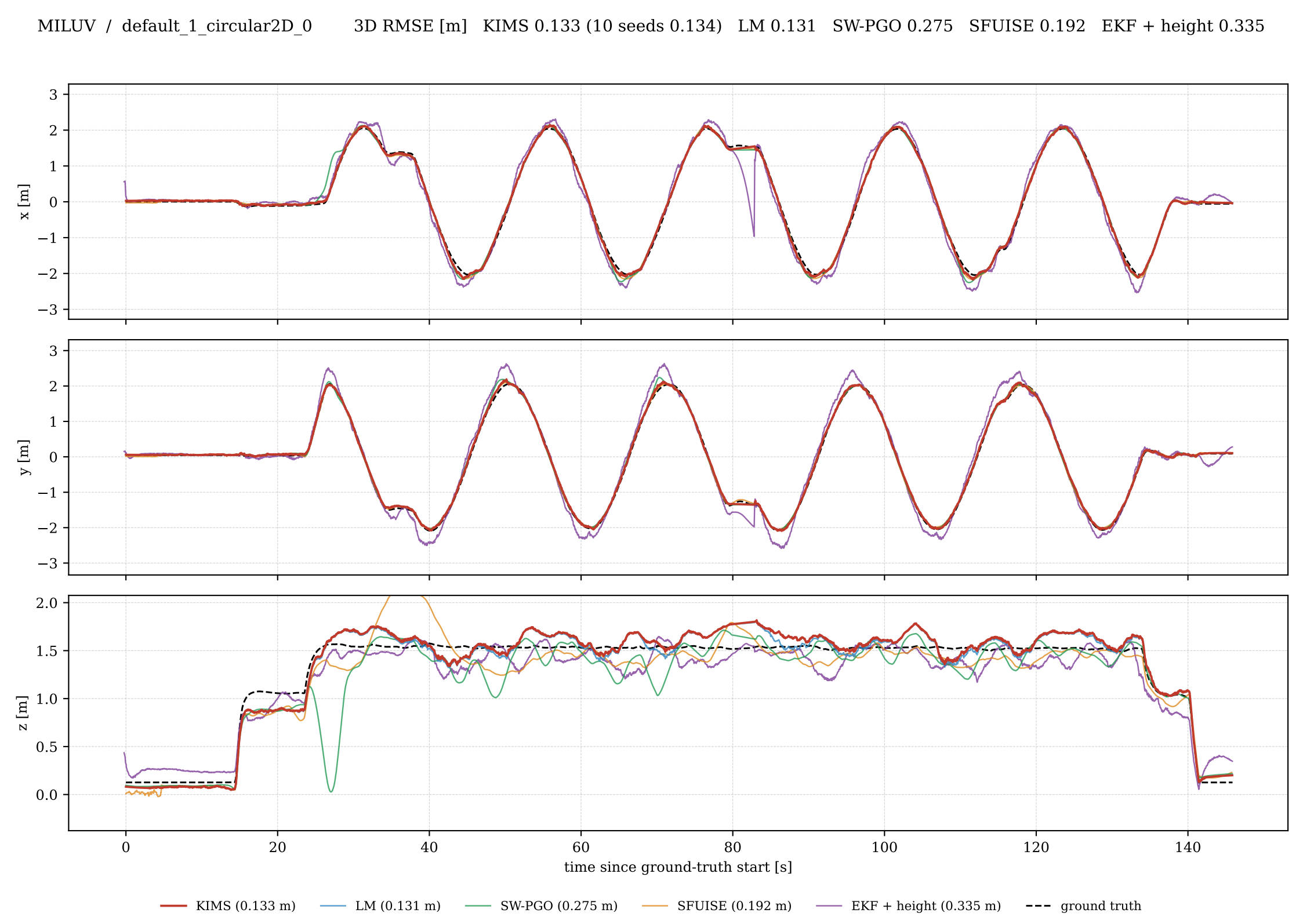

MILUV / default_1_random3_1 3D RMSE [m] KIMS 0.232 (10 seeds 0.232) LM 0.265 SW-PGO 0.541 SFUISE 0.488 EKF + height 0.255
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.232 m)
LM (0.265 m)
SW-PGO (0.541 m)
SFUISE (0.488 m)
EKF + height (0.255 m)
ground truth
anchor plane (1.42 m)

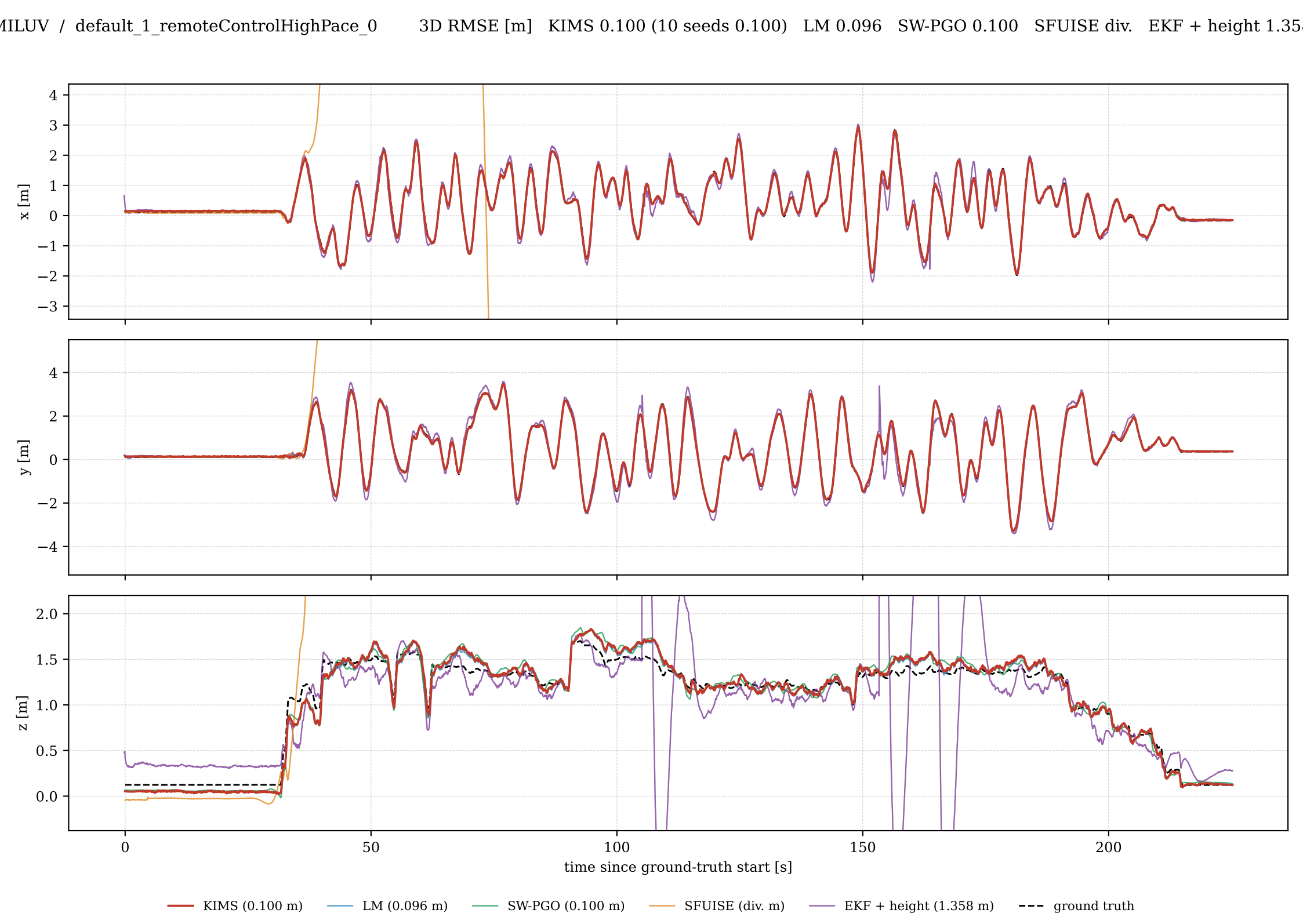

MILUV / default_1_random3_0 3D RMSE [m] KIMS 0.166 (10 seeds 0.165) LM 0.682 SW-PGO 0.627 SFUISE 0.704 EKF + height 0.217
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.166 m)
LM (0.682 m)
SW-PGO (0.627 m)
SFUISE (0.704 m)
EKF + height (0.217 m)
ground truth

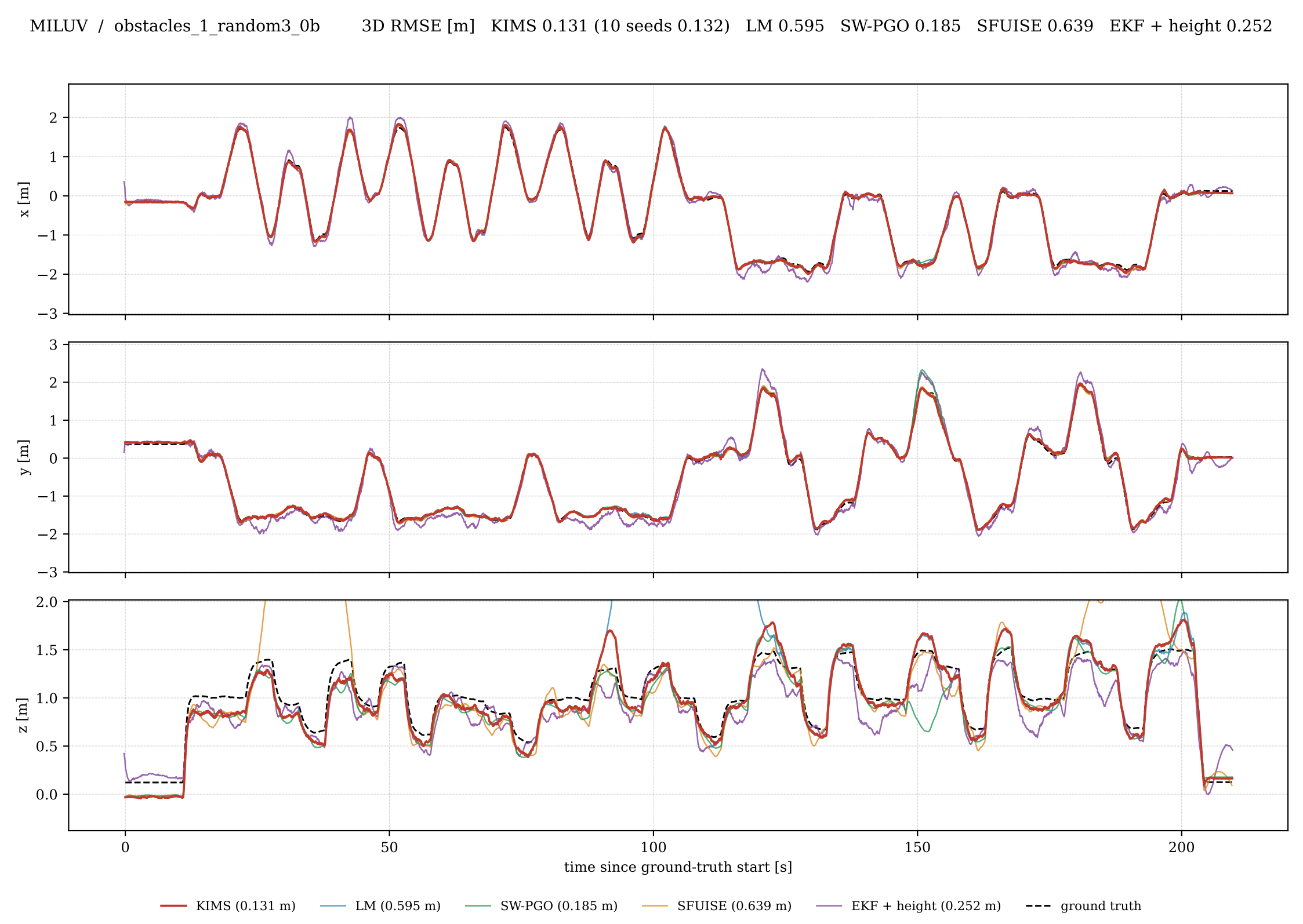

MILUV / default_1_circular3D_0 3D RMSE [m] KIMS 0.093 (10 seeds 0.093) LM 0.091 SW-PGO 0.144 SFUISE 0.123 EKF + height 0.303
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.093 m)
LM (0.091 m)
SW-PGO (0.144 m)
SFUISE (0.123 m)
EKF + height (0.303 m)
ground truth

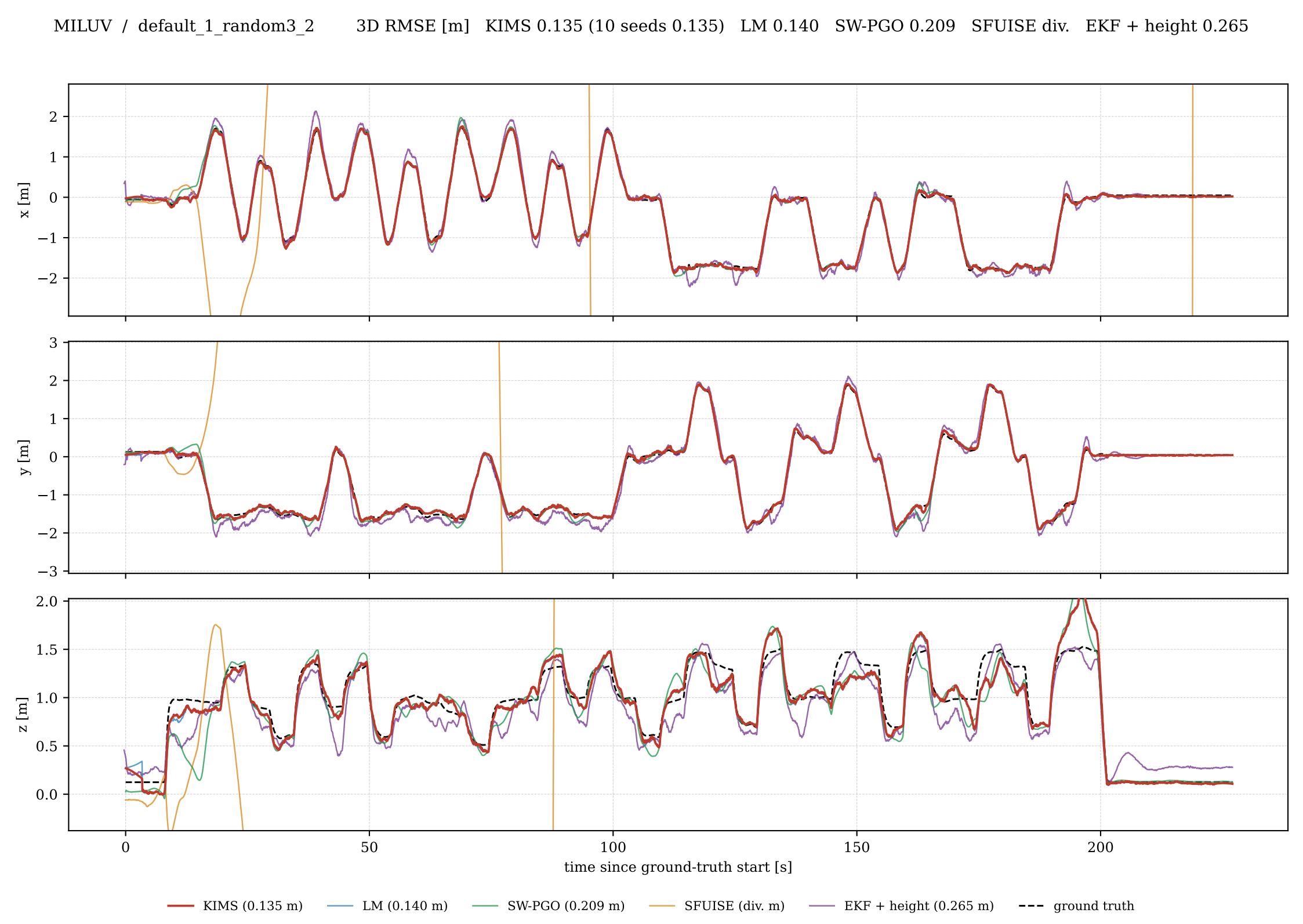

LUV / default_1_remoteControlLowPace_0_v2 3D RMSE [m] KIMS 0.086 (10 seeds 0.086) LM 0.088 SW-PGO 0.099 SFUISE 0.114 EKF + height 0.
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.086 m)
LM (0.088 m)
SW-PGO (0.099 m)
SFUISE (0.114 m)
EKF + height (0.219 m)
ground truth

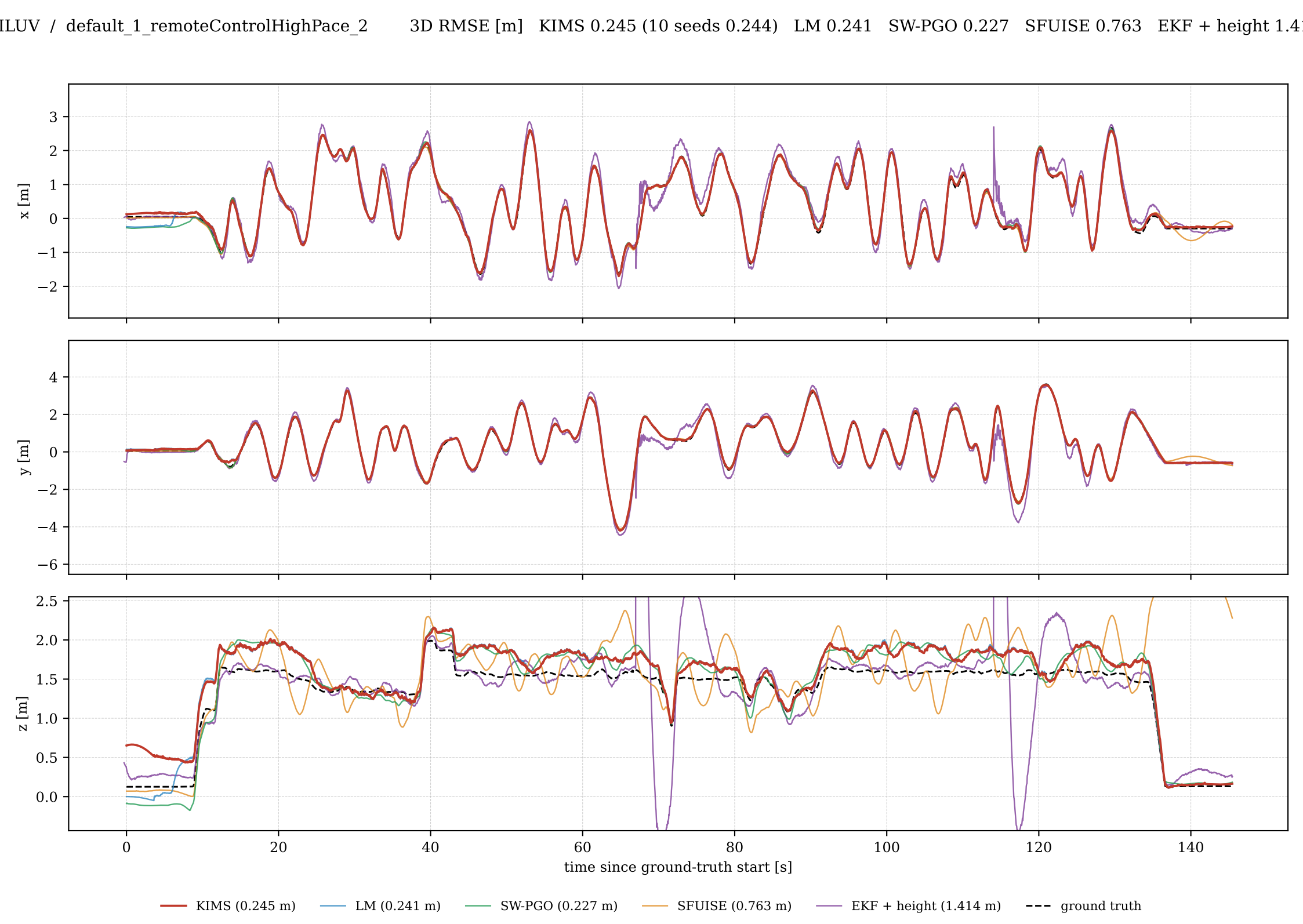

ILUV / default_1_remoteControlLowPace_0 3D RMSE [m] KIMS 0.201 (10 seeds 0.201) LM 0.216 SW-PGO 0.328 SFUISE 0.955 EKF + height 0.24
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.201 m)
LM (0.216 m)
SW-PGO (0.328 m)
SFUISE (0.955 m)
EKF + height (0.247 m)
ground truth

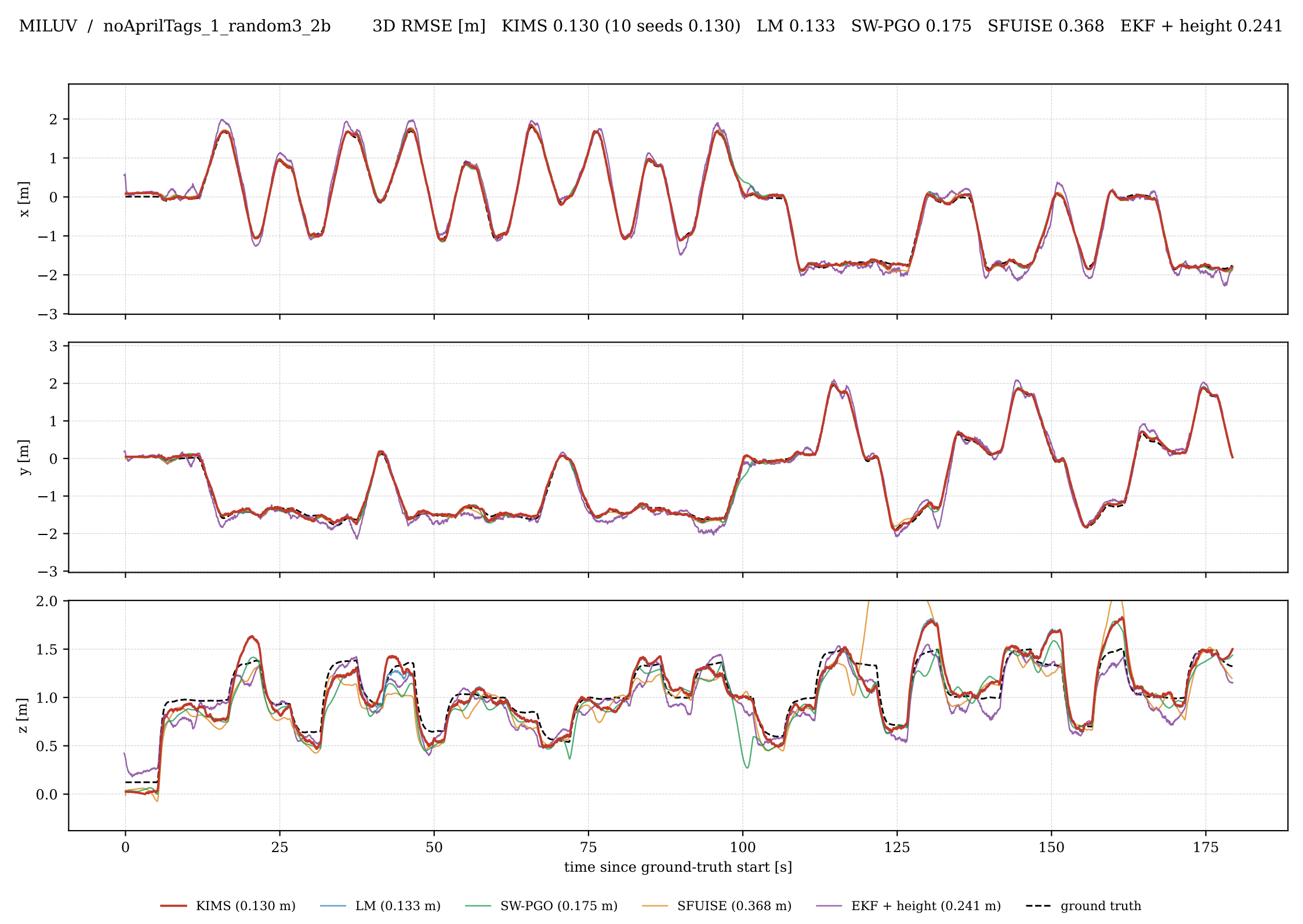

MILUV / default_1_circular2D_0 3D RMSE [m] KIMS 0.133 (10 seeds 0.134) LM 0.131 SW-PGO 0.275 SFUISE 0.192 EKF + height 0.335
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.133 m)
LM (0.131 m)
SW-PGO (0.275 m)
SFUISE (0.192 m)
EKF + height (0.335 m)
ground truth

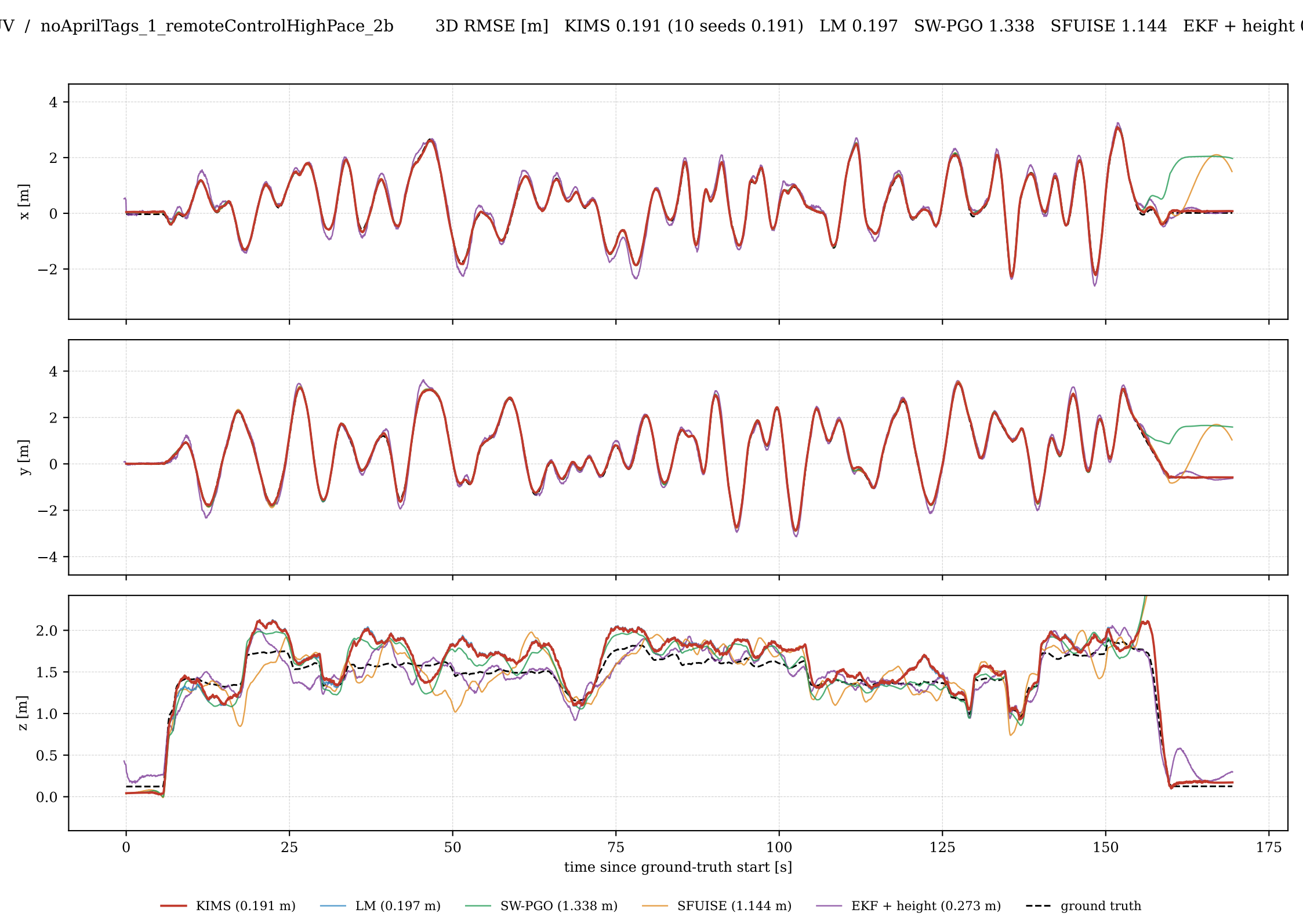

MILUV / default_1_remoteControlHighPace_0 3D RMSE [m] KIMS 0.100 (10 seeds 0.100) LM 0.096 SW-PGO 0.100 SFUISE div. EKF + height 1.35
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.100 m)
LM (0.096 m)
SW-PGO (0.100 m)
SFUISE (div. m)
EKF + height (1.358 m)
ground truth

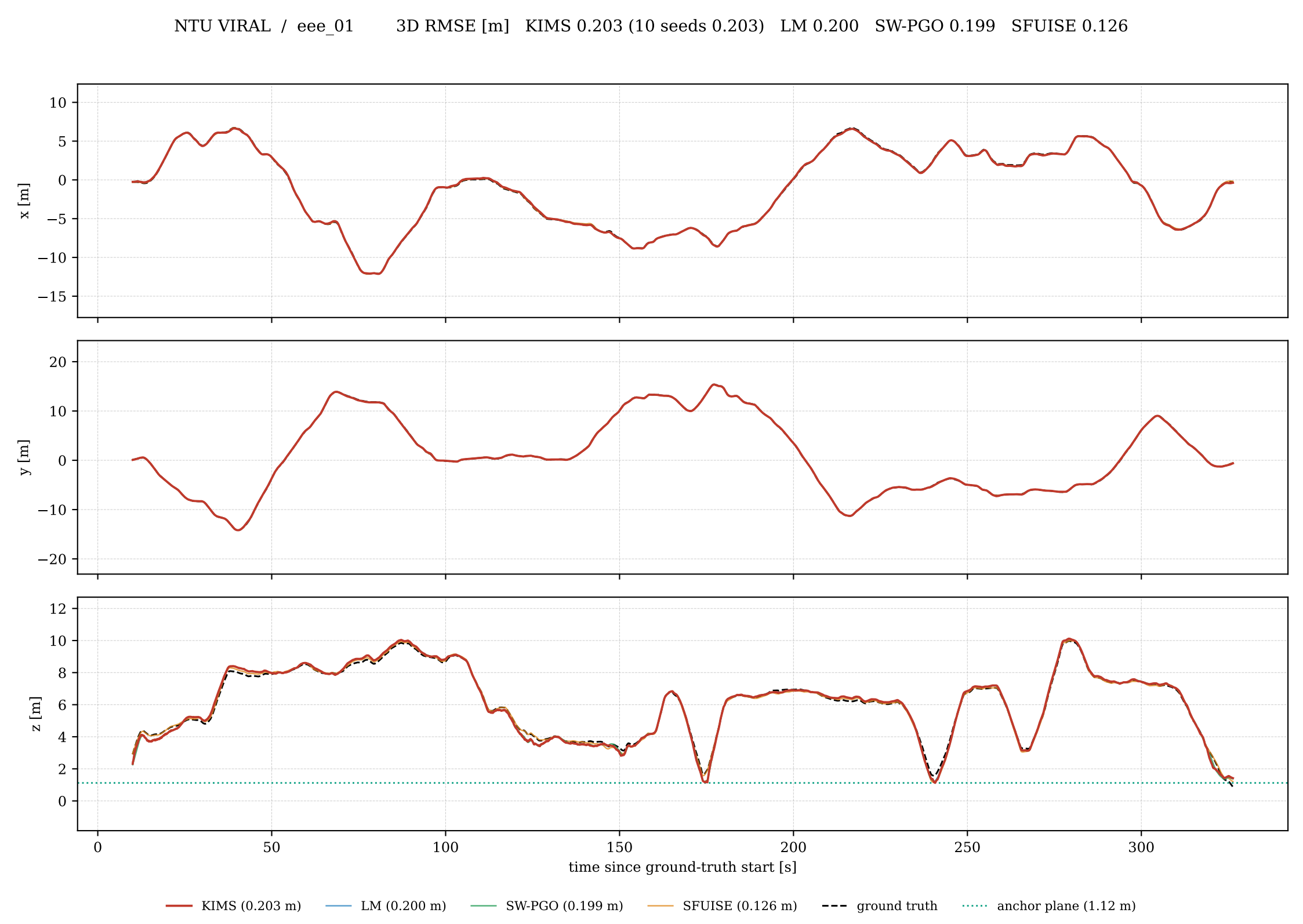

MILUV / obstacles_1_random3_0b 3D RMSE [m] KIMS 0.131 (10 seeds 0.132) LM 0.595 SW-PGO 0.185 SFUISE 0.639 EKF + height 0.252
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.131 m)
LM (0.595 m)
SW-PGO (0.185 m)
SFUISE (0.639 m)
EKF + height (0.252 m)
ground truth

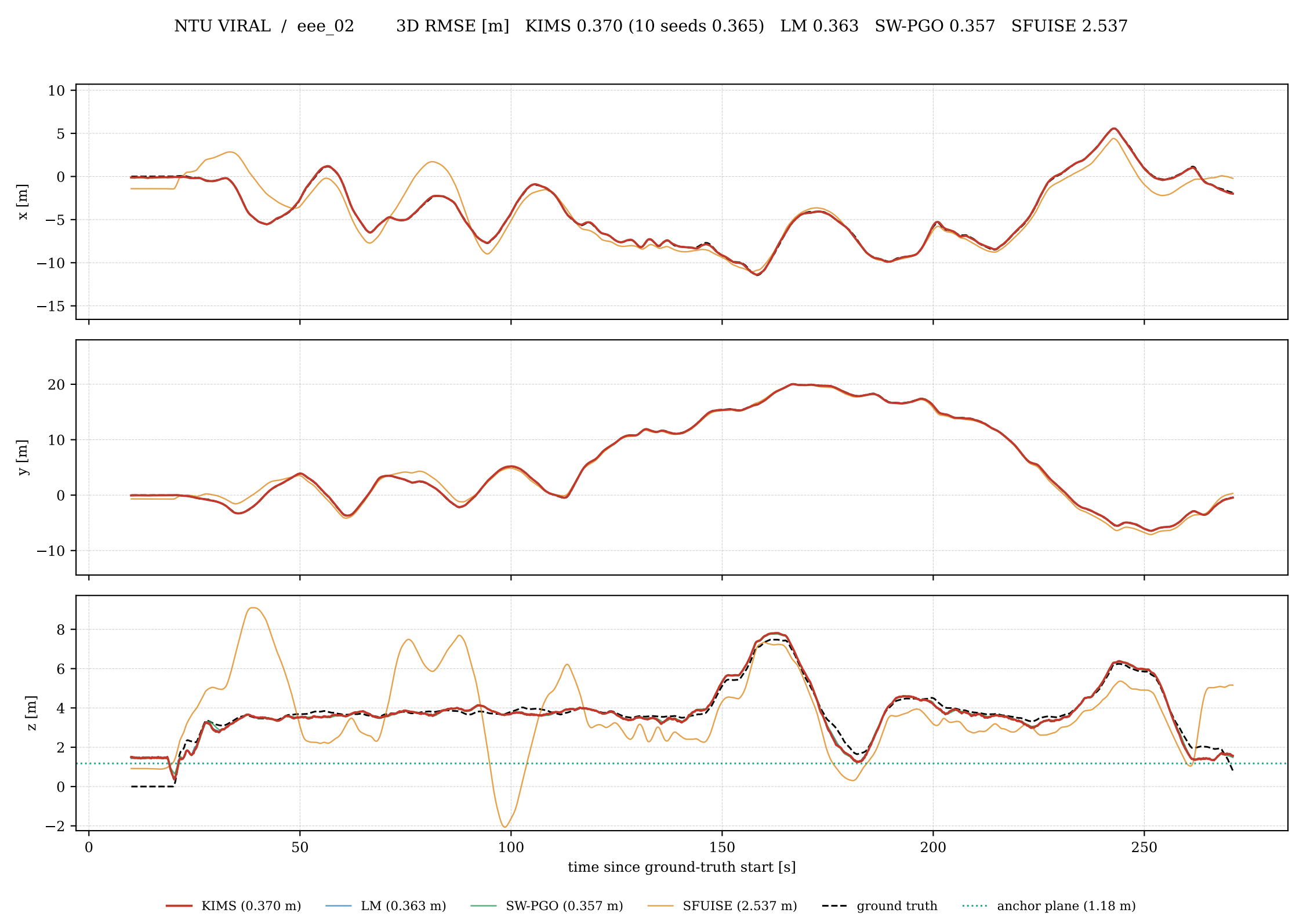

MILUV / default_1_random3_2 3D RMSE [m] KIMS 0.135 (10 seeds 0.135) LM 0.140 SW-PGO 0.209 SFUISE div. EKF + height 0.265
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.135 m)
LM (0.140 m)
SW-PGO (0.209 m)
SFUISE (div. m)
EKF + height (0.265 m)
ground truth

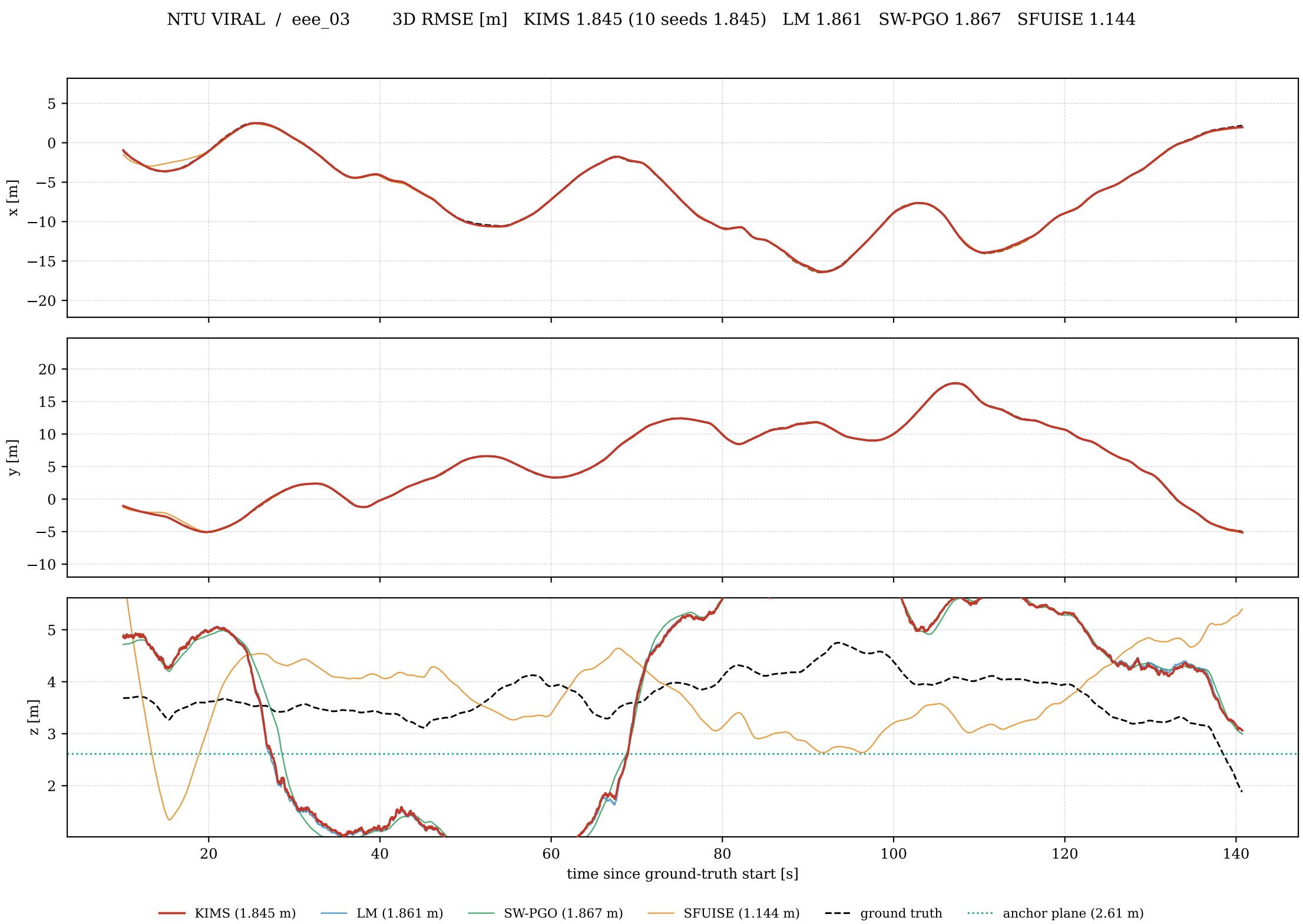

ILUV / default_1_remoteControlHighPace_2 3D RMSE [m] KIMS 0.245 (10 seeds 0.244) LM 0.241 SW-PGO 0.227 SFUISE 0.763 EKF + height 1.41
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.245 m)
LM (0.241 m)
SW-PGO (0.227 m)
SFUISE (0.763 m)
EKF + height (1.414 m)
ground truth

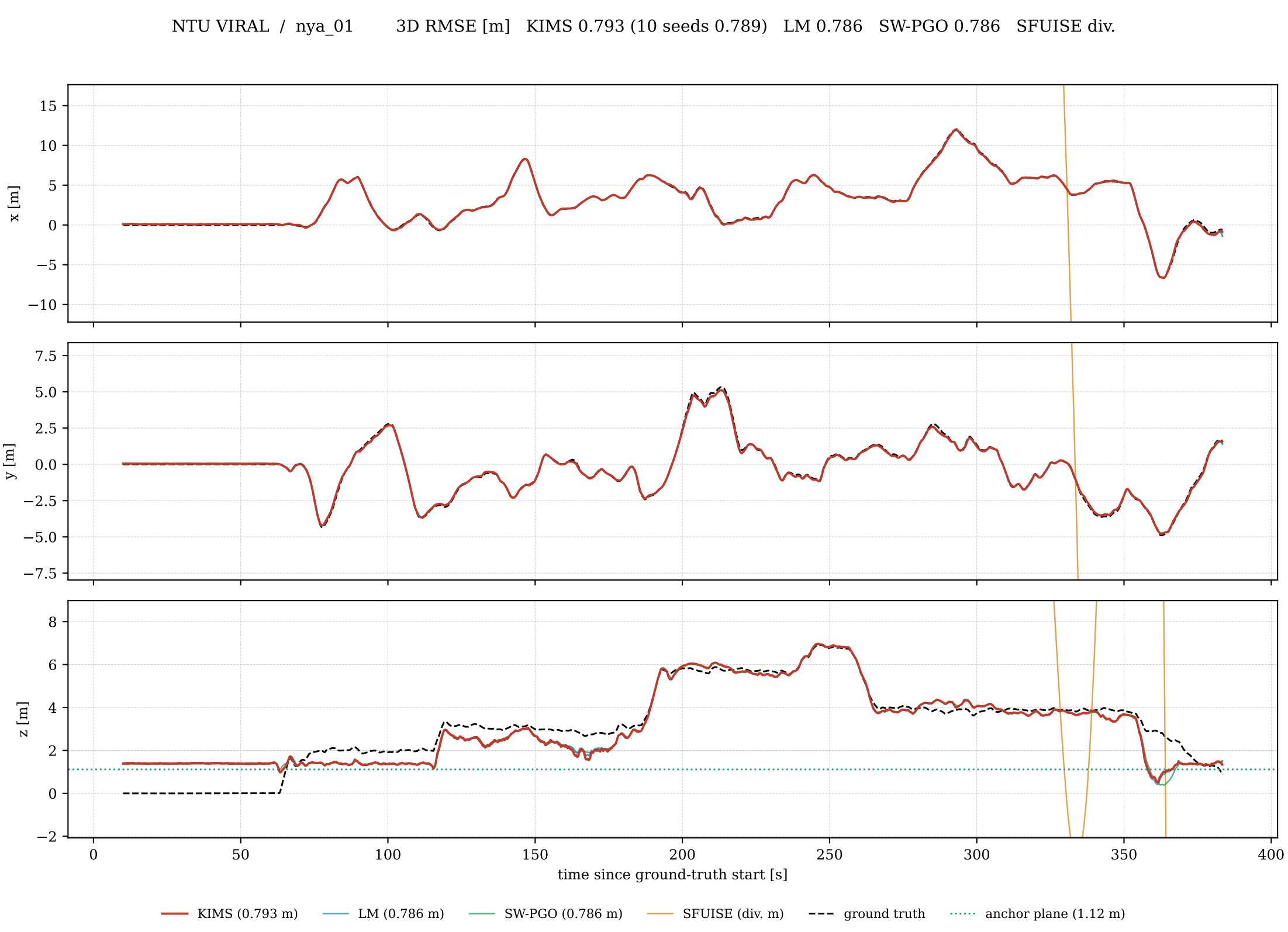

MILUV / noAprilTags_1_random3_2b 3D RMSE [m] KIMS 0.130 (10 seeds 0.130) LM 0.133 SW-PGO 0.175 SFUISE 0.368 EKF + height 0.241
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.130 m)
LM (0.133 m)
SW-PGO (0.175 m)
SFUISE (0.368 m)
EKF + height (0.241 m)
ground truth

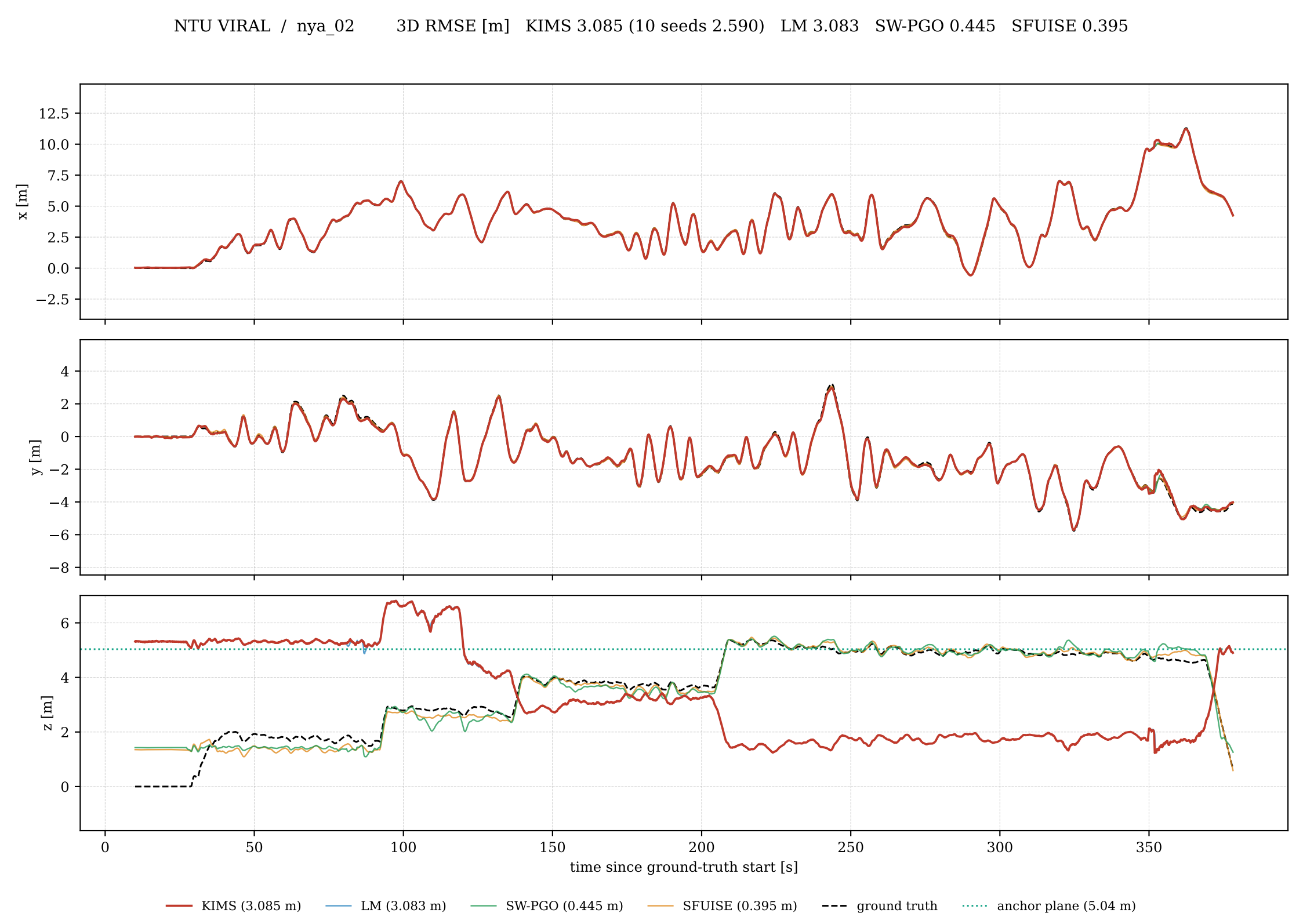

UV / noAprilTags_1_remoteControlHighPace_2b 3D RMSE [m] KIMS 0.191 (10 seeds 0.191) LM 0.197 SW-PGO 1.338 SFUISE 1.144 EKF + height
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.191 m)
LM (0.197 m)
SW-PGO (1.338 m)
SFUISE (1.144 m)
EKF + height (0.273 m)
ground truth

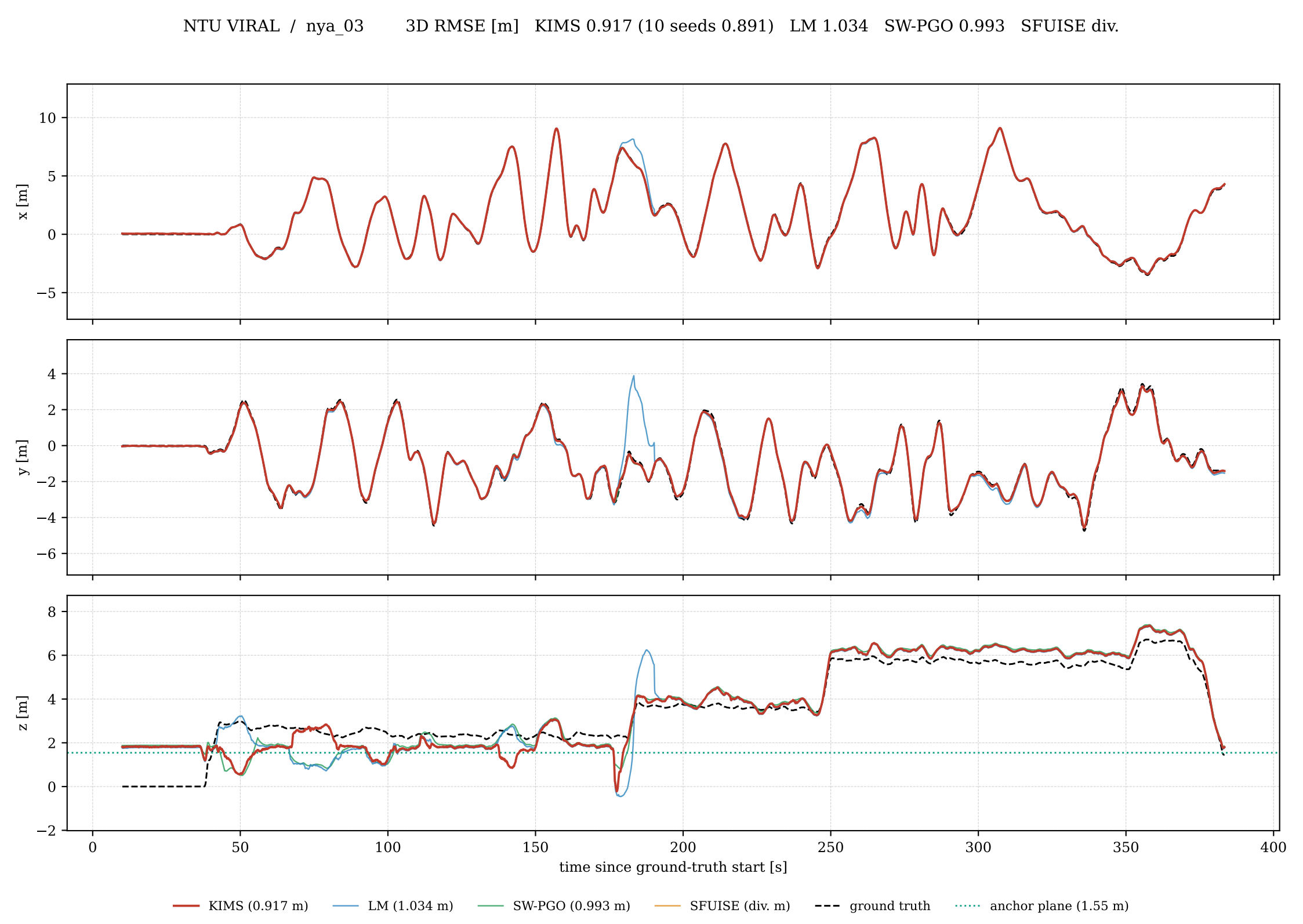

NTU VIRAL / eee_01 3D RMSE [m] KIMS 0.203 (10 seeds 0.203) LM 0.200 SW-PGO 0.199 SFUISE 0.126
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.203 m)
LM (0.200 m)
SW-PGO (0.199 m)
SFUISE (0.126 m)
ground truth
anchor plane (1.12 m)

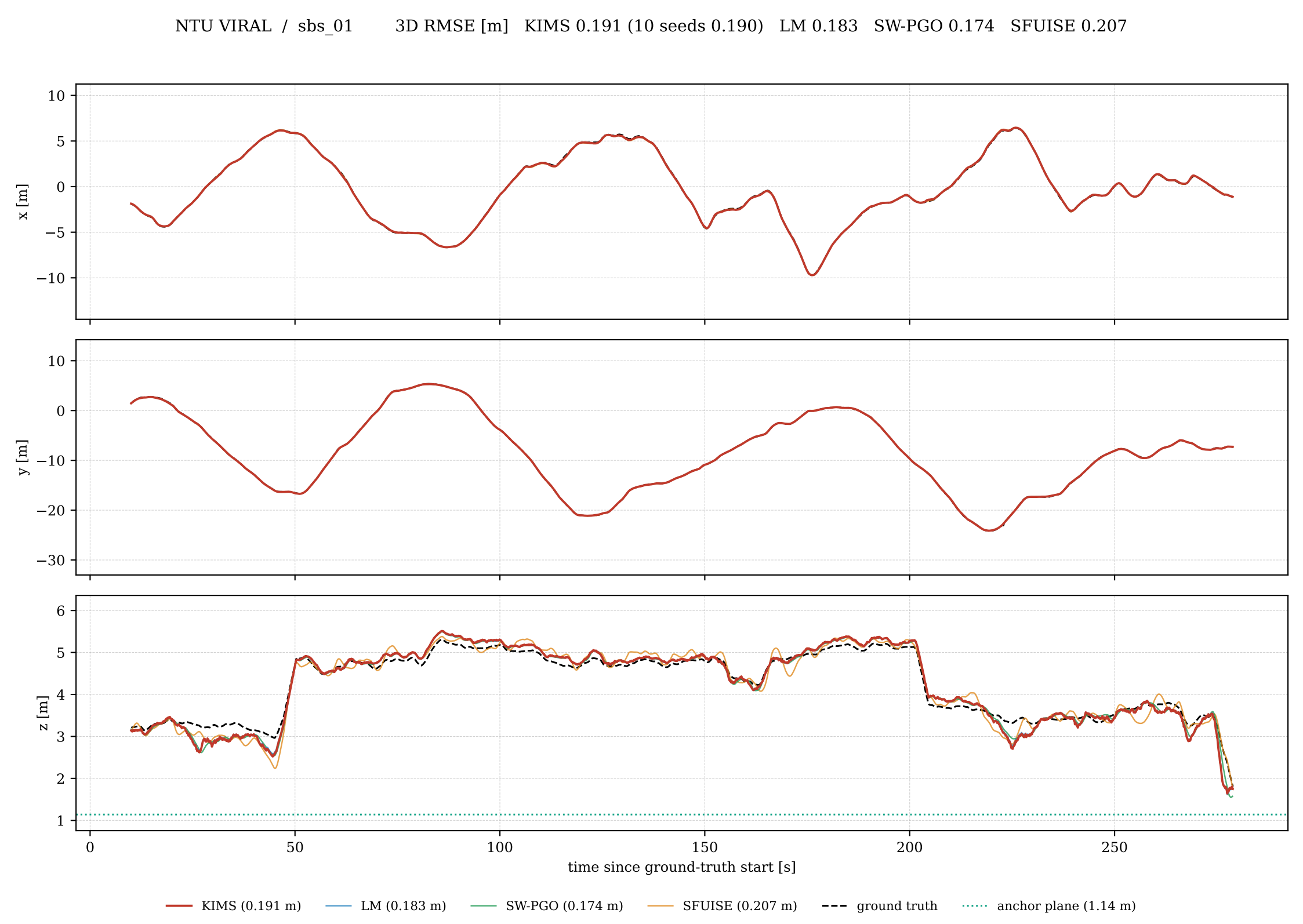

NTU VIRAL / eee_02 3D RMSE [m] KIMS 0.370 (10 seeds 0.365) LM 0.363 SW-PGO 0.357 SFUISE 2.537
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.370 m)
LM (0.363 m)
SW-PGO (0.357 m)
SFUISE (2.537 m)
ground truth
anchor plane (1.18 m)

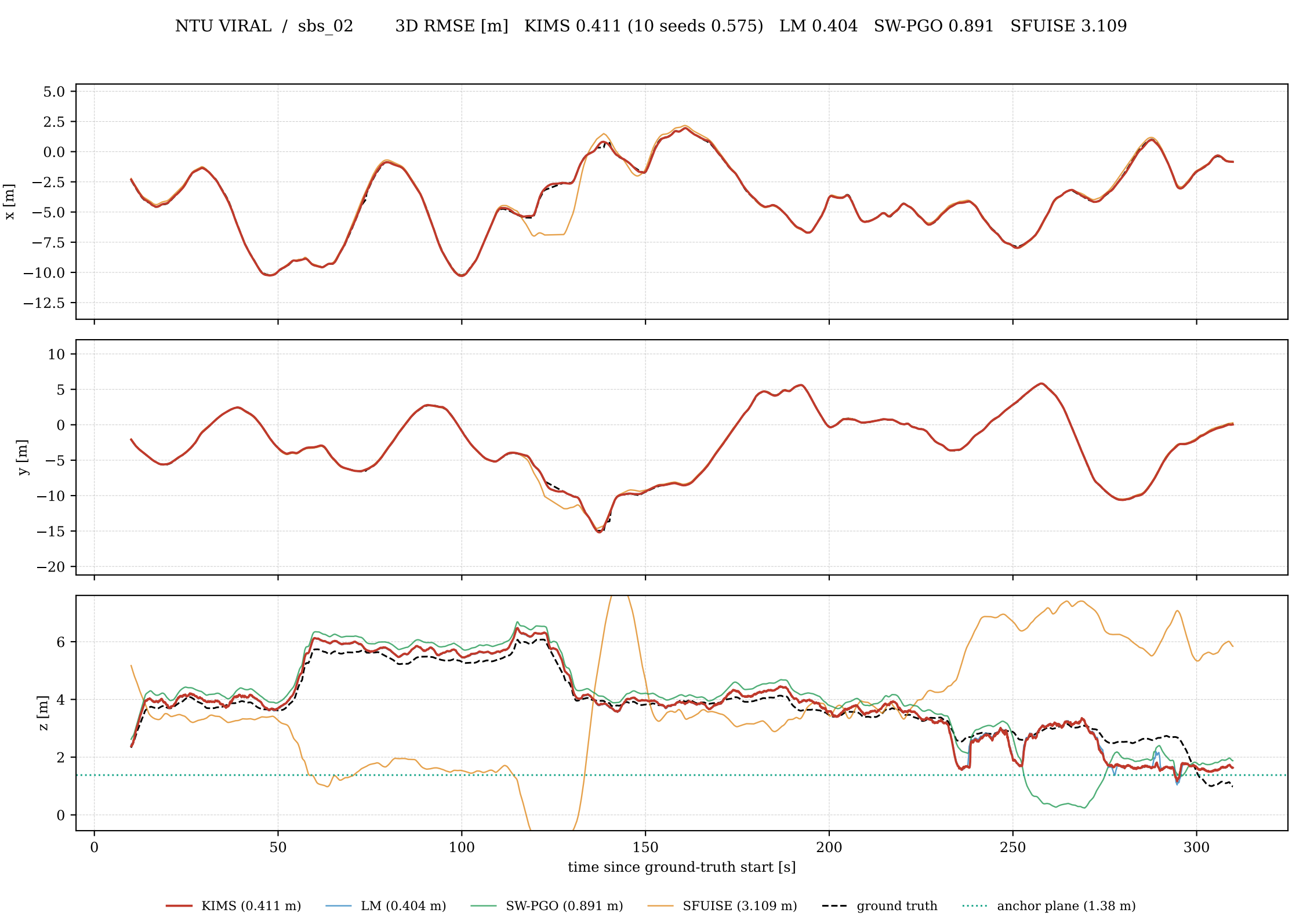

NTU VIRAL / eee_03 3D RMSE [m] KIMS 1.845 (10 seeds 1.845) LM 1.861 SW-PGO 1.867 SFUISE 1.144
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (1.845 m)
LM (1.861 m)
SW-PGO (1.867 m)
SFUISE (1.144 m)
ground truth
anchor plane (2.61 m)

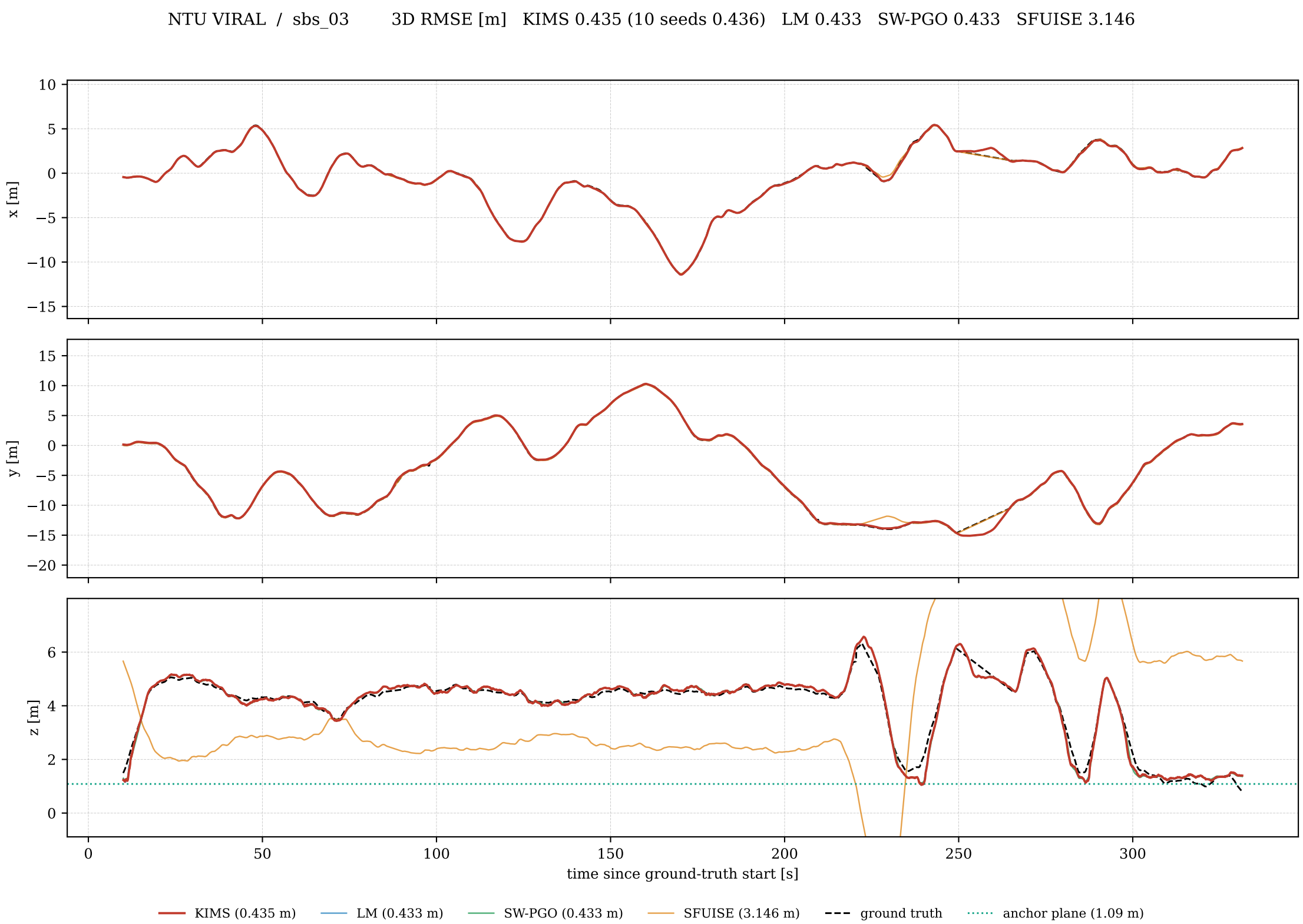

NTU VIRAL / nya_01 3D RMSE [m] KIMS 0.793 (10 seeds 0.789) LM 0.786 SW-PGO 0.786 SFUISE div.
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.793 m)
LM (0.786 m)
SW-PGO (0.786 m)
SFUISE (div. m)
ground truth
anchor plane (1.12 m)

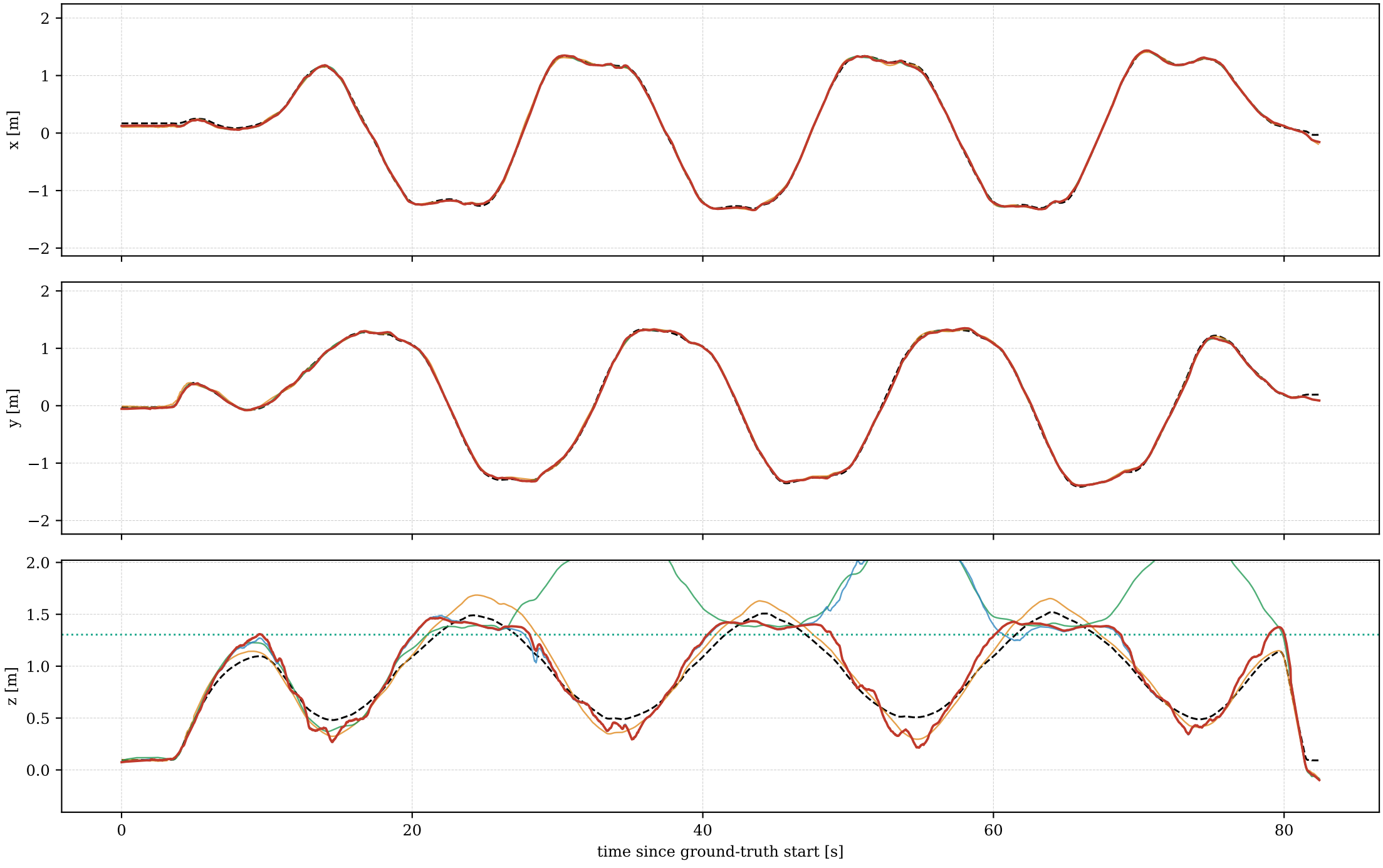

NTU VIRAL / nya_02 3D RMSE [m] KIMS 3.085 (10 seeds 2.590) LM 3.083 SW-PGO 0.445 SFUISE 0.395
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (3.085 m)
LM (3.083 m)
SW-PGO (0.445 m)
SFUISE (0.395 m)
ground truth
anchor plane (5.04 m)

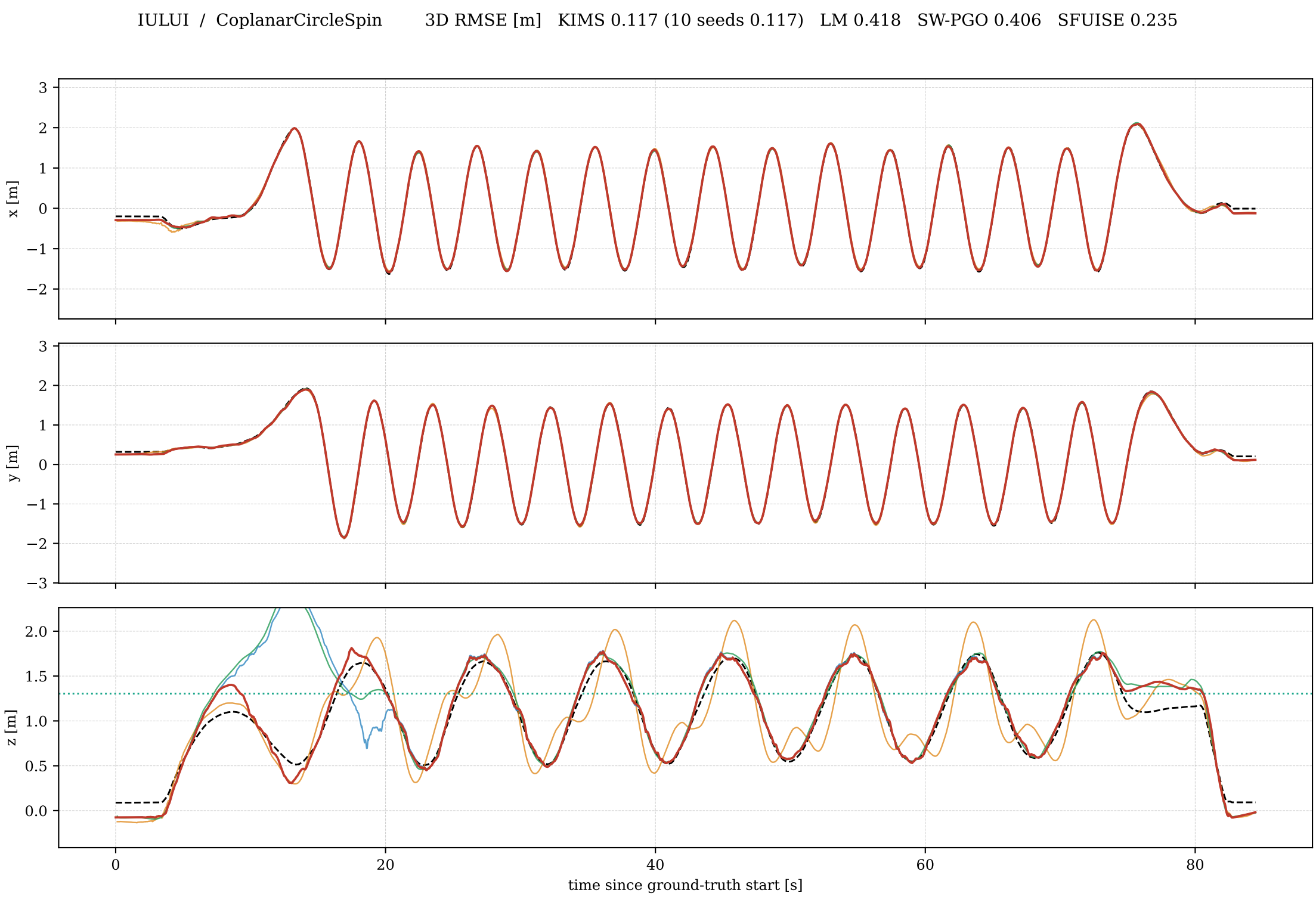

NTU VIRAL / nya_03 3D RMSE [m] KIMS 0.917 (10 seeds 0.891) LM 1.034 SW-PGO 0.993 SFUISE div.
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.917 m)
LM (1.034 m)
SW-PGO (0.993 m)
SFUISE (div. m)
ground truth
anchor plane (1.55 m)

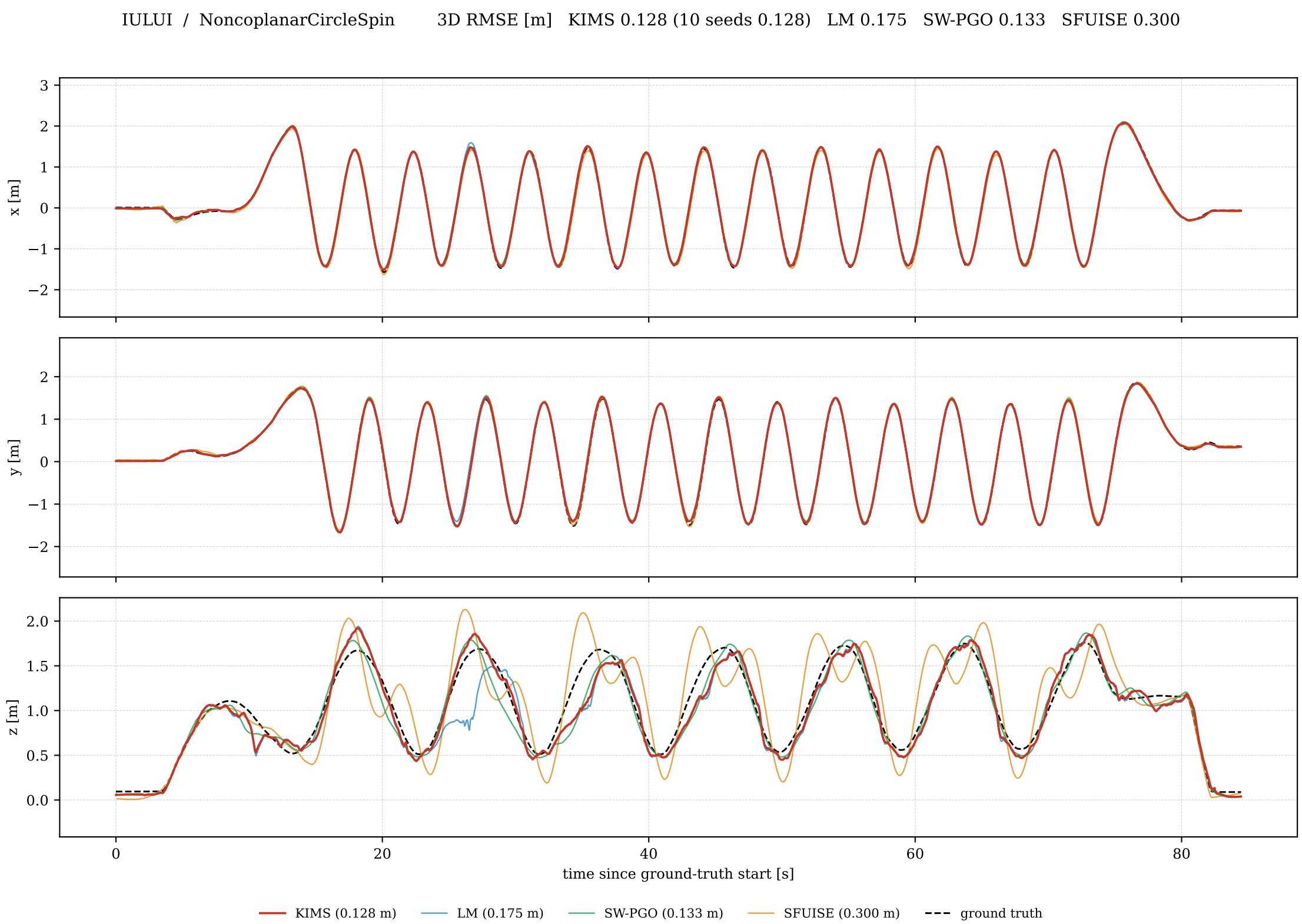

NTU VIRAL / sbs_01 3D RMSE [m] KIMS 0.191 (10 seeds 0.190) LM 0.183 SW-PGO 0.174 SFUISE 0.207
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.191 m)
LM (0.183 m)
SW-PGO (0.174 m)
SFUISE (0.207 m)
ground truth
anchor plane (1.14 m)

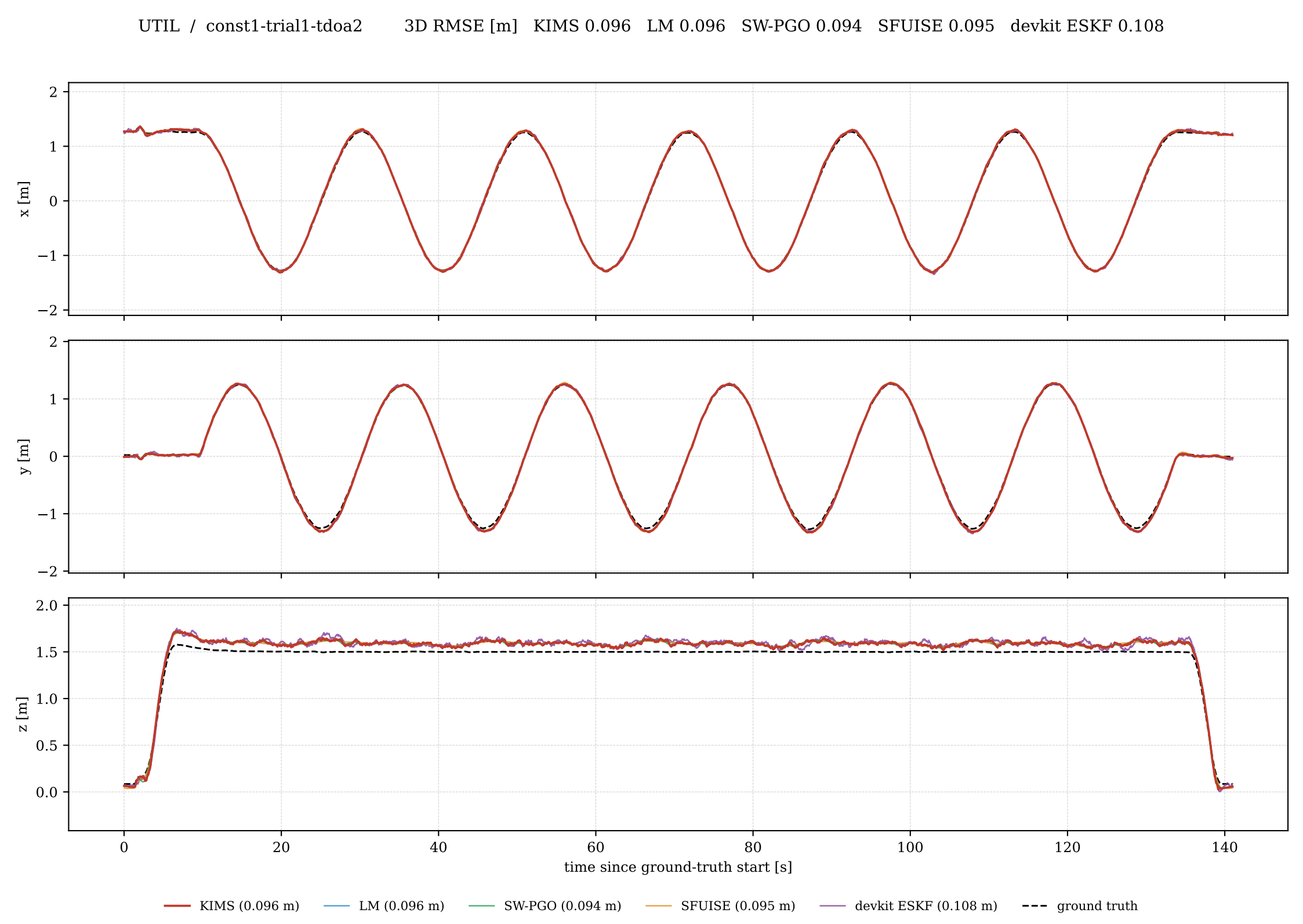

NTU VIRAL / sbs_02 3D RMSE [m] KIMS 0.411 (10 seeds 0.575) LM 0.404 SW-PGO 0.891 SFUISE 3.109
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.411 m)
LM (0.404 m)
SW-PGO (0.891 m)
SFUISE (3.109 m)
ground truth
anchor plane (1.38 m)

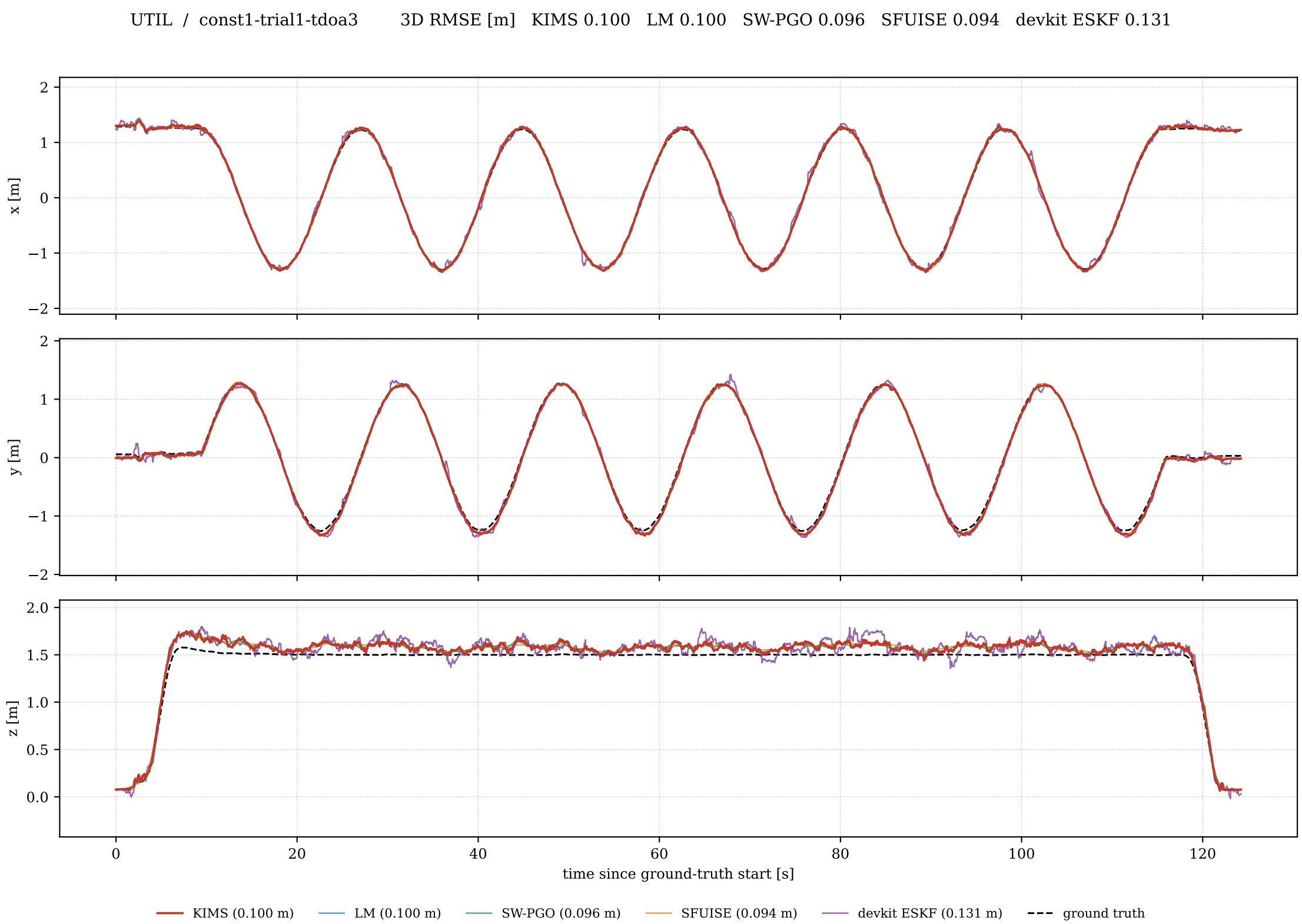

NTU VIRAL / sbs_03 3D RMSE [m] KIMS 0.435 (10 seeds 0.436) LM 0.433 SW-PGO 0.433 SFUISE 3.146
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.435 m)
LM (0.433 m)
SW-PGO (0.433 m)
SFUISE (3.146 m)
ground truth
anchor plane (1.09 m)

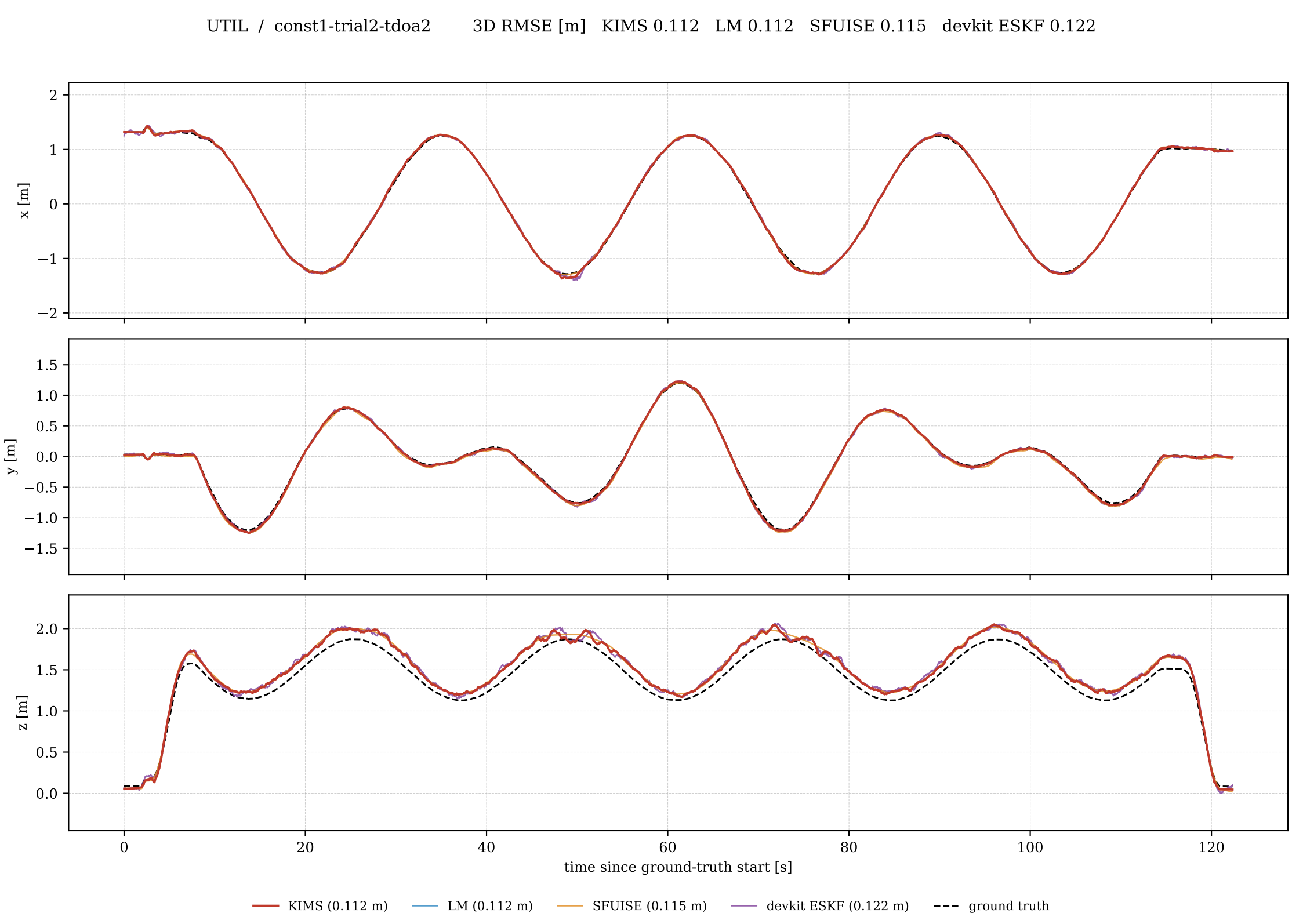

IULUI / CoplanarSquareTurn 3D RMSE [m] KIMS 0.120 (10 seeds 0.120) LM 0.548 SW-PGO 0.912 SFUISE 0.106
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.120 m)
LM (0.548 m)
SW-PGO (0.912 m)
SFUISE (0.106 m)
ground truth
anchor plane (1.30 m)

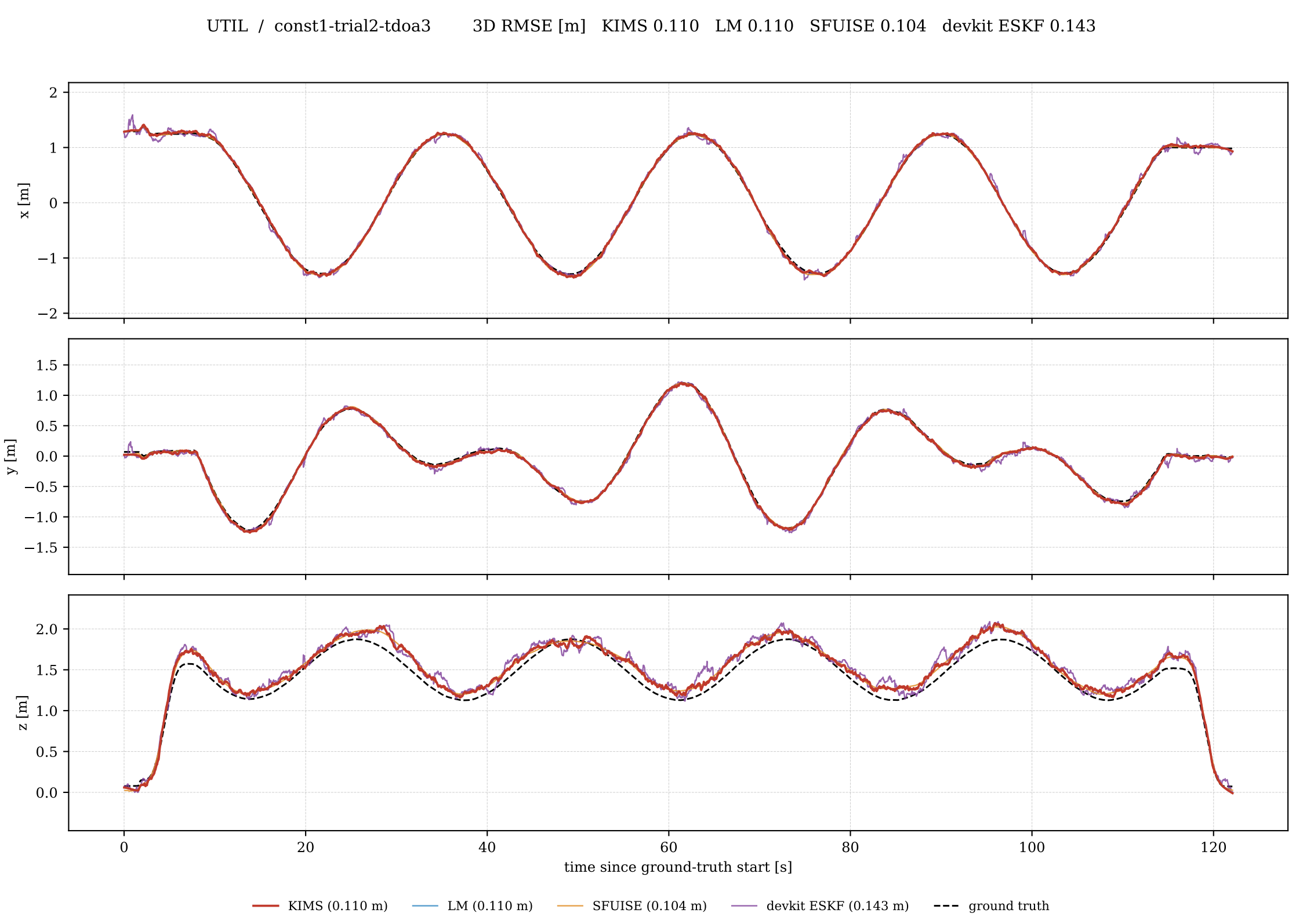

IULUI / CoplanarCircleSpin 3D RMSE [m] KIMS 0.117 (10 seeds 0.117) LM 0.418 SW-PGO 0.406 SFUISE 0.235
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.117 m)
LM (0.418 m)
SW-PGO (0.406 m)
SFUISE (0.235 m)
ground truth
anchor plane (1.30 m)

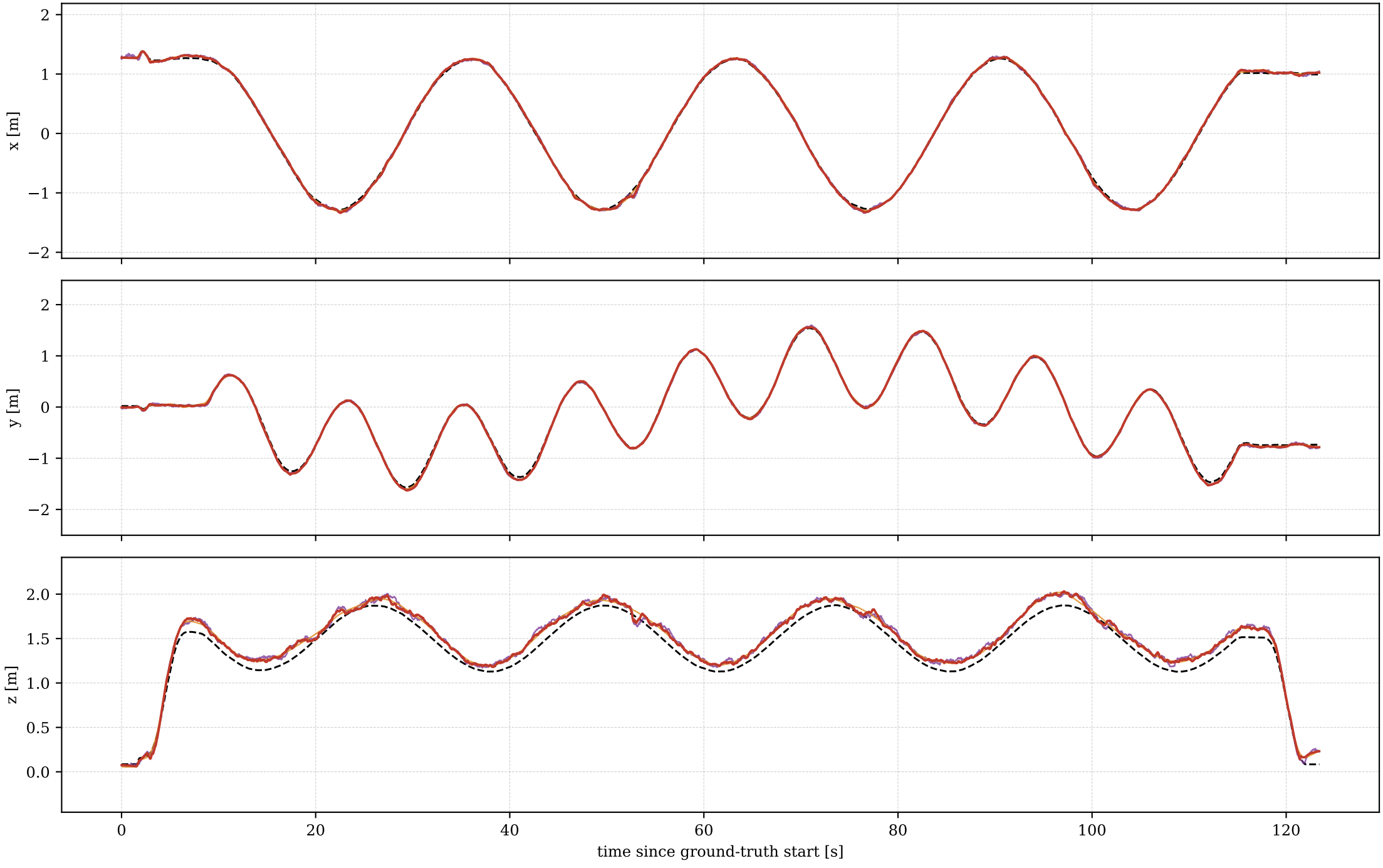

IULUI / NoncoplanarCircleSpin 3D RMSE [m] KIMS 0.128 (10 seeds 0.128) LM 0.175 SW-PGO 0.133 SFUISE 0.300
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.128 m)
LM (0.175 m)
SW-PGO (0.133 m)
SFUISE (0.300 m)
ground truth

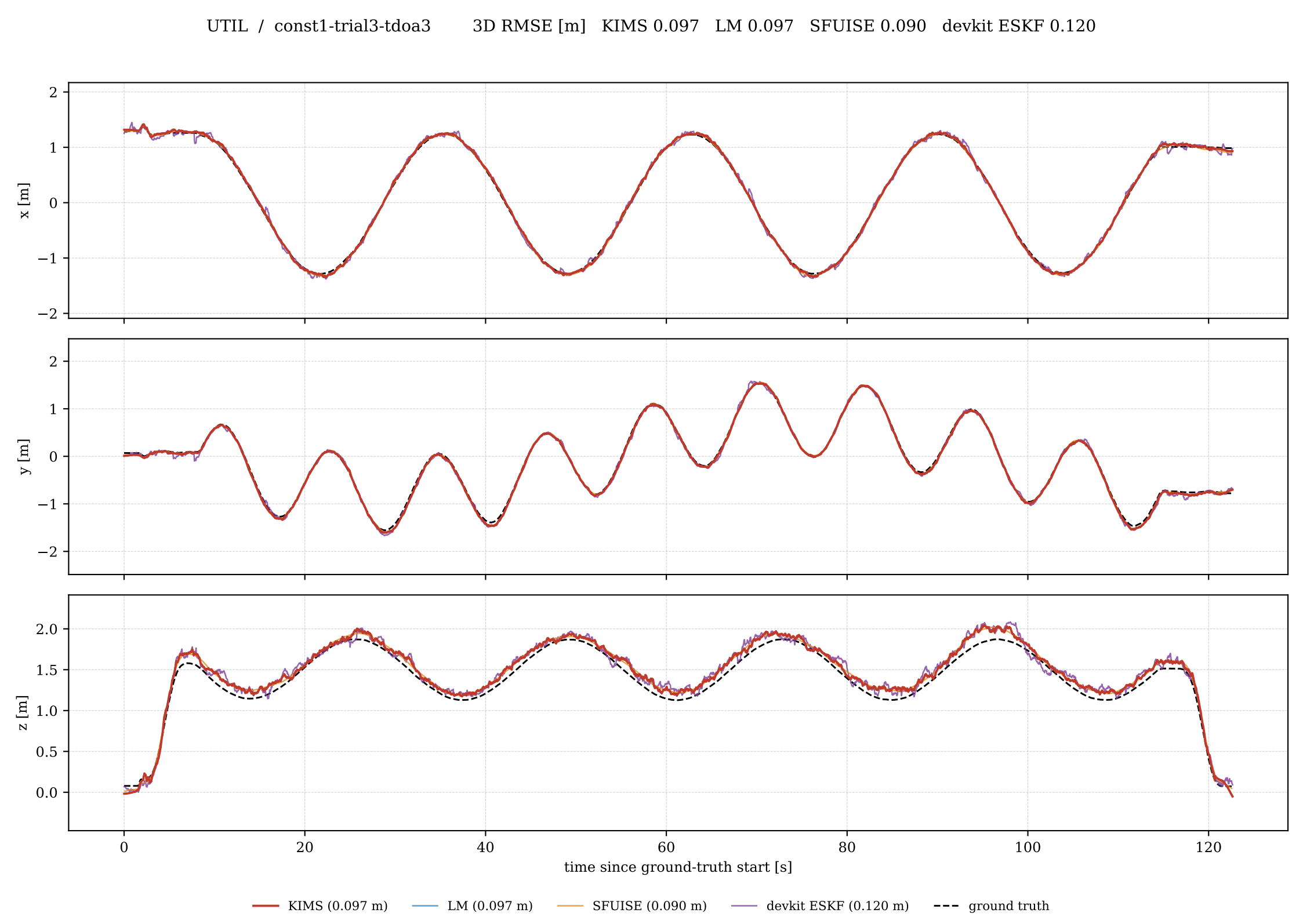

UTIL / const1-trial1-tdoa2 3D RMSE [m] KIMS 0.096 LM 0.096 SW-PGO 0.094 SFUISE 0.095 devkit ESKF 0.108
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.096 m)
LM (0.096 m)
SW-PGO (0.094 m)
SFUISE (0.095 m)
devkit ESKF (0.108 m)
ground truth

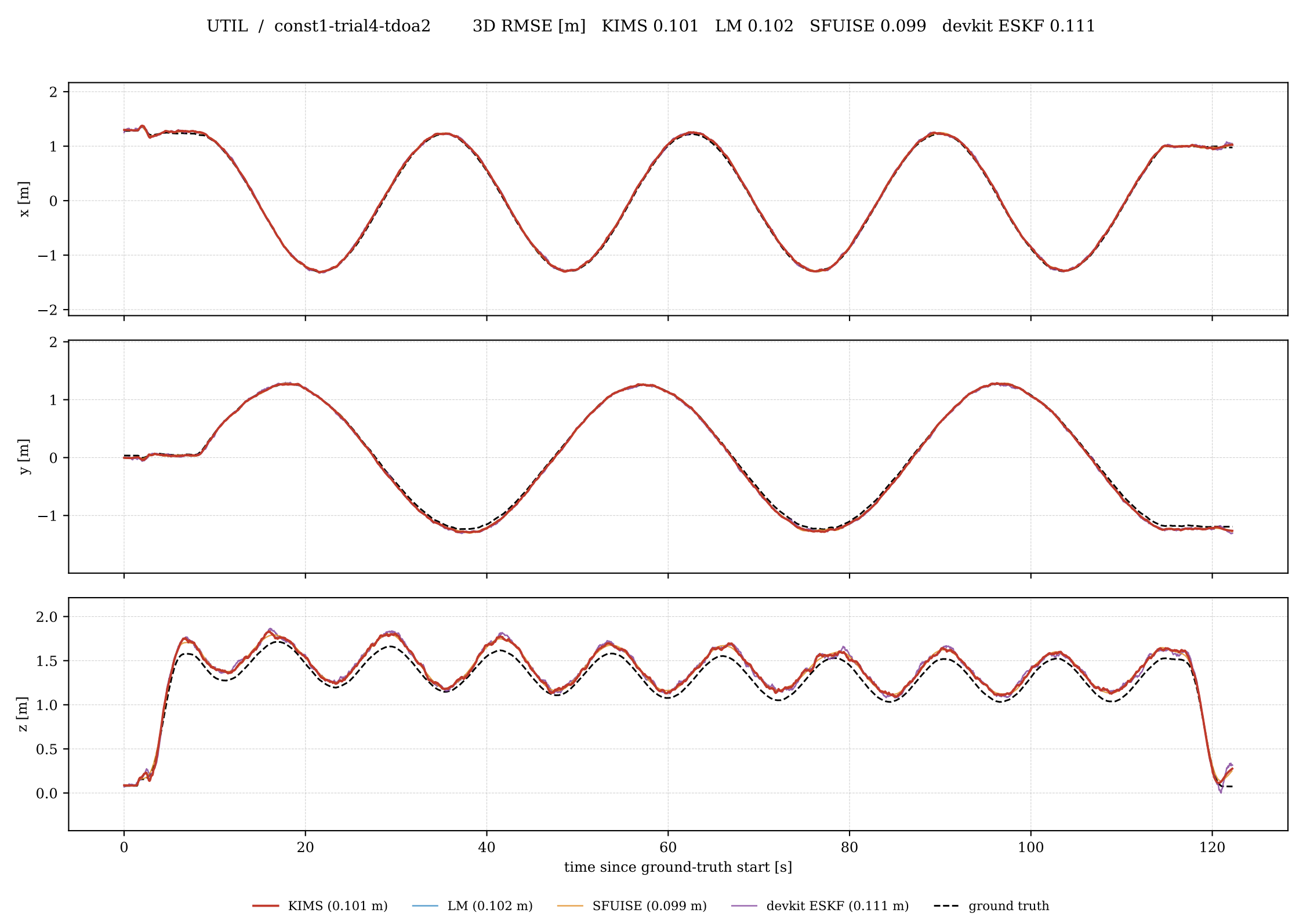

UTIL / const1-trial1-tdoa3 3D RMSE [m] KIMS 0.100 LM 0.100 SW-PGO 0.096 SFUISE 0.094 devkit ESKF 0.131
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.100 m)
LM (0.100 m)
SW-PGO (0.096 m)
SFUISE (0.094 m)
devkit ESKF (0.131 m)
ground truth

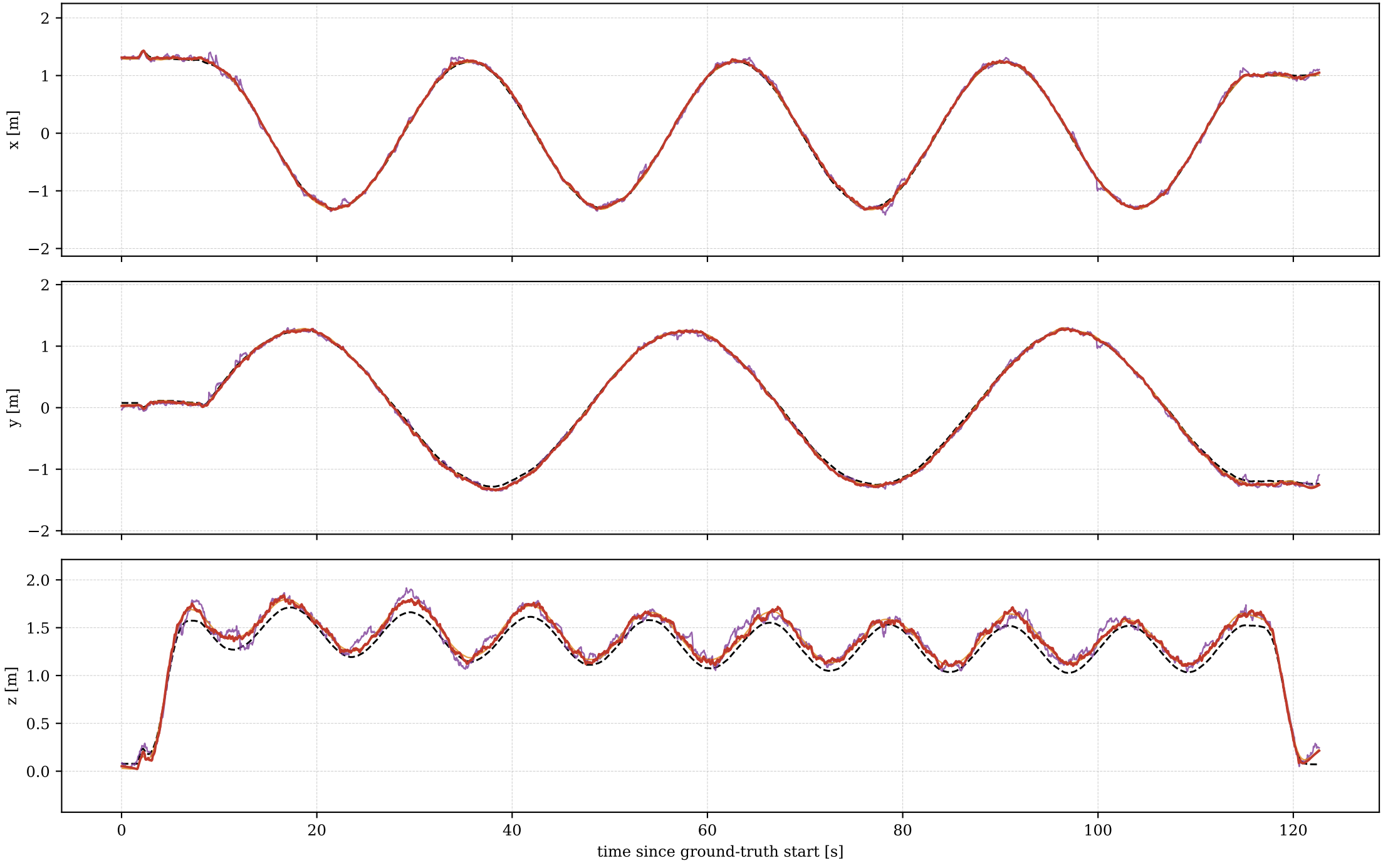

UTIL / const1-trial2-tdoa2 3D RMSE [m] KIMS 0.112 LM 0.112 SFUISE 0.115 devkit ESKF 0.122
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.112 m)
LM (0.112 m)
SFUISE (0.115 m)
devkit ESKF (0.122 m)
ground truth

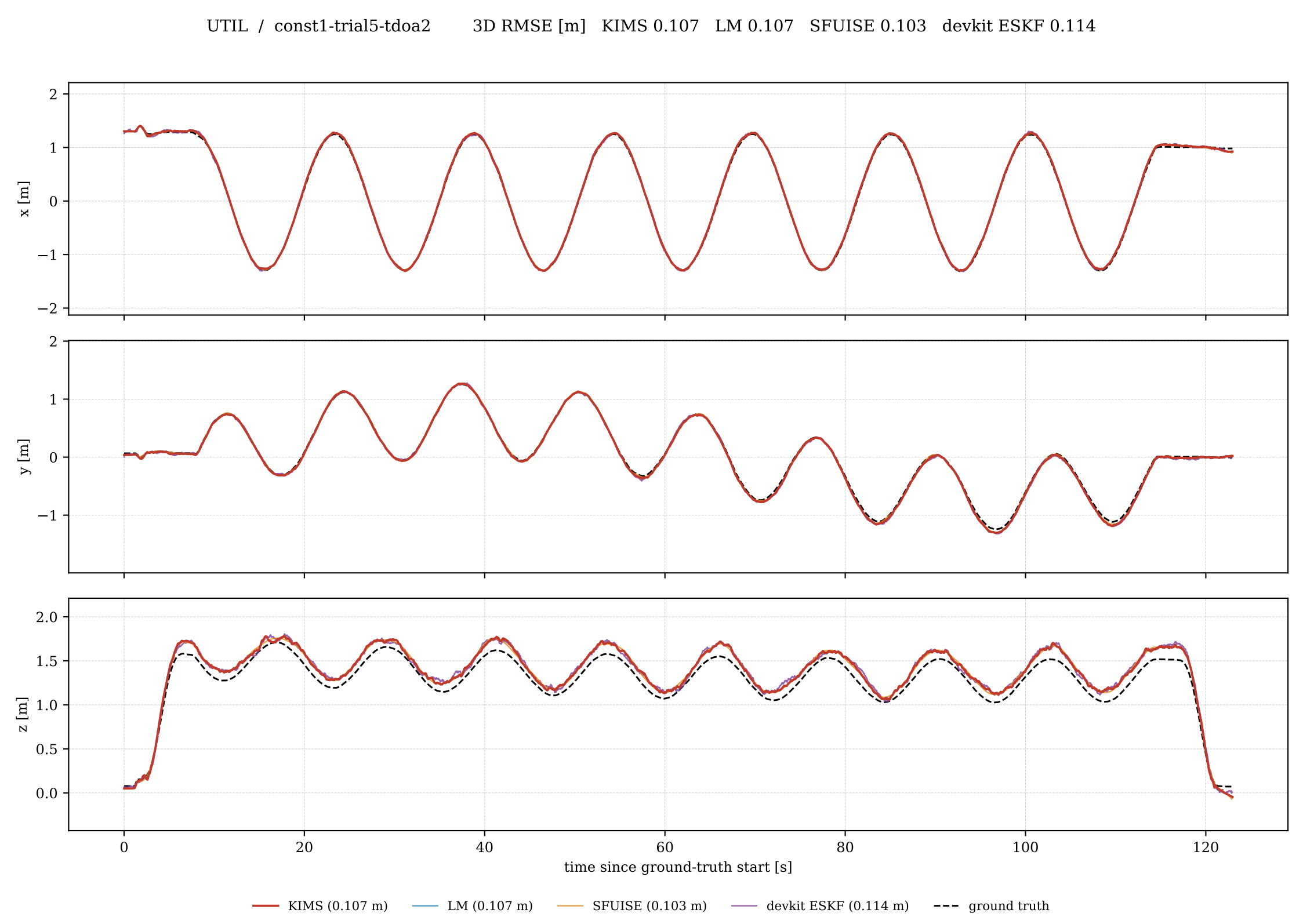

UTIL / const1-trial2-tdoa3 3D RMSE [m] KIMS 0.110 LM 0.110 SFUISE 0.104 devkit ESKF 0.143
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.110 m)
LM (0.110 m)
SFUISE (0.104 m)
devkit ESKF (0.143 m)
ground truth

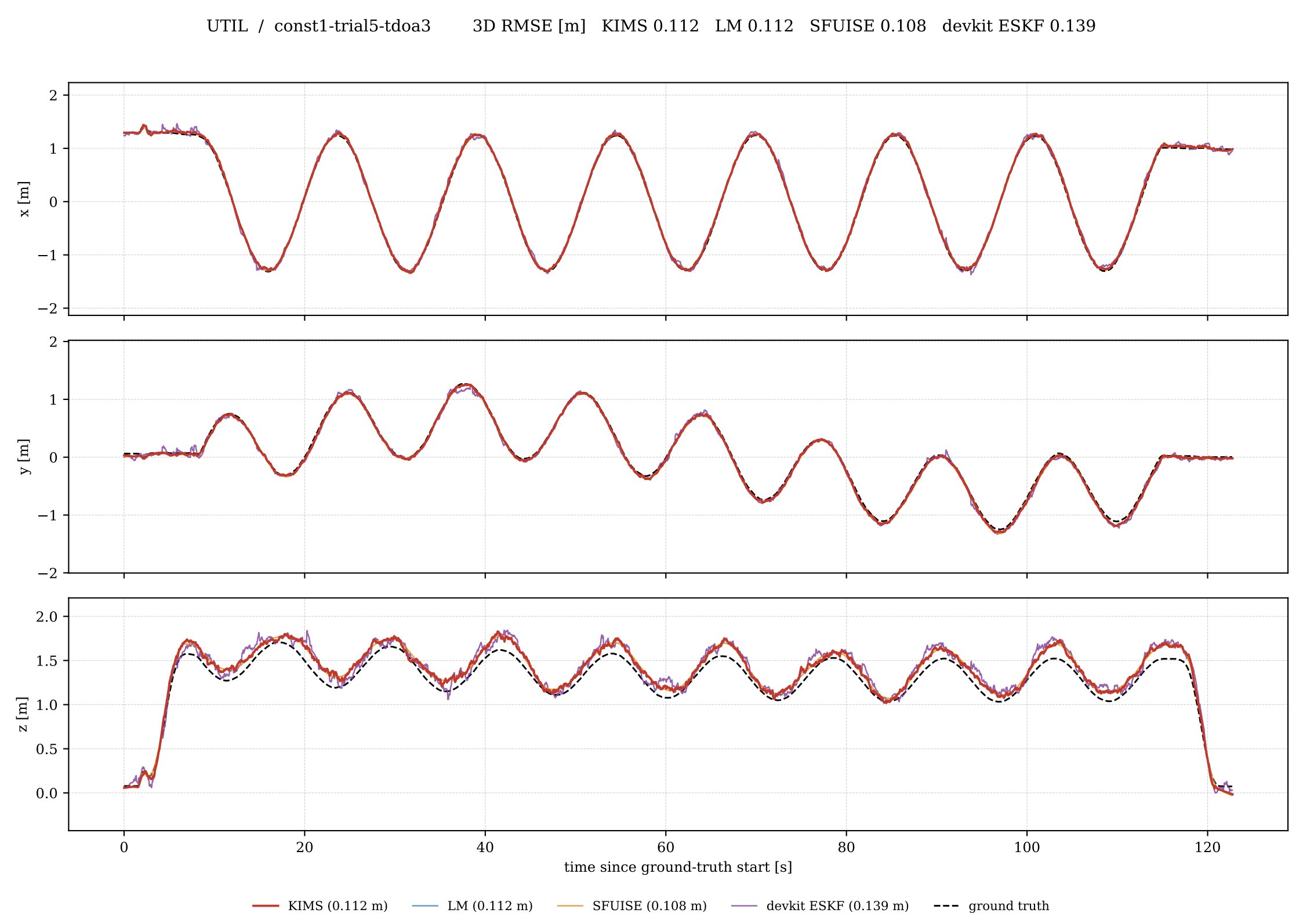

UTIL / const1-trial3-tdoa2 3D RMSE [m] KIMS 0.098 LM 0.098 SFUISE 0.094 devkit ESKF 0.108
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.098 m)
LM (0.098 m)
SFUISE (0.094 m)
devkit ESKF (0.108 m)
ground truth

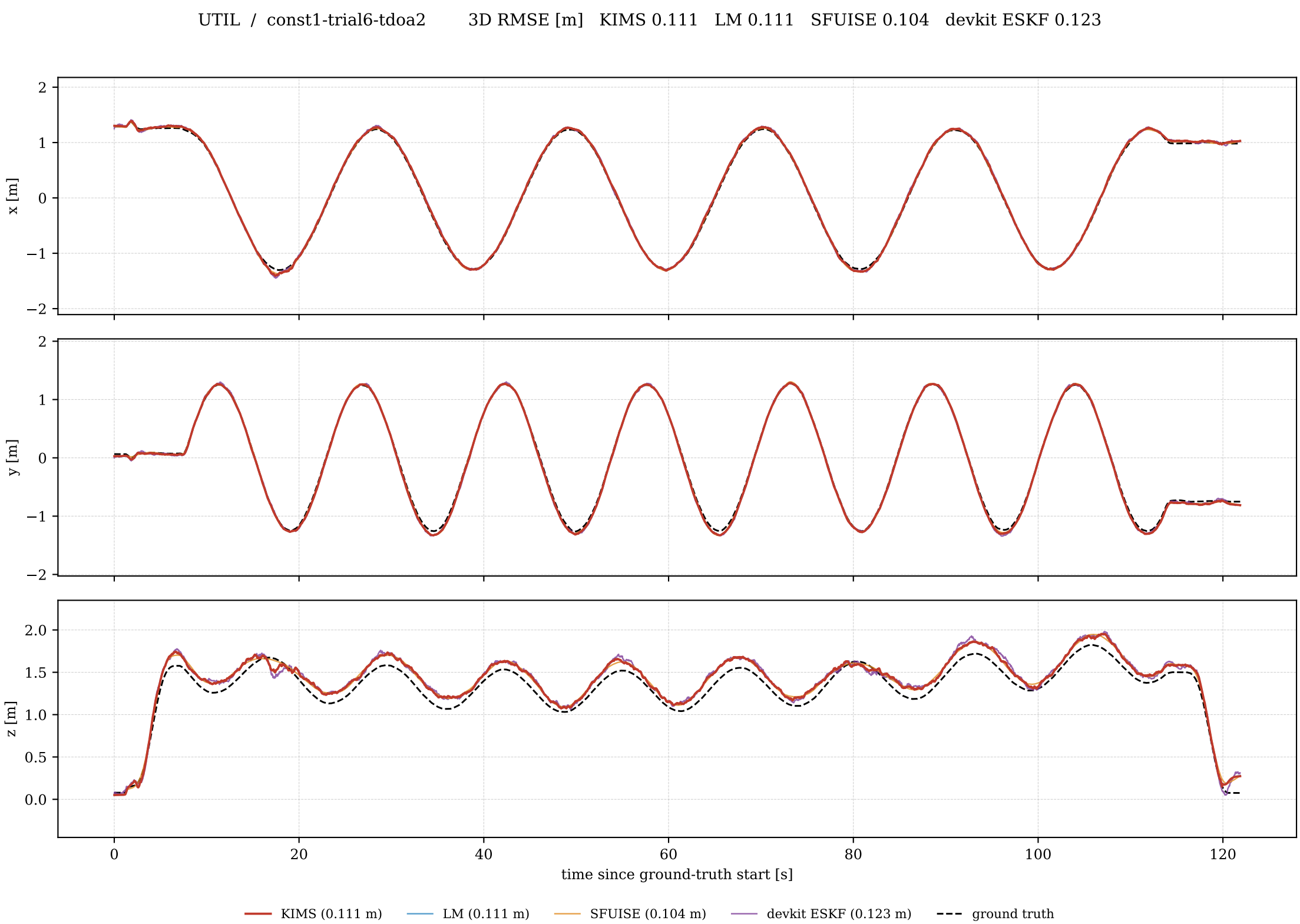

UTIL / const1-trial3-tdoa3 3D RMSE [m] KIMS 0.097 LM 0.097 SFUISE 0.090 devkit ESKF 0.120
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.097 m)
LM (0.097 m)
SFUISE (0.090 m)
devkit ESKF (0.120 m)
ground truth

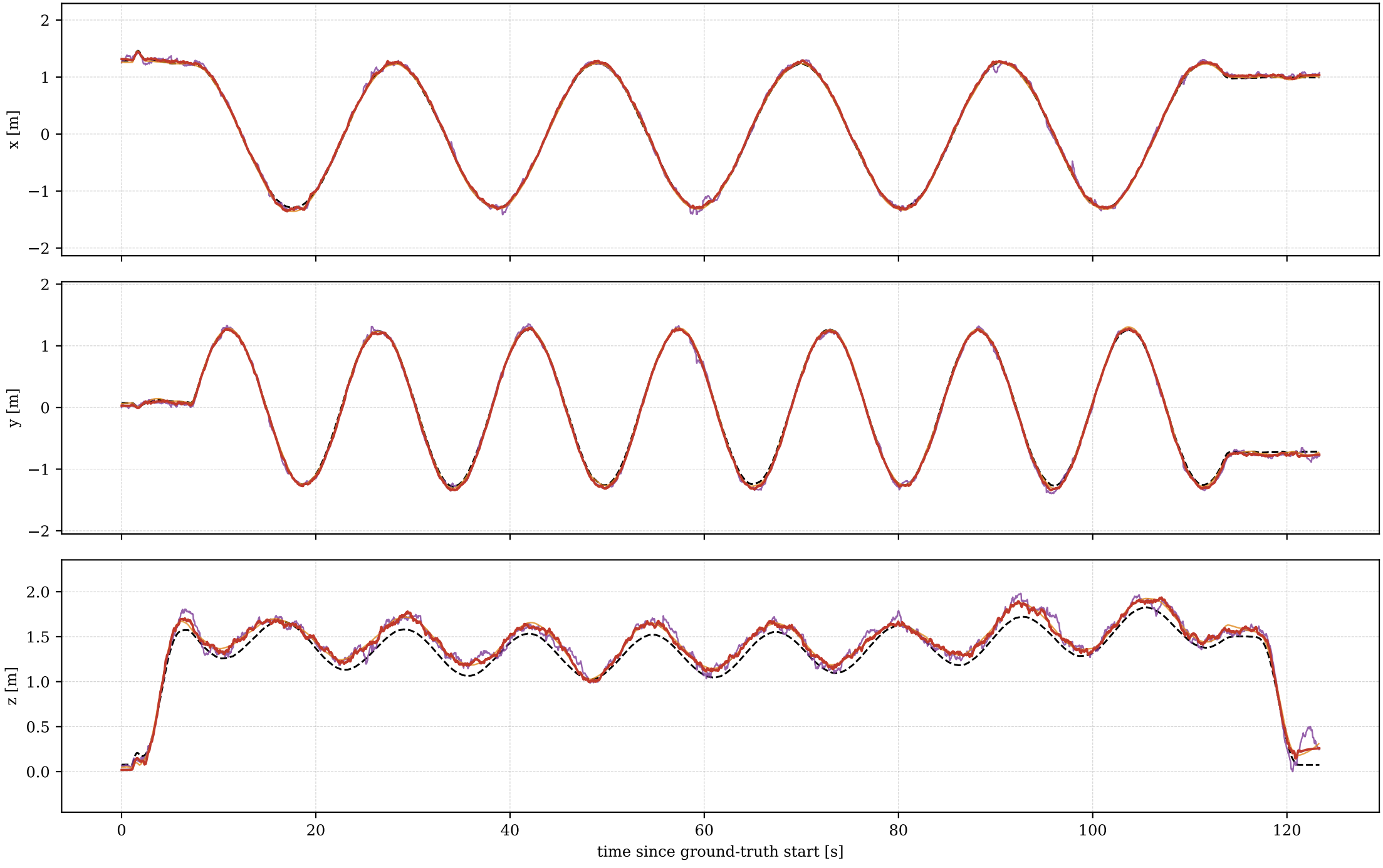

UTIL / const1-trial4-tdoa2 3D RMSE [m] KIMS 0.101 LM 0.102 SFUISE 0.099 devkit ESKF 0.111
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.101 m)
LM (0.102 m)
SFUISE (0.099 m)
devkit ESKF (0.111 m)
ground truth

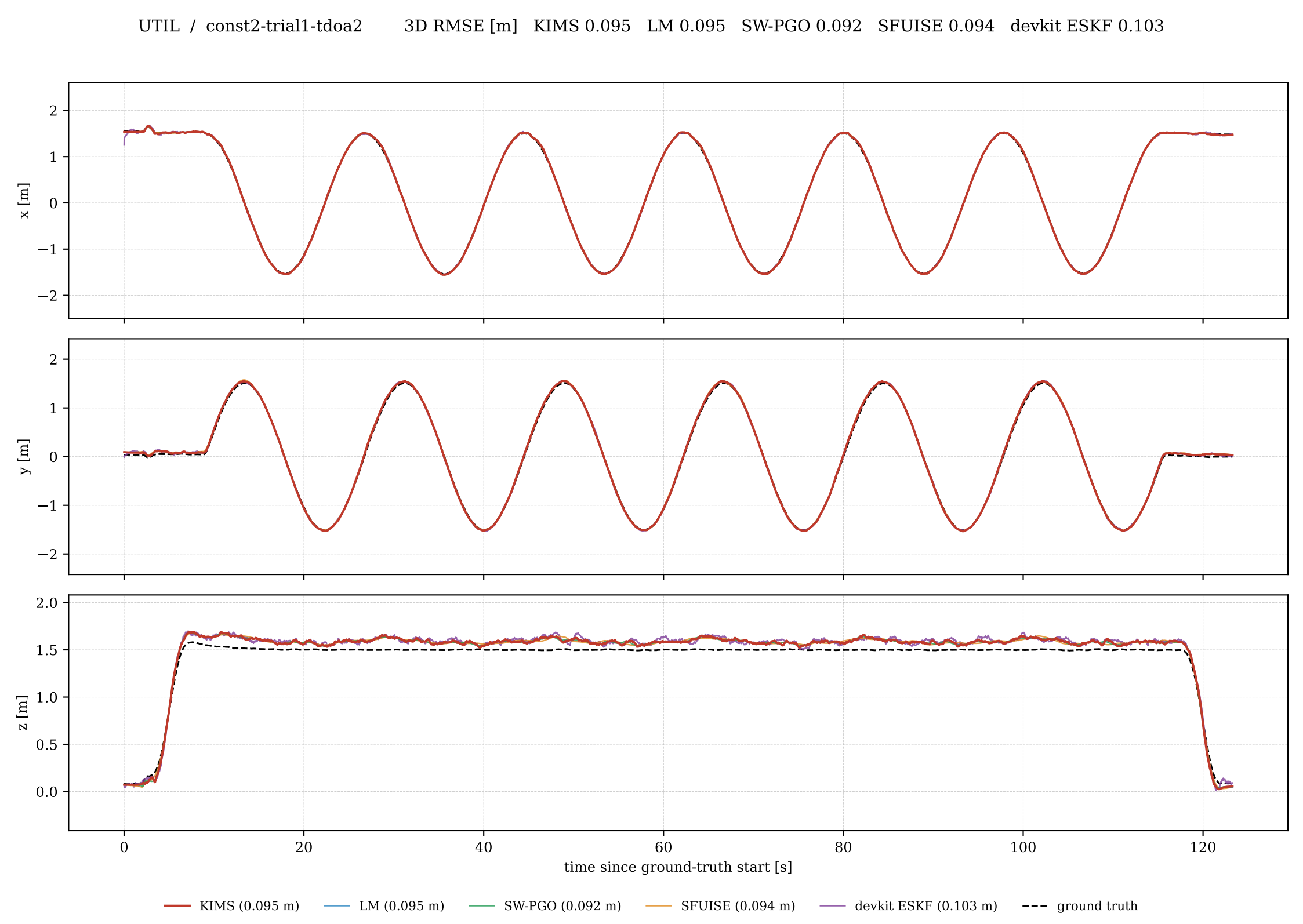

UTIL / const1-trial4-tdoa3 3D RMSE [m] KIMS 0.096 LM 0.097 SFUISE 0.090 devkit ESKF 0.124
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.096 m)
LM (0.097 m)
SFUISE (0.090 m)
devkit ESKF (0.124 m)
ground truth

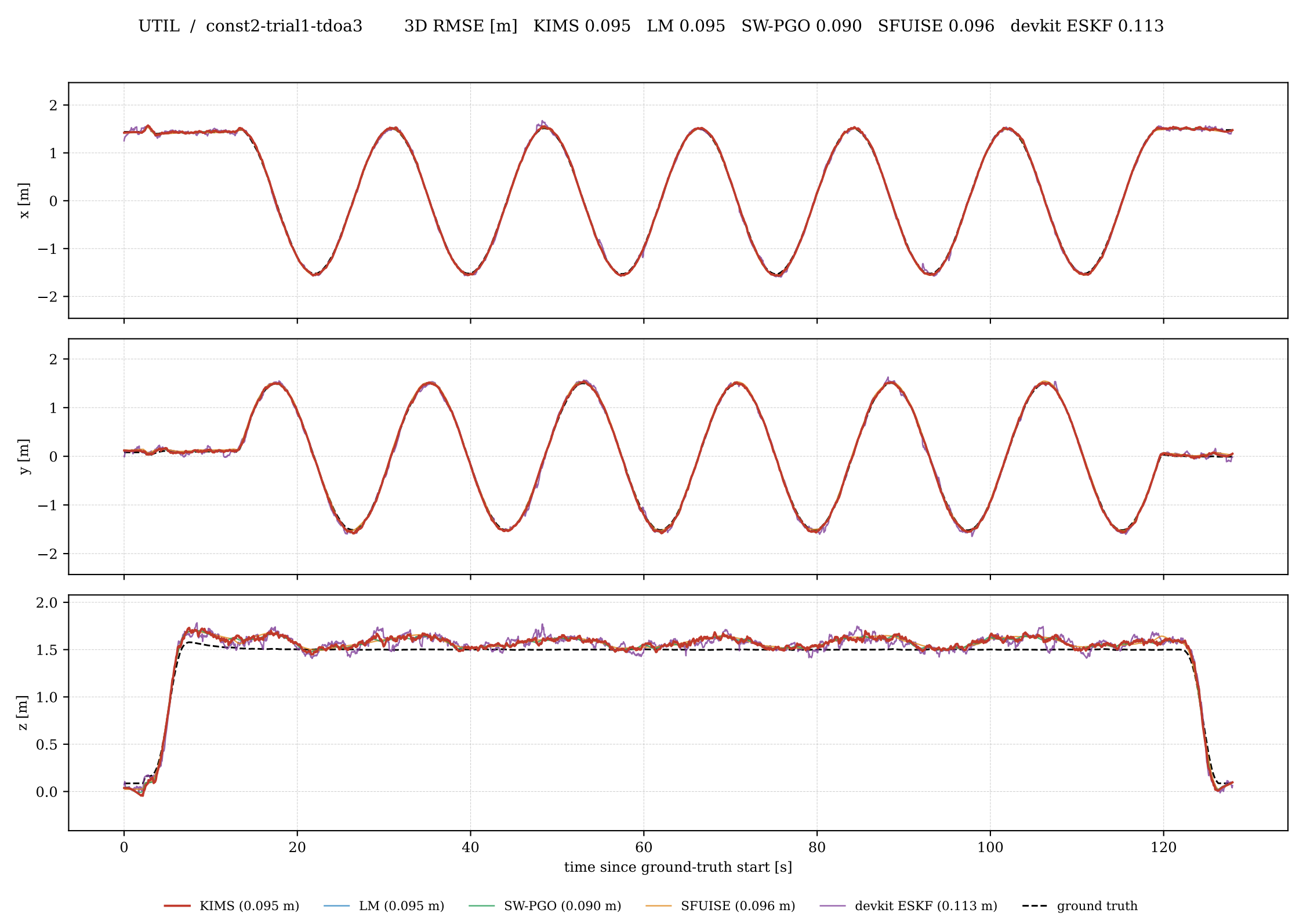

UTIL / const1-trial5-tdoa2 3D RMSE [m] KIMS 0.107 LM 0.107 SFUISE 0.103 devkit ESKF 0.114
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.107 m)
LM (0.107 m)
SFUISE (0.103 m)
devkit ESKF (0.114 m)
ground truth

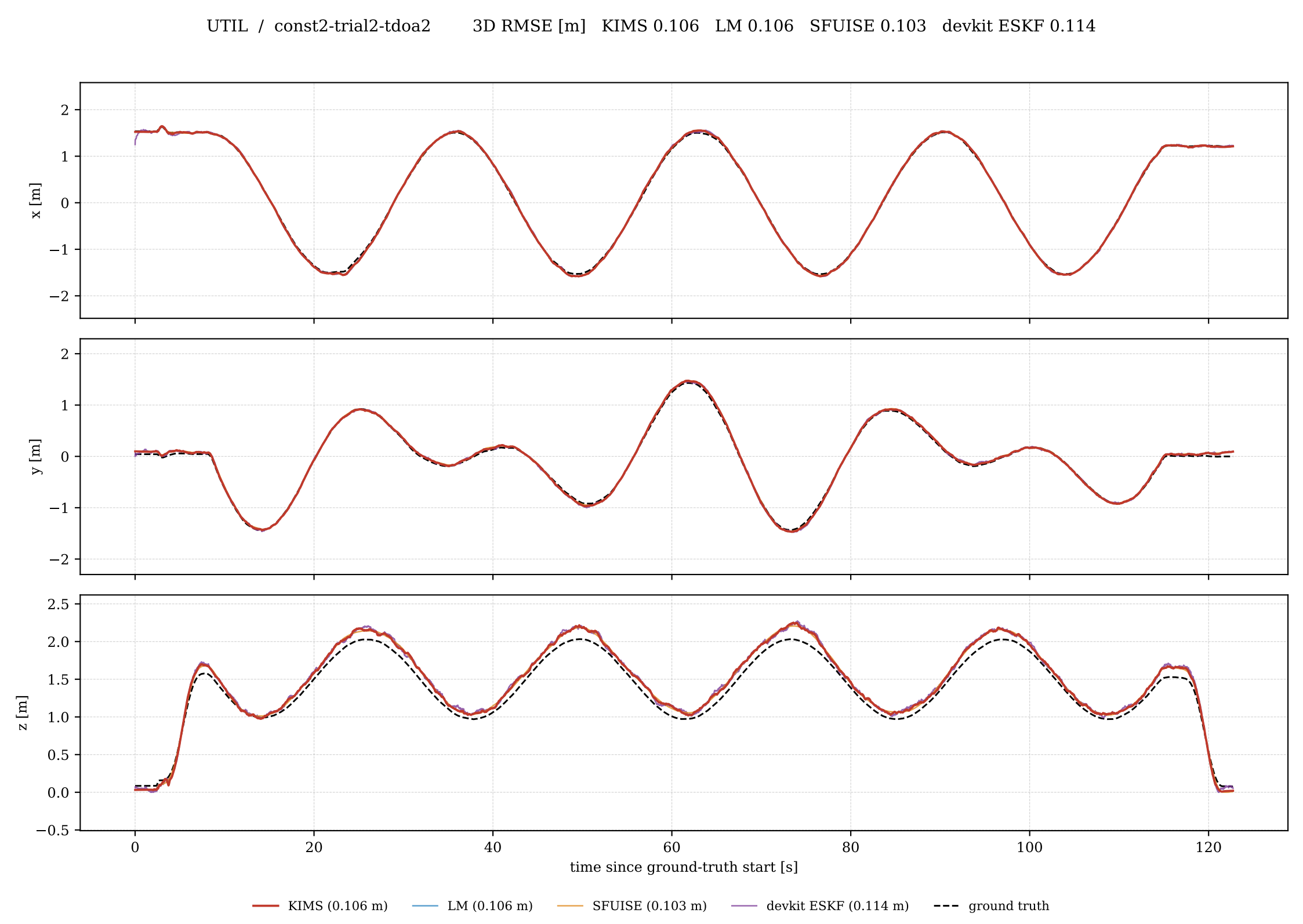

UTIL / const1-trial5-tdoa3 3D RMSE [m] KIMS 0.112 LM 0.112 SFUISE 0.108 devkit ESKF 0.139
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.112 m)
LM (0.112 m)
SFUISE (0.108 m)
devkit ESKF (0.139 m)
ground truth

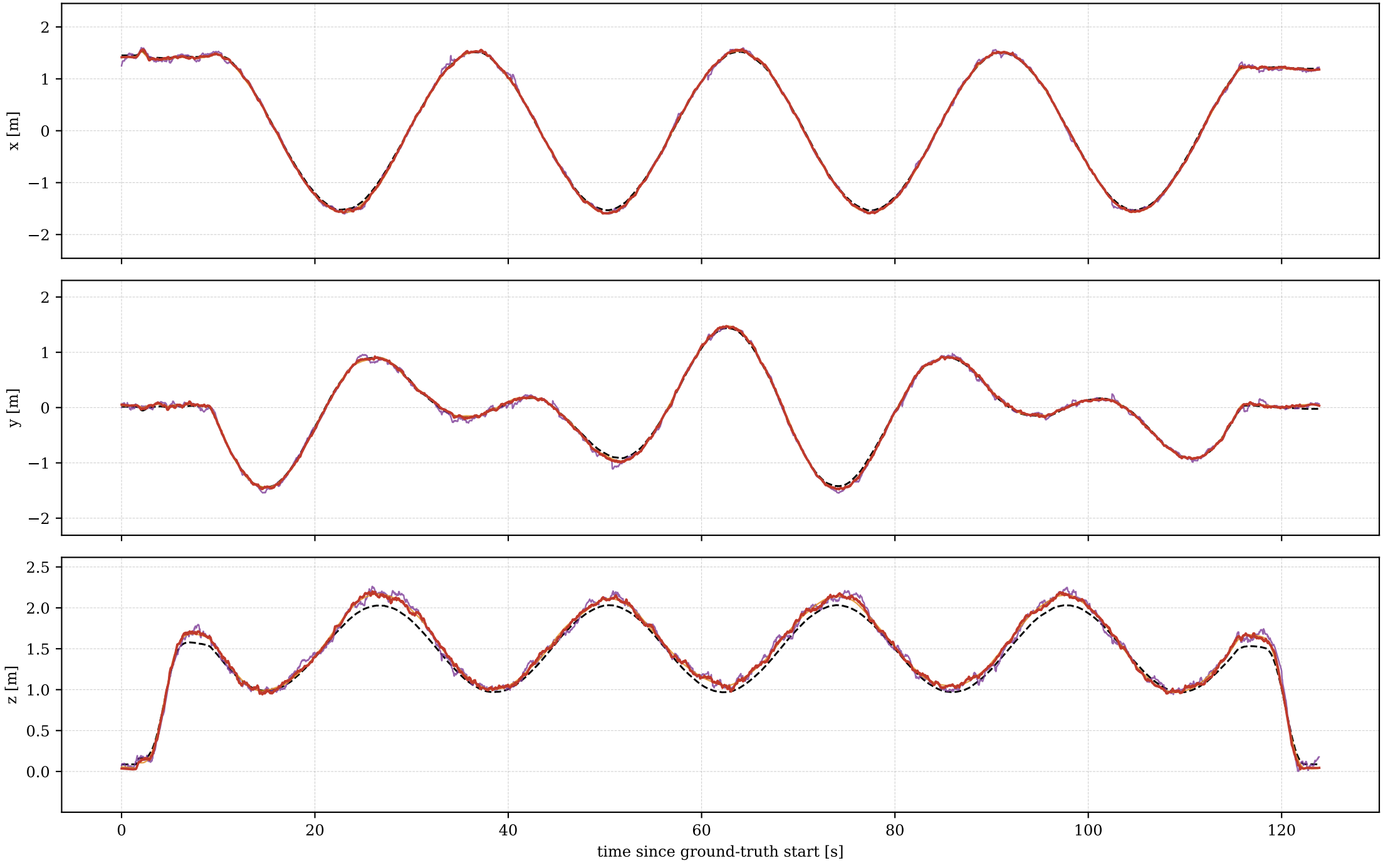

UTIL / const1-trial6-tdoa2 3D RMSE [m] KIMS 0.111 LM 0.111 SFUISE 0.104 devkit ESKF 0.123
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.111 m)
LM (0.111 m)
SFUISE (0.104 m)
devkit ESKF (0.123 m)
ground truth

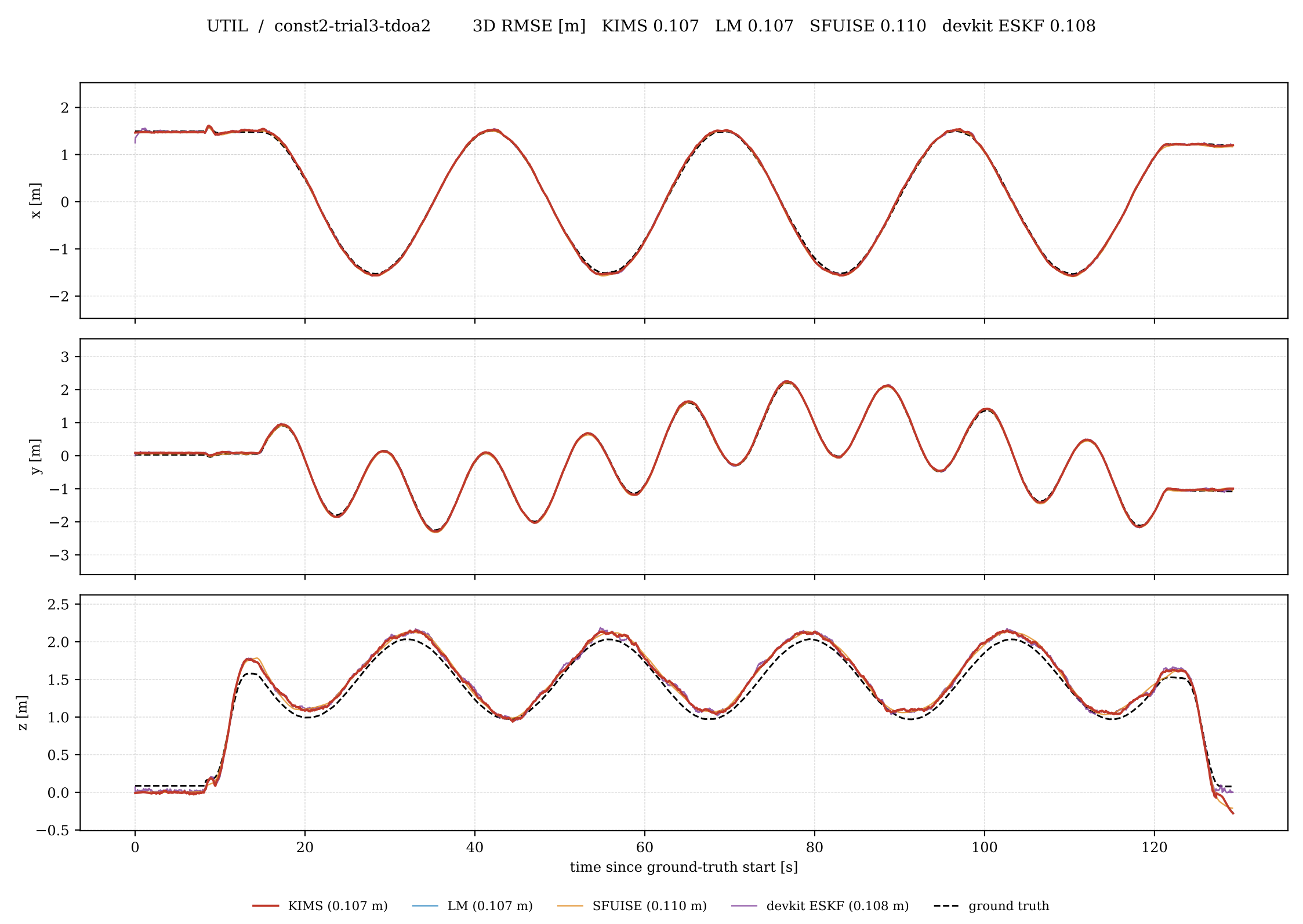

UTIL / const1-trial6-tdoa3 3D RMSE [m] KIMS 0.108 LM 0.108 SFUISE 0.103 devkit ESKF 0.134
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.108 m)
LM (0.108 m)
SFUISE (0.103 m)
devkit ESKF (0.134 m)
ground truth

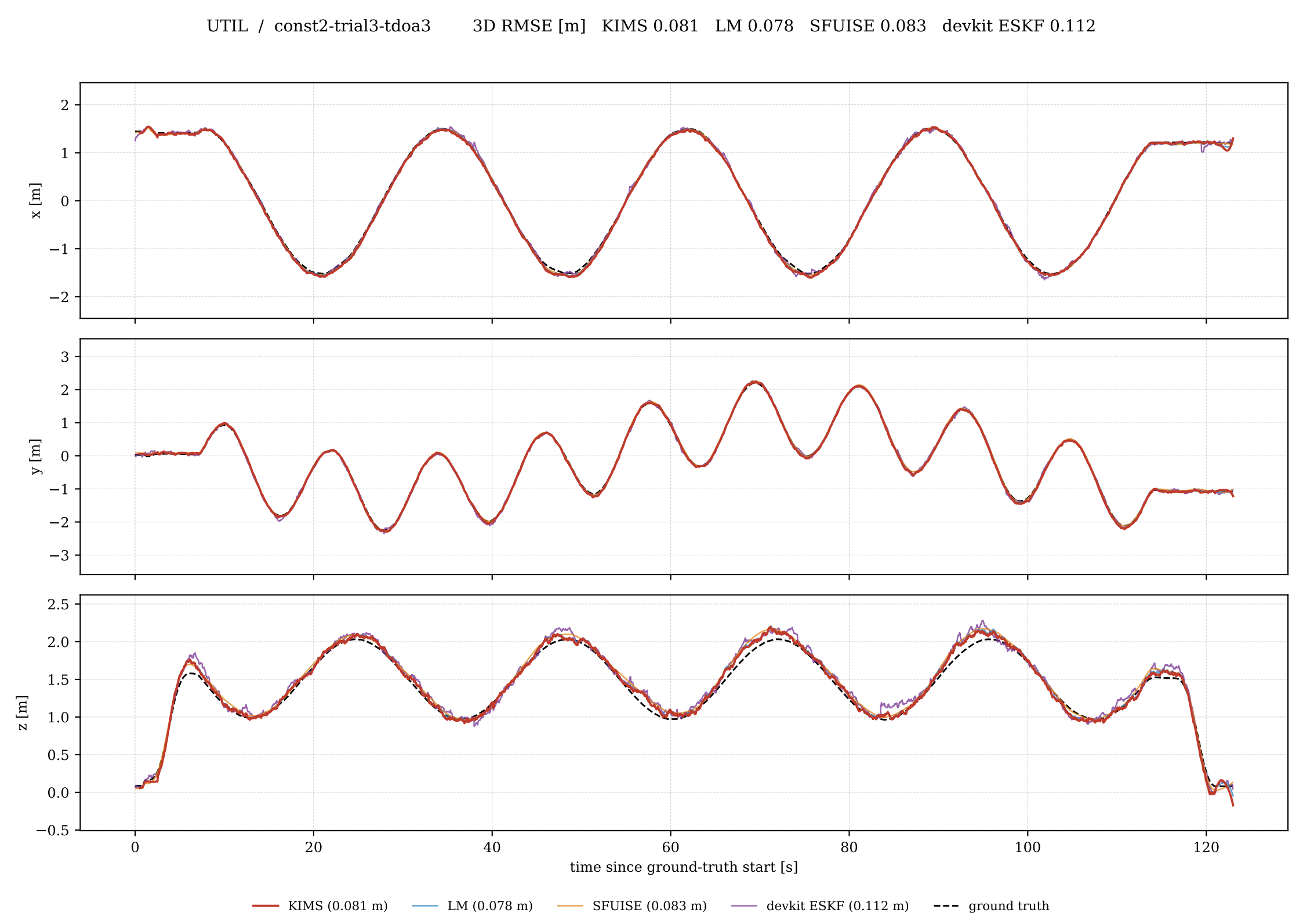

UTIL / const2-trial1-tdoa2 3D RMSE [m] KIMS 0.095 LM 0.095 SW-PGO 0.092 SFUISE 0.094 devkit ESKF 0.103
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.095 m)
LM (0.095 m)
SW-PGO (0.092 m)
SFUISE (0.094 m)
devkit ESKF (0.103 m)
ground truth

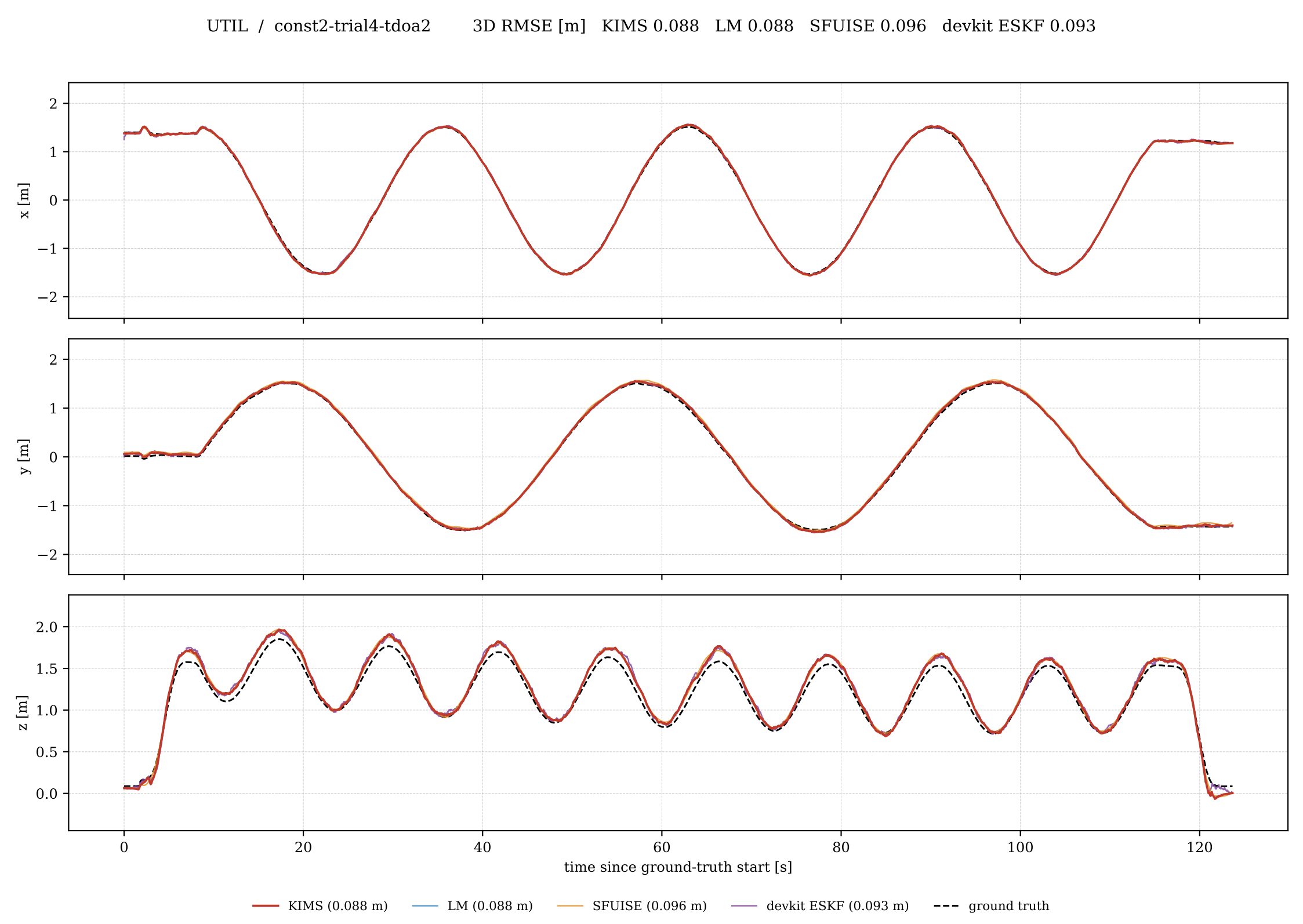

UTIL / const2-trial1-tdoa3 3D RMSE [m] KIMS 0.095 LM 0.095 SW-PGO 0.090 SFUISE 0.096 devkit ESKF 0.113
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.095 m)
LM (0.095 m)
SW-PGO (0.090 m)
SFUISE (0.096 m)
devkit ESKF (0.113 m)
ground truth

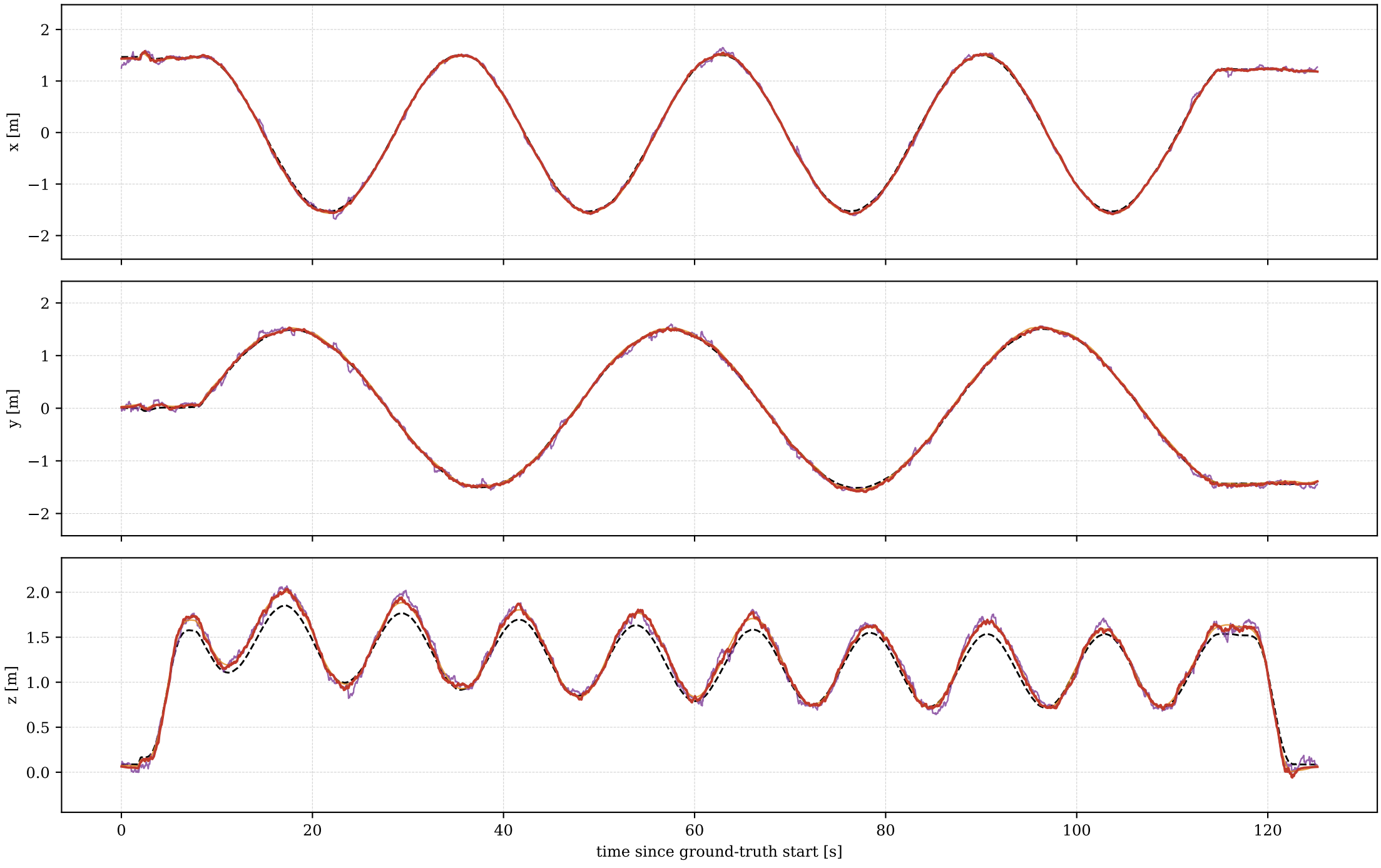

UTIL / const2-trial2-tdoa2 3D RMSE [m] KIMS 0.106 LM 0.106 SFUISE 0.103 devkit ESKF 0.114
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.106 m)
LM (0.106 m)
SFUISE (0.103 m)
devkit ESKF (0.114 m)
ground truth

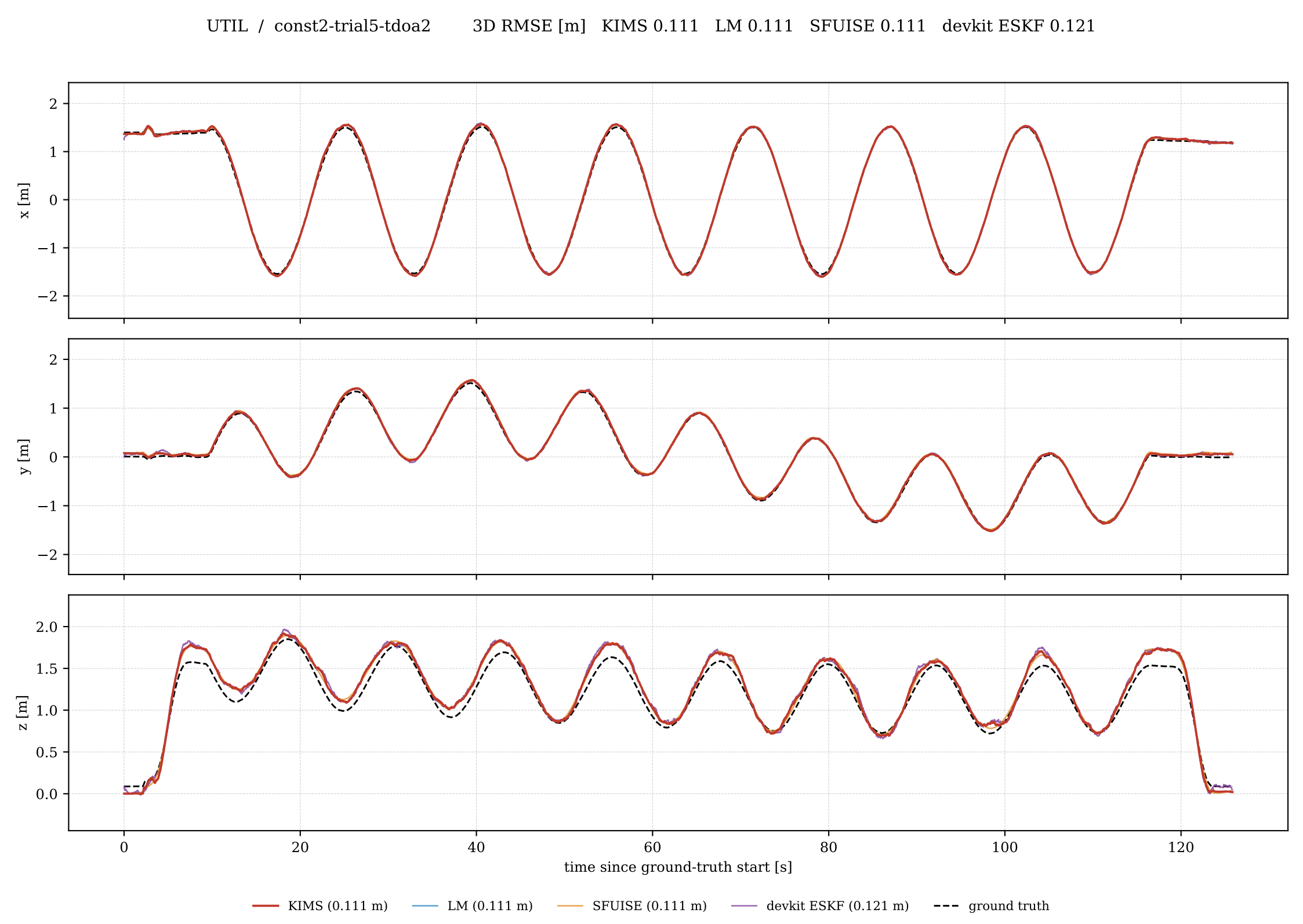

UTIL / const2-trial2-tdoa3 3D RMSE [m] KIMS 0.088 LM 0.088 SFUISE 0.084 devkit ESKF 0.113
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.088 m)
LM (0.088 m)
SFUISE (0.084 m)
devkit ESKF (0.113 m)
ground truth

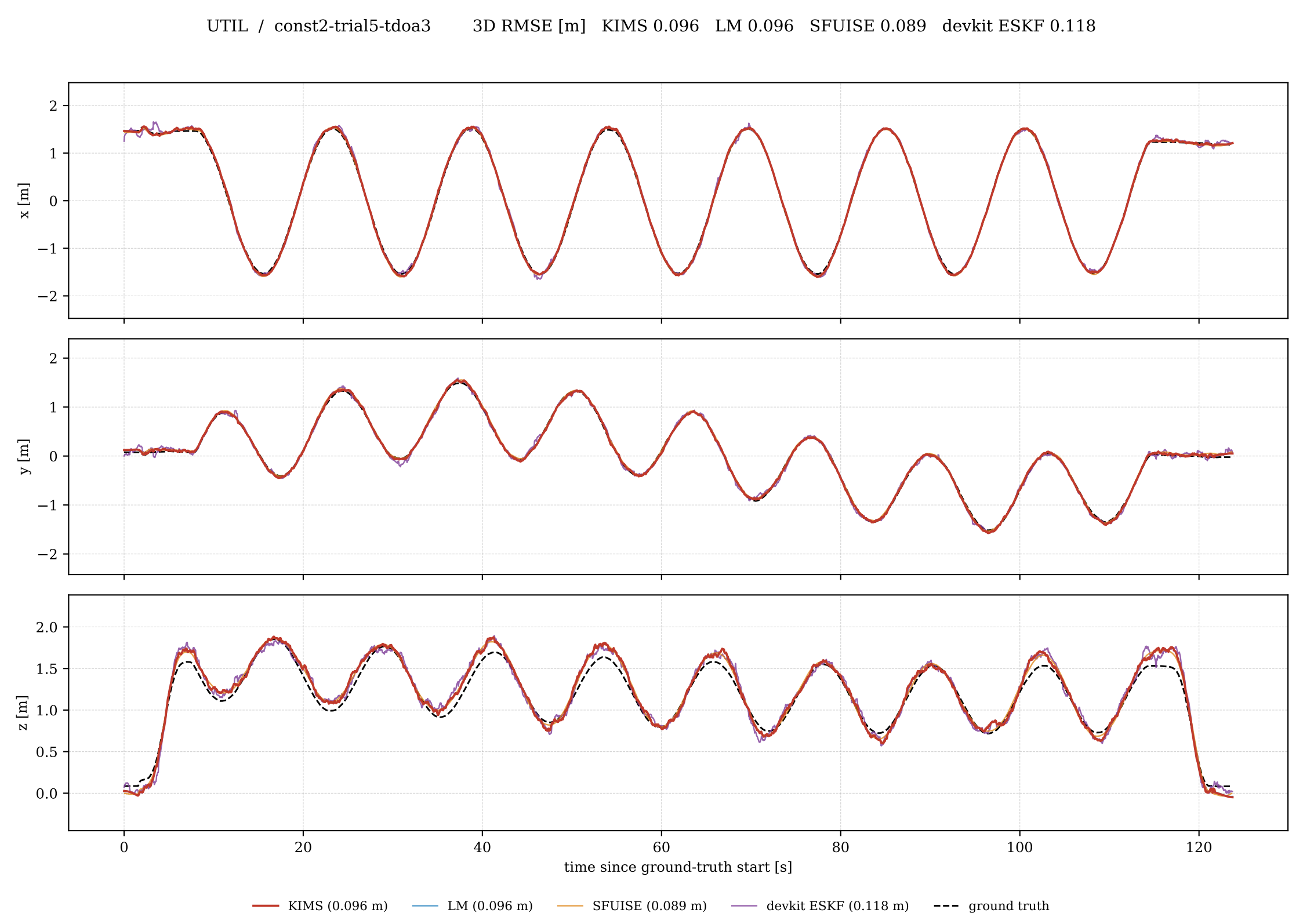

UTIL / const2-trial3-tdoa2 3D RMSE [m] KIMS 0.107 LM 0.107 SFUISE 0.110 devkit ESKF 0.108
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.107 m)
LM (0.107 m)
SFUISE (0.110 m)
devkit ESKF (0.108 m)
ground truth

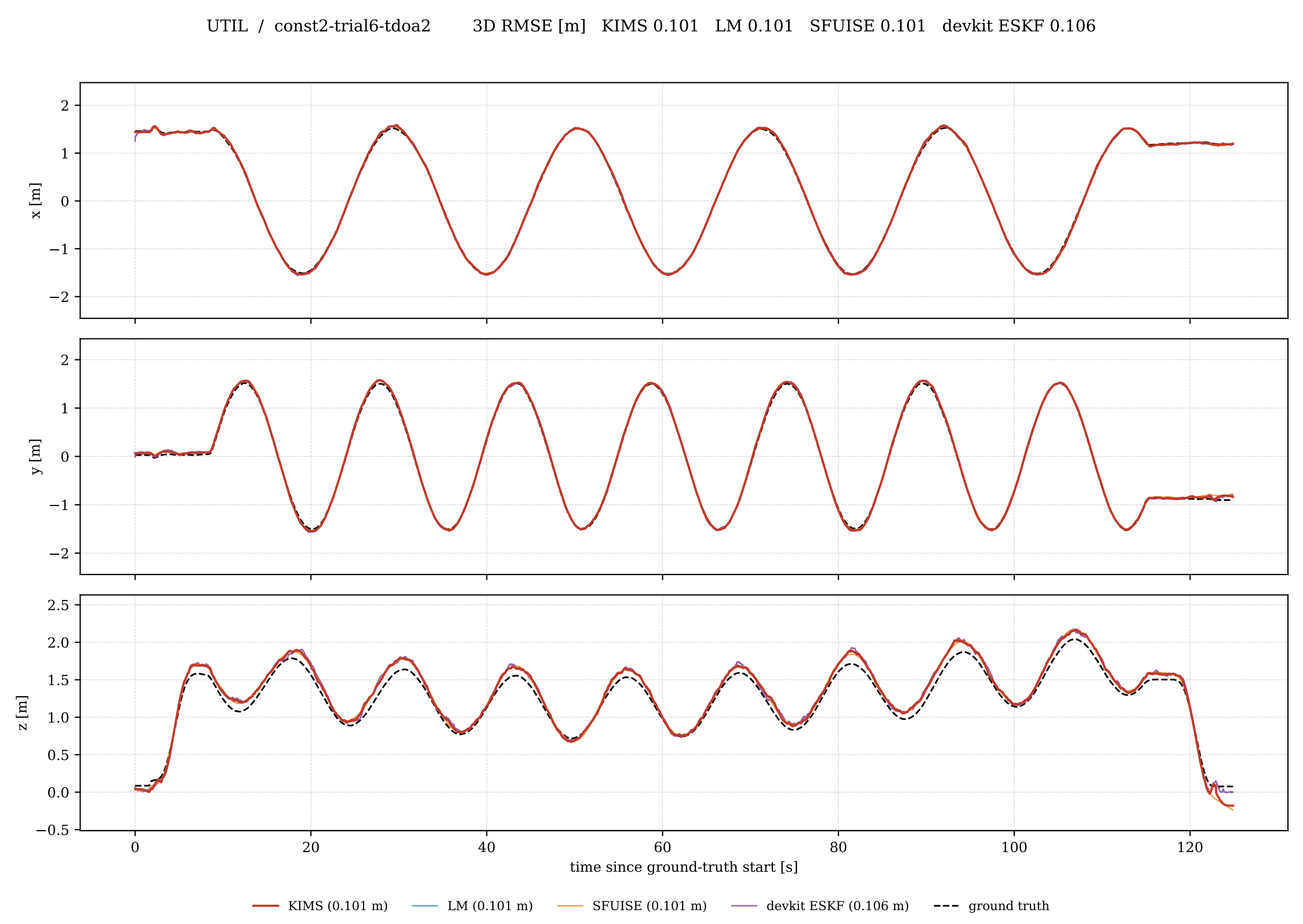

UTIL / const2-trial3-tdoa3 3D RMSE [m] KIMS 0.081 LM 0.078 SFUISE 0.083 devkit ESKF 0.112
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.081 m)
LM (0.078 m)
SFUISE (0.083 m)
devkit ESKF (0.112 m)
ground truth

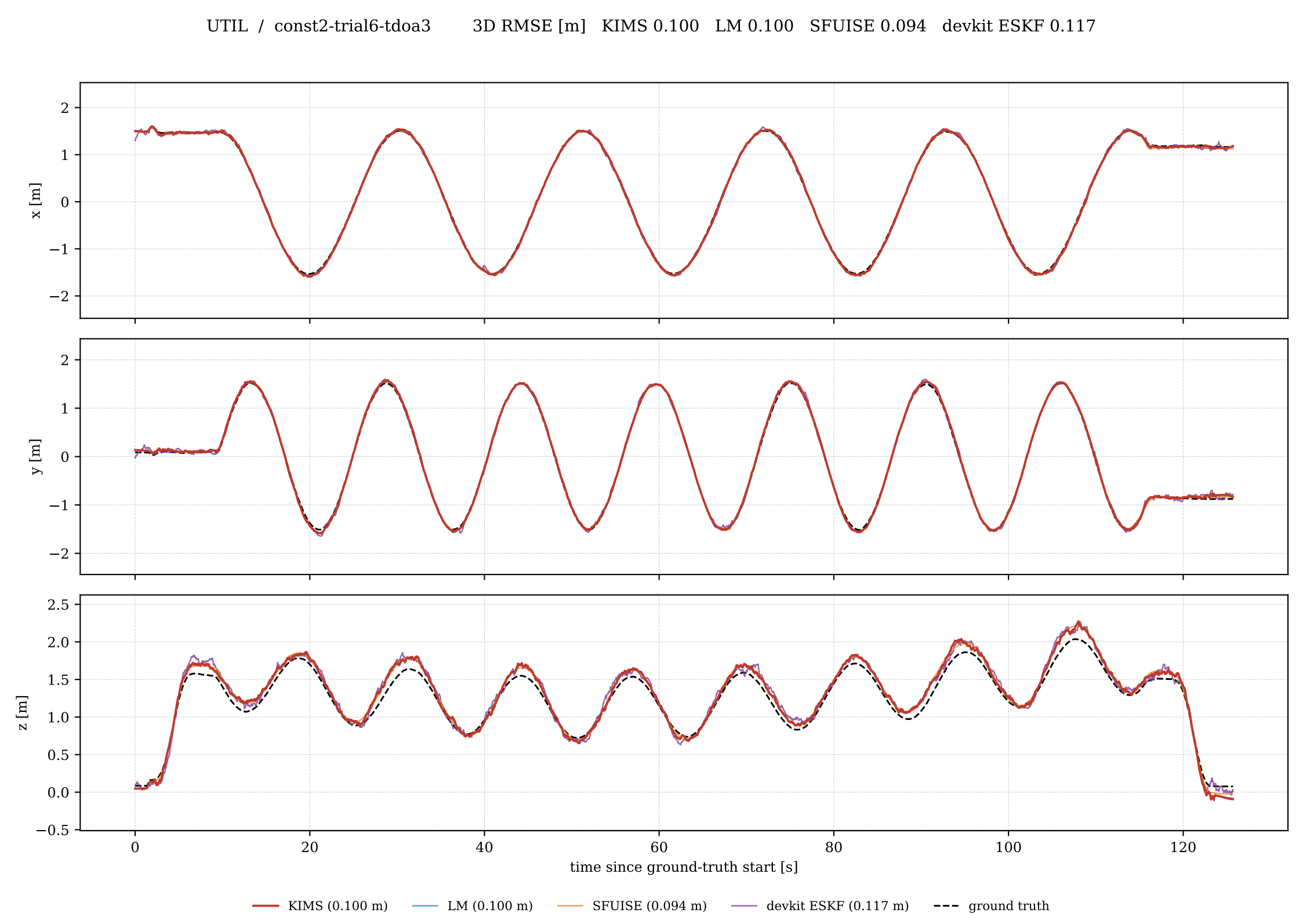

UTIL / const2-trial4-tdoa2 3D RMSE [m] KIMS 0.088 LM 0.088 SFUISE 0.096 devkit ESKF 0.093
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.088 m)
LM (0.088 m)
SFUISE (0.096 m)
devkit ESKF (0.093 m)
ground truth

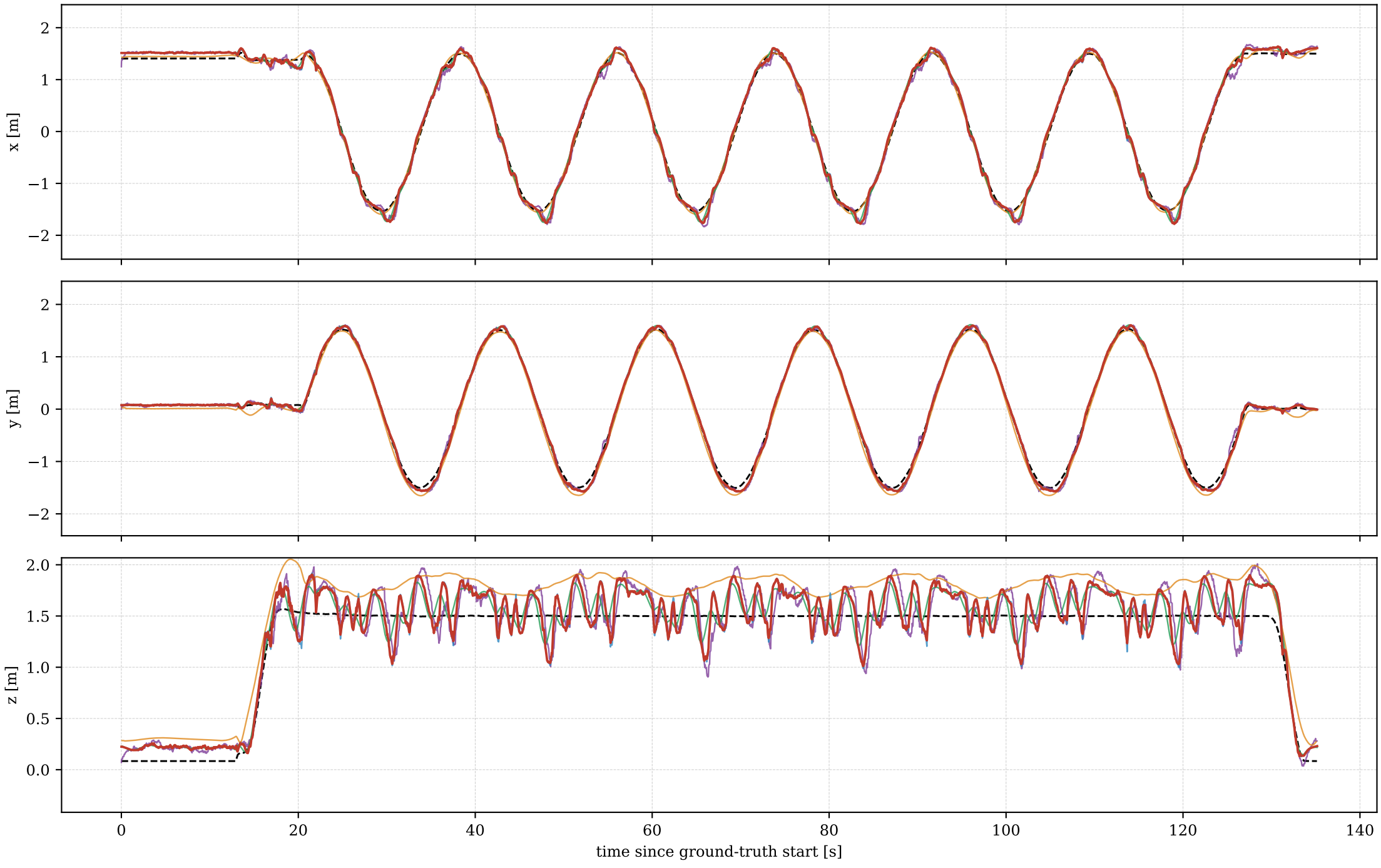

UTIL / const2-trial4-tdoa3 3D RMSE [m] KIMS 0.088 LM 0.088 SFUISE 0.086 devkit ESKF 0.117
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.088 m)
LM (0.088 m)
SFUISE (0.086 m)
devkit ESKF (0.117 m)
ground truth

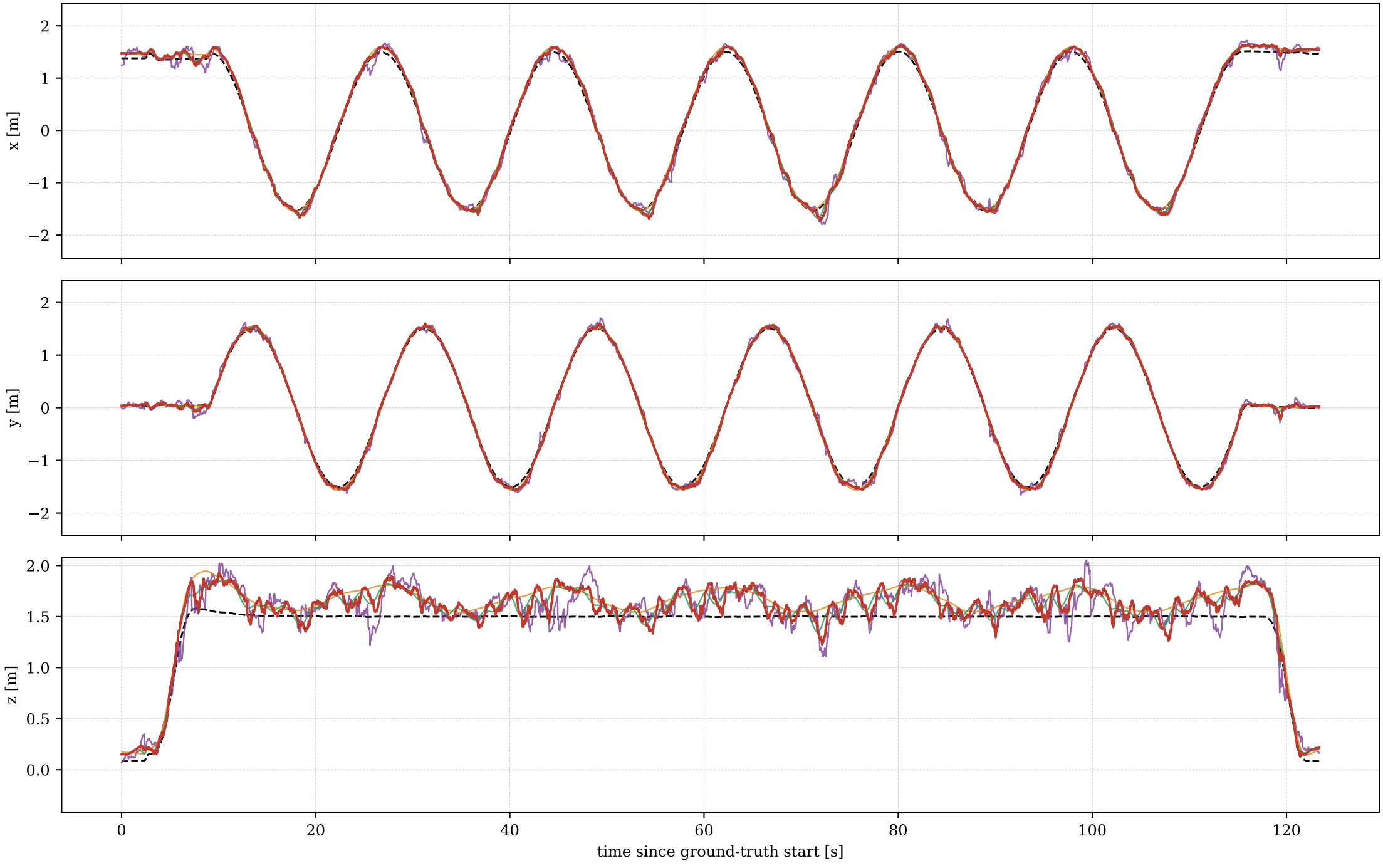

UTIL / const2-trial5-tdoa2 3D RMSE [m] KIMS 0.111 LM 0.111 SFUISE 0.111 devkit ESKF 0.121
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.111 m)
LM (0.111 m)
SFUISE (0.111 m)
devkit ESKF (0.121 m)
ground truth

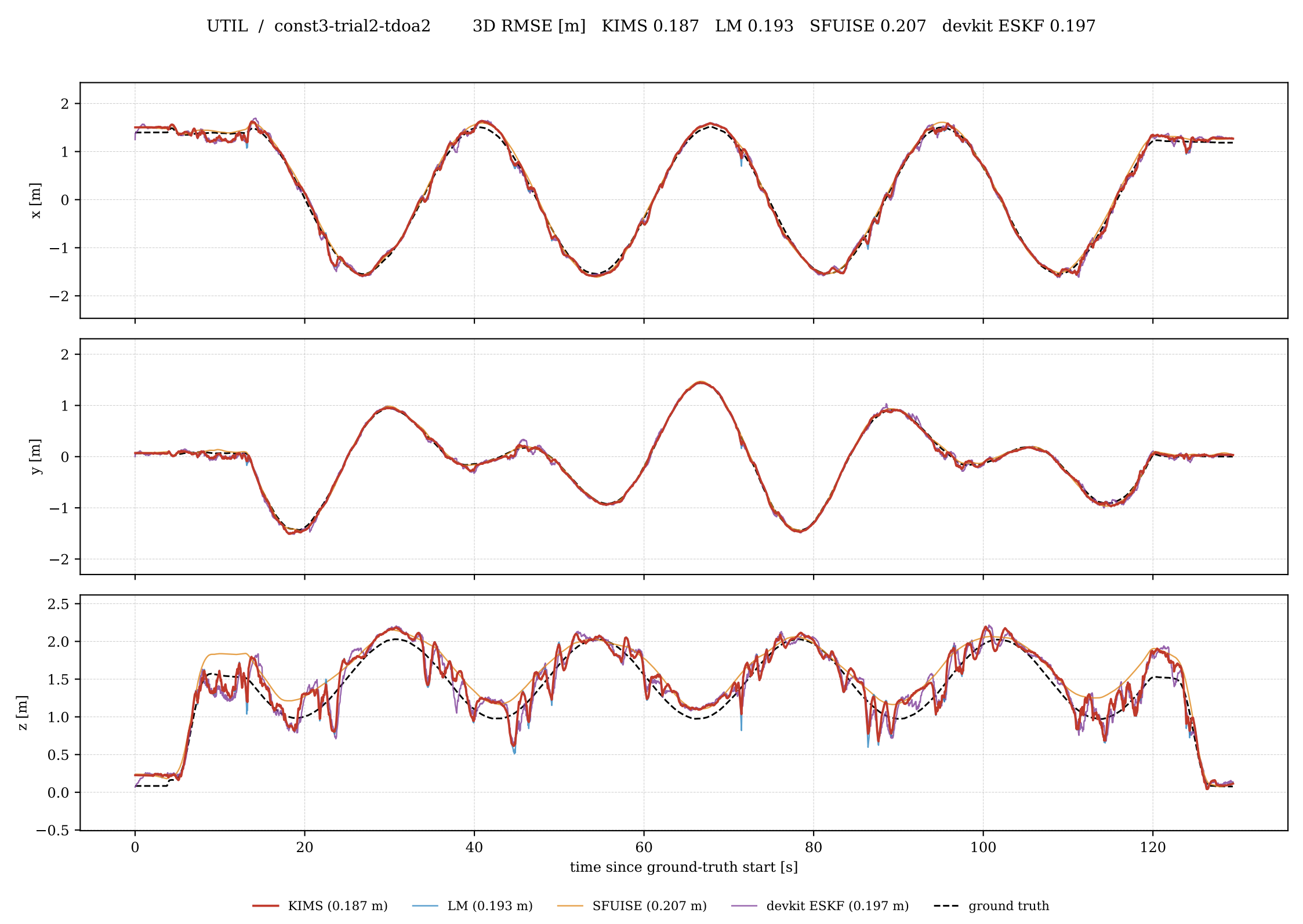

UTIL / const2-trial5-tdoa3 3D RMSE [m] KIMS 0.096 LM 0.096 SFUISE 0.089 devkit ESKF 0.118
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.096 m)
LM (0.096 m)
SFUISE (0.089 m)
devkit ESKF (0.118 m)
ground truth

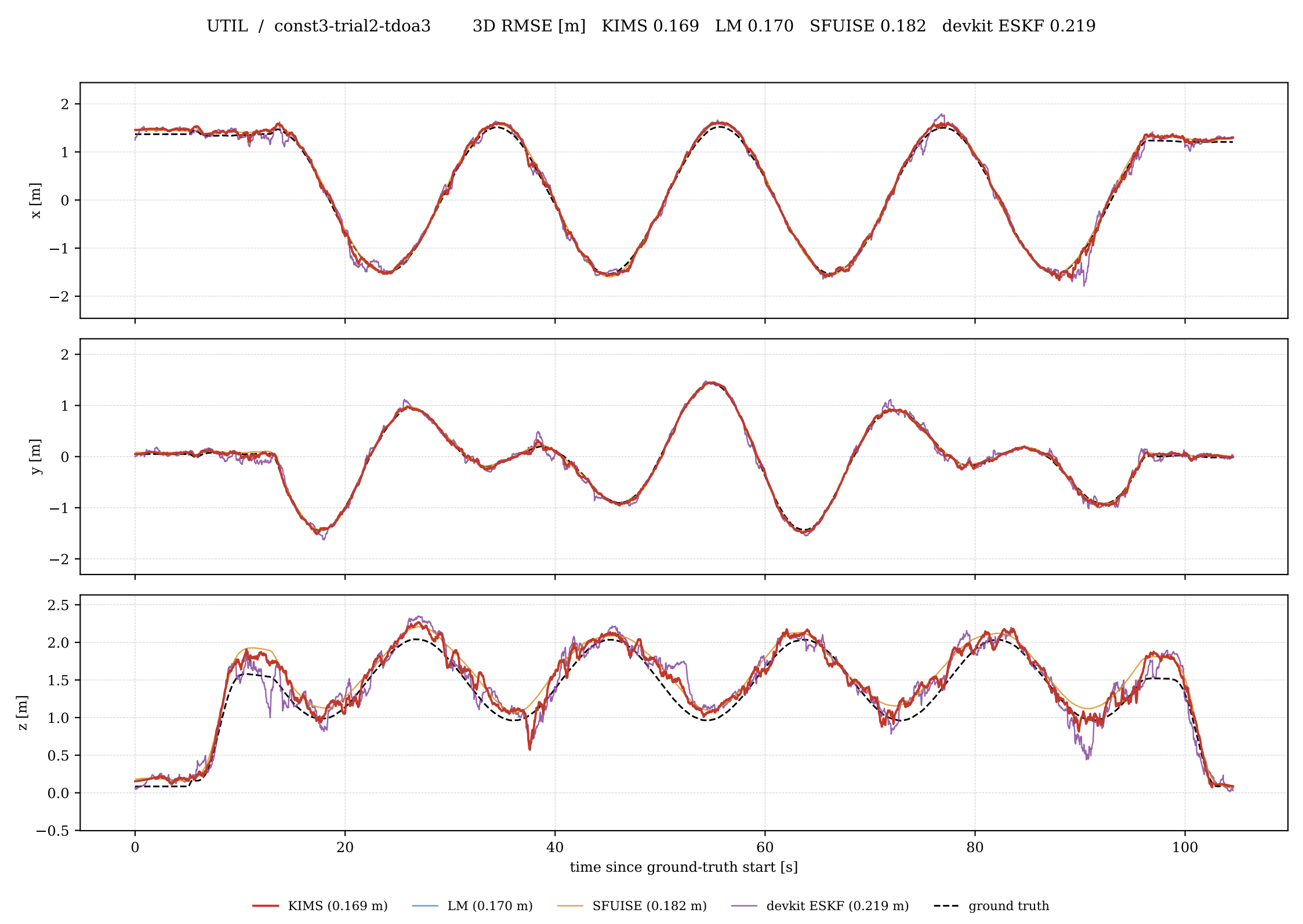

UTIL / const2-trial6-tdoa2 3D RMSE [m] KIMS 0.101 LM 0.101 SFUISE 0.101 devkit ESKF 0.106
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.101 m)
LM (0.101 m)
SFUISE (0.101 m)
devkit ESKF (0.106 m)
ground truth

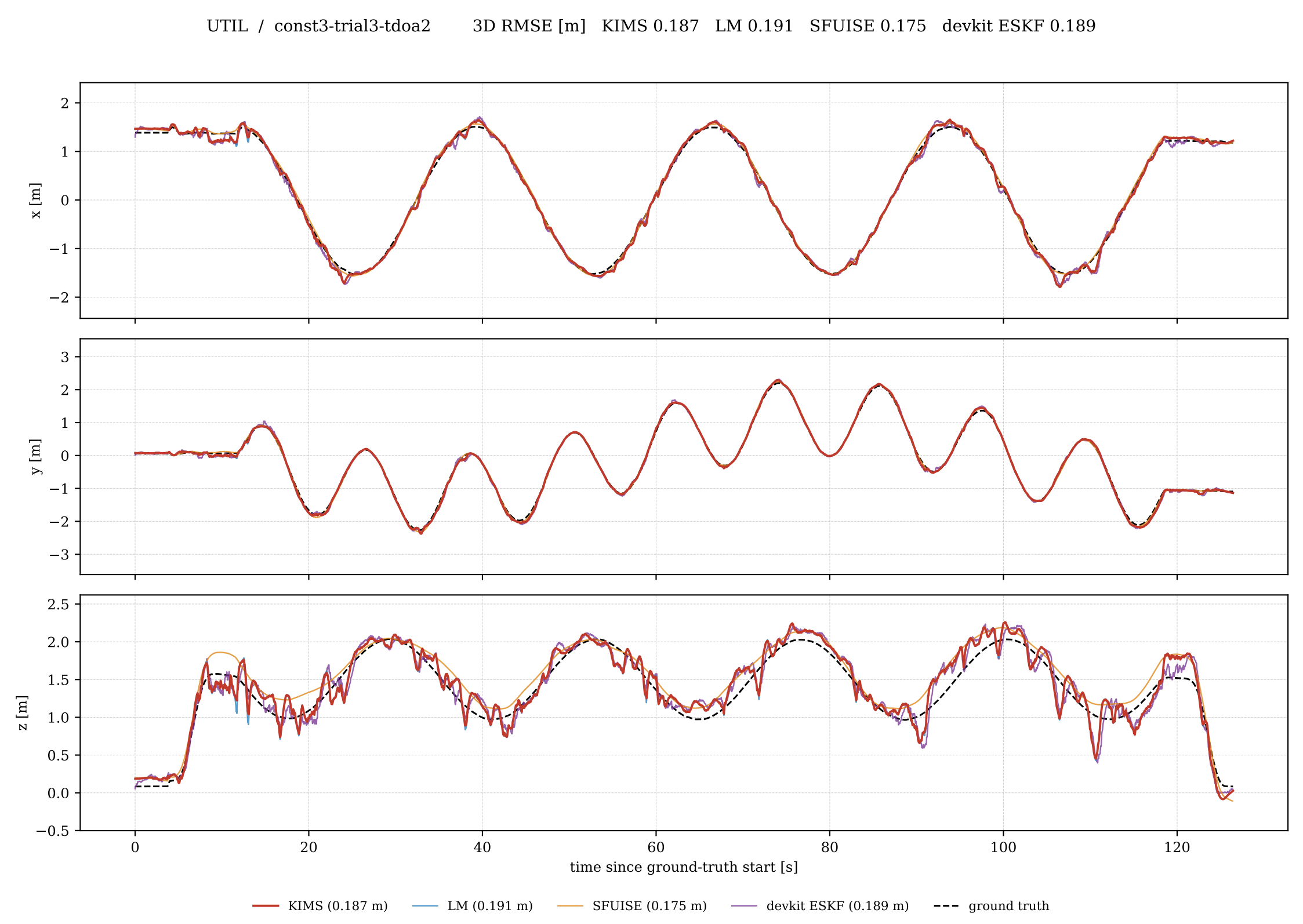

UTIL / const2-trial6-tdoa3 3D RMSE [m] KIMS 0.100 LM 0.100 SFUISE 0.094 devkit ESKF 0.117
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.100 m)
LM (0.100 m)
SFUISE (0.094 m)
devkit ESKF (0.117 m)
ground truth

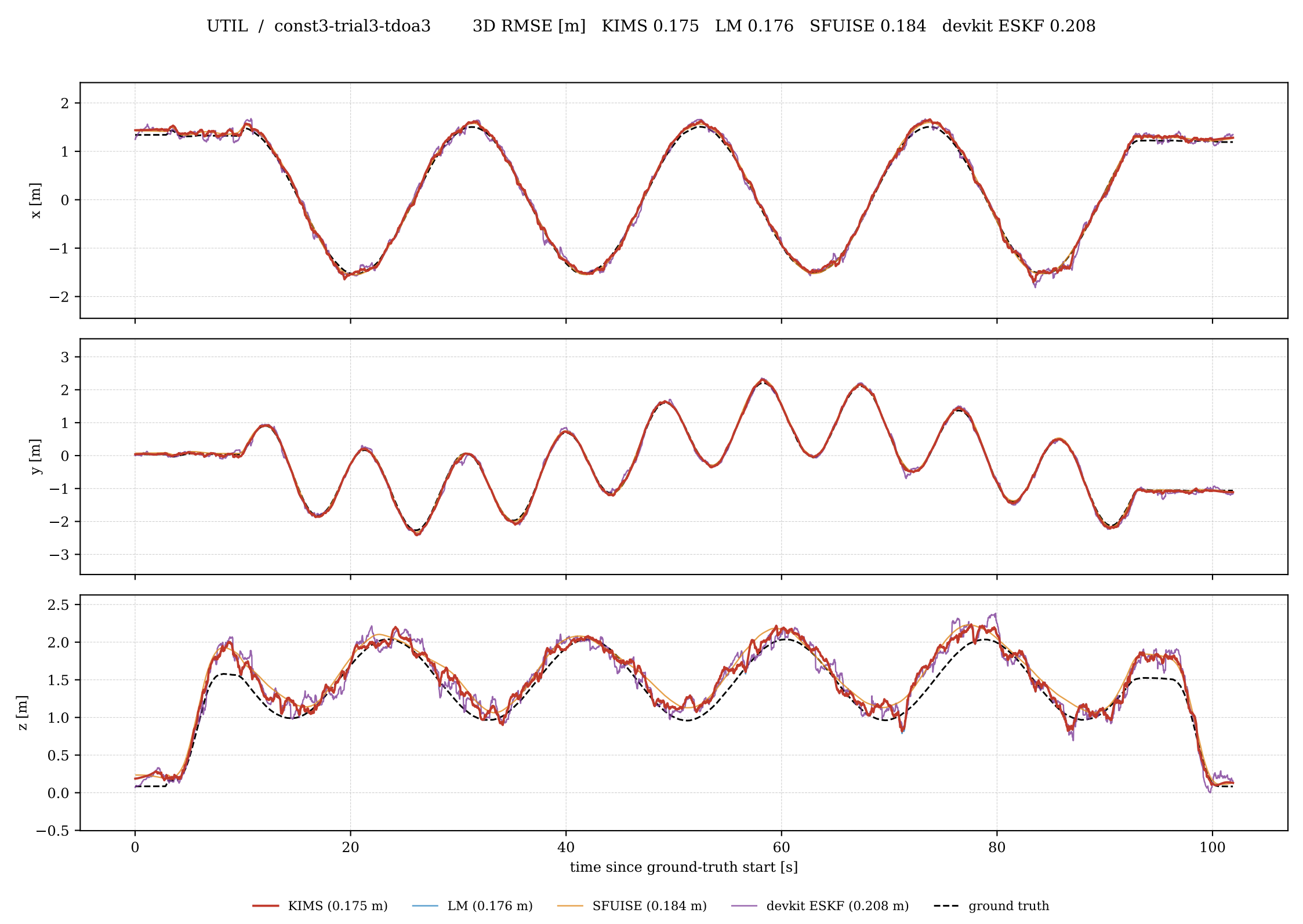

UTIL / const3-trial1-tdoa2 3D RMSE [m] KIMS 0.230 LM 0.234 SW-PGO 0.190 SFUISE 0.330 devkit ESKF 0.245
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.230 m)
LM (0.234 m)
SW-PGO (0.190 m)
SFUISE (0.330 m)
devkit ESKF (0.245 m)
ground truth

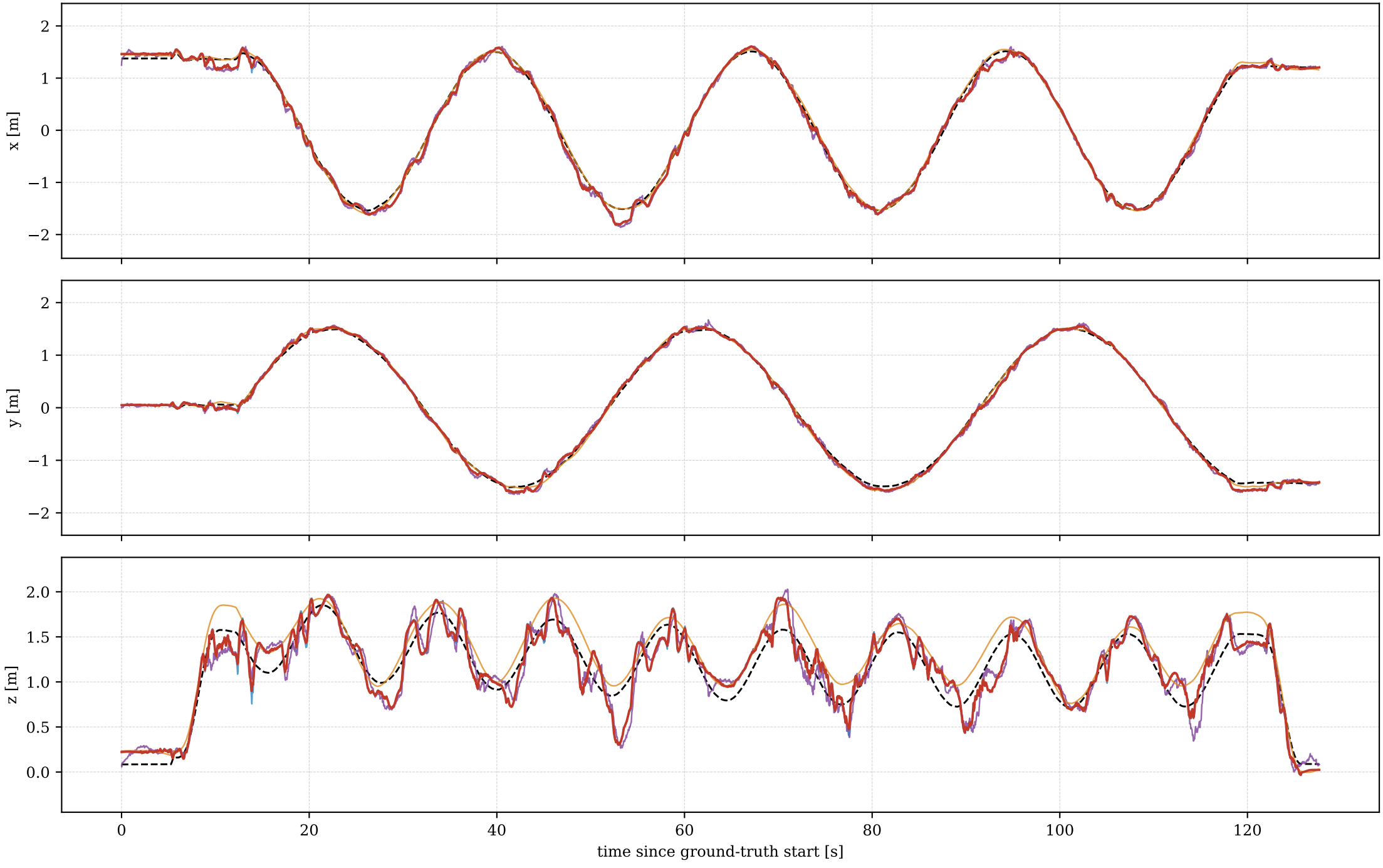

UTIL / const3-trial1-tdoa3 3D RMSE [m] KIMS 0.198 LM 0.200 SW-PGO 0.181 SFUISE 0.207 devkit ESKF 0.226
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.198 m)
LM (0.200 m)
SW-PGO (0.181 m)
SFUISE (0.207 m)
devkit ESKF (0.226 m)
ground truth

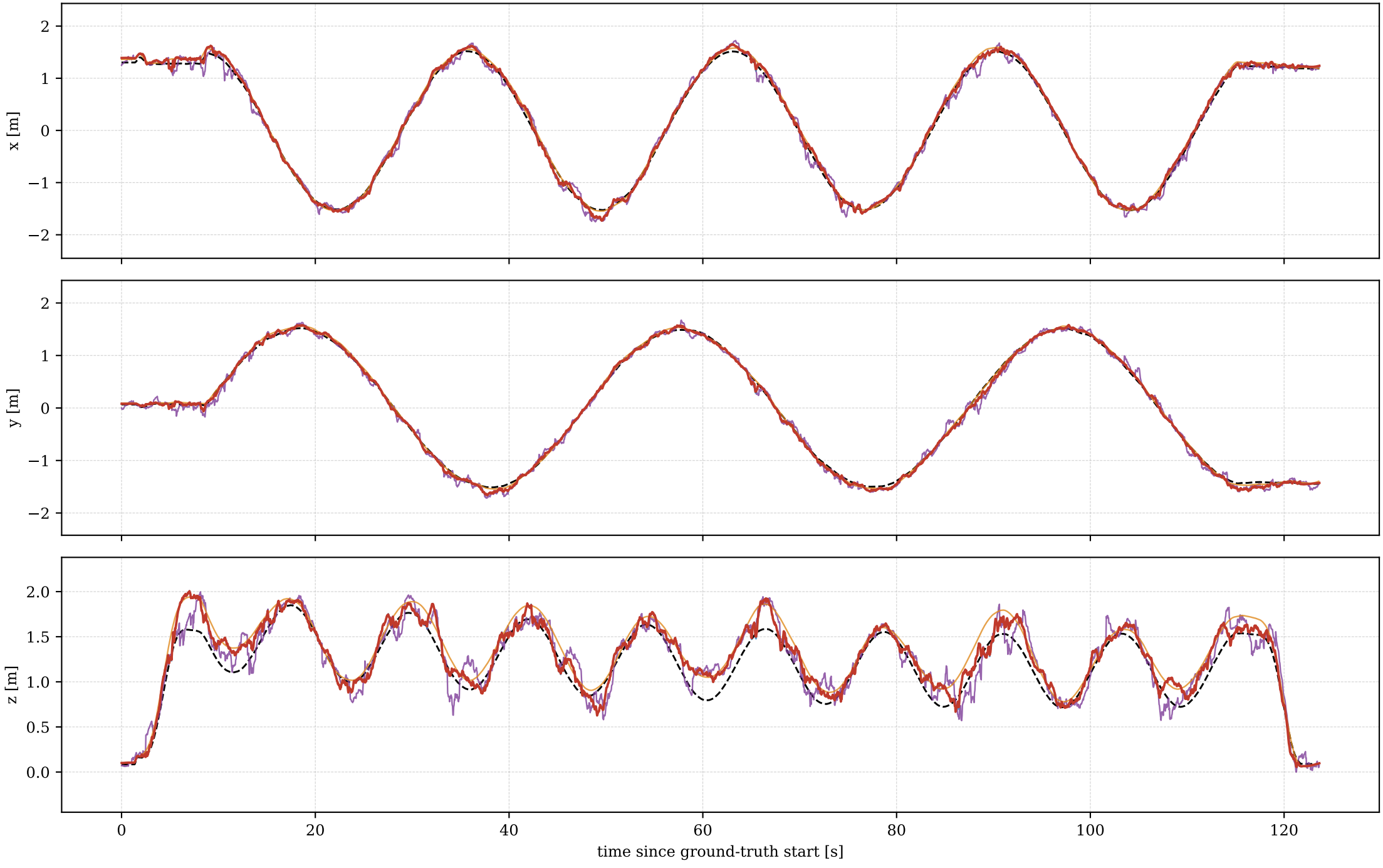

UTIL / const3-trial2-tdoa2 3D RMSE [m] KIMS 0.187 LM 0.193 SFUISE 0.207 devkit ESKF 0.197
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.187 m)
LM (0.193 m)
SFUISE (0.207 m)
devkit ESKF (0.197 m)
ground truth

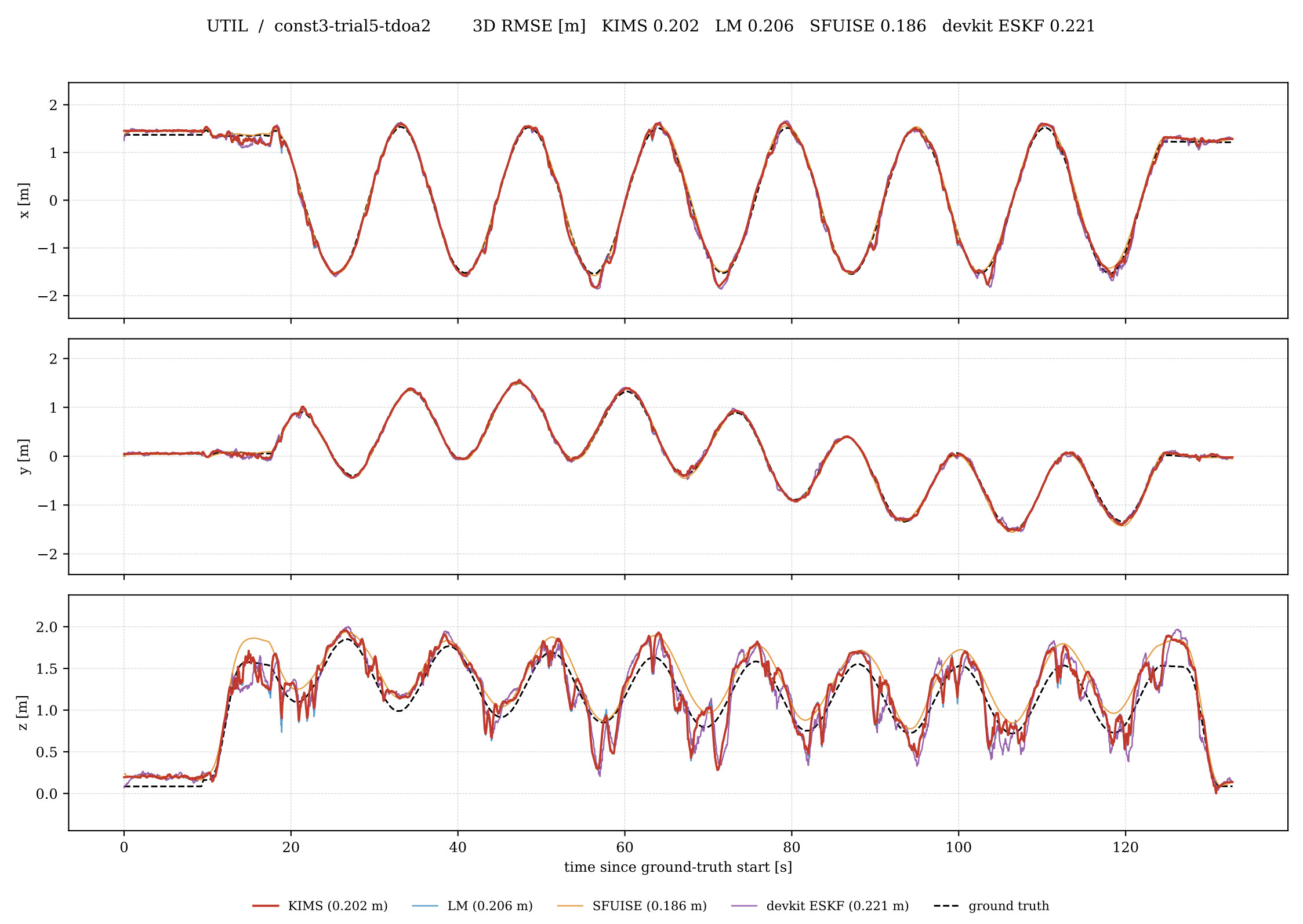

UTIL / const3-trial2-tdoa3 3D RMSE [m] KIMS 0.169 LM 0.170 SFUISE 0.182 devkit ESKF 0.219
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.169 m)
LM (0.170 m)
SFUISE (0.182 m)
devkit ESKF (0.219 m)
ground truth

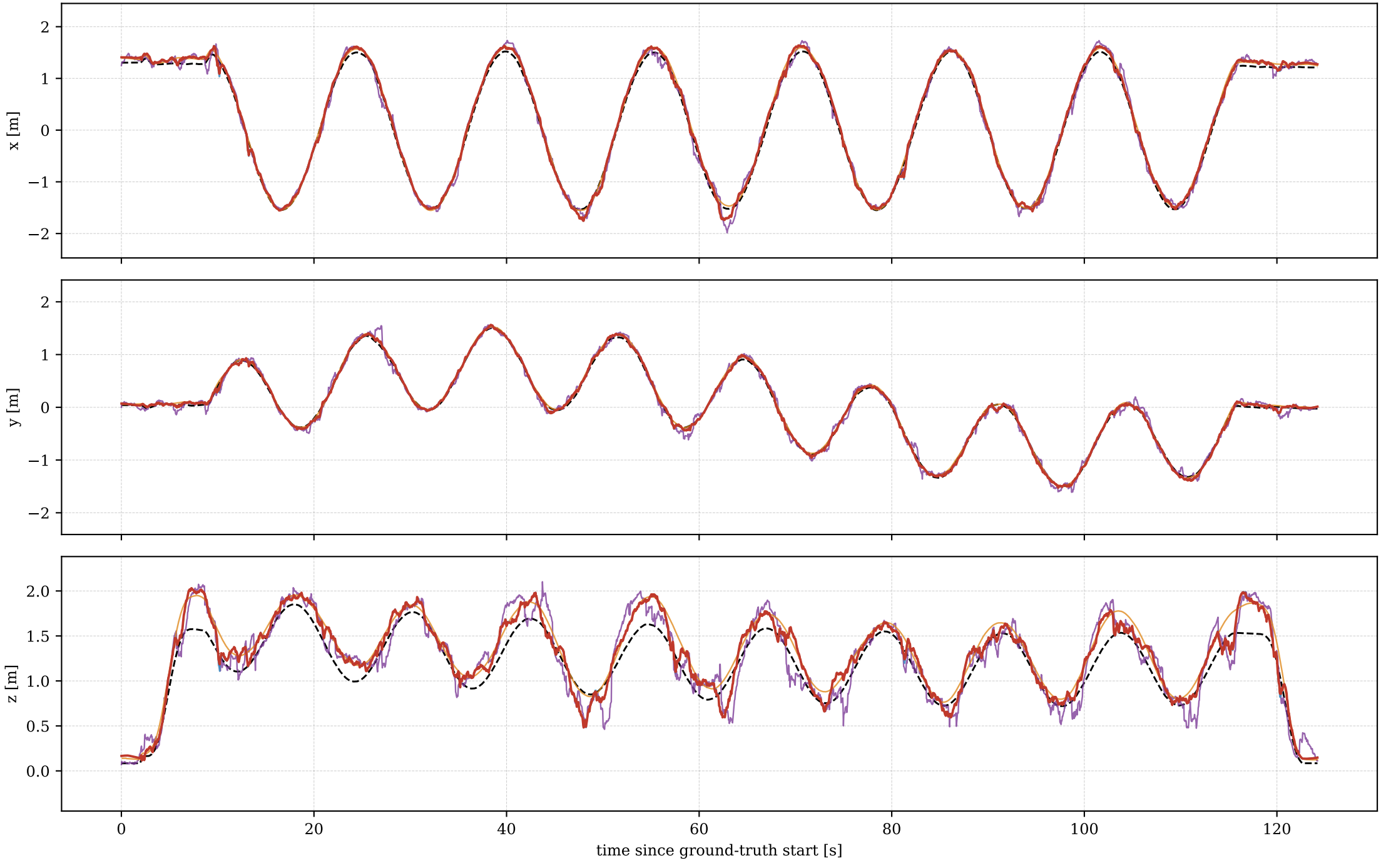

UTIL / const3-trial3-tdoa2 3D RMSE [m] KIMS 0.187 LM 0.191 SFUISE 0.175 devkit ESKF 0.189
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.187 m)
LM (0.191 m)
SFUISE (0.175 m)
devkit ESKF (0.189 m)
ground truth

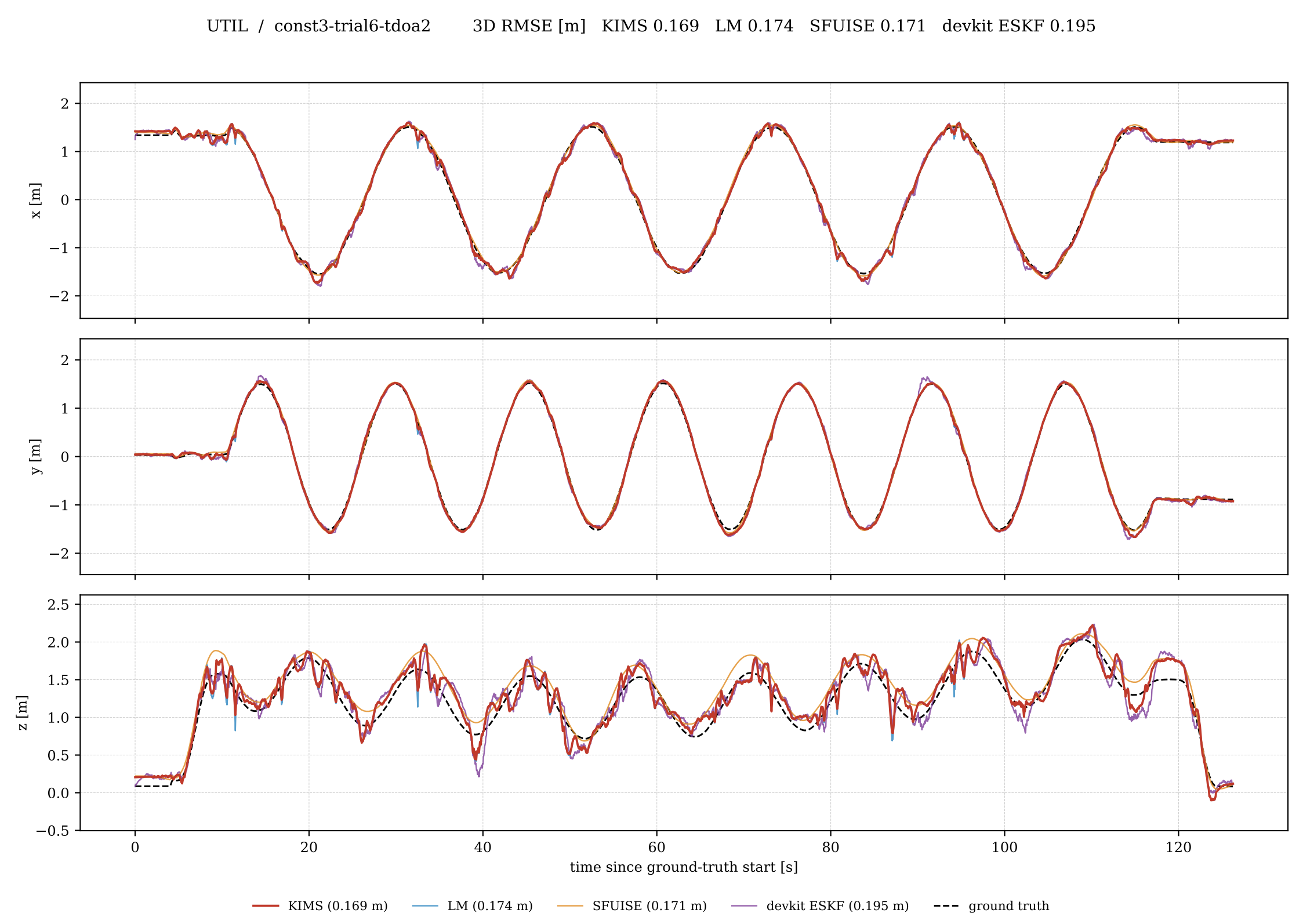

UTIL / const3-trial3-tdoa3 3D RMSE [m] KIMS 0.175 LM 0.176 SFUISE 0.184 devkit ESKF 0.208
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.175 m)
LM (0.176 m)
SFUISE (0.184 m)
devkit ESKF (0.208 m)
ground truth

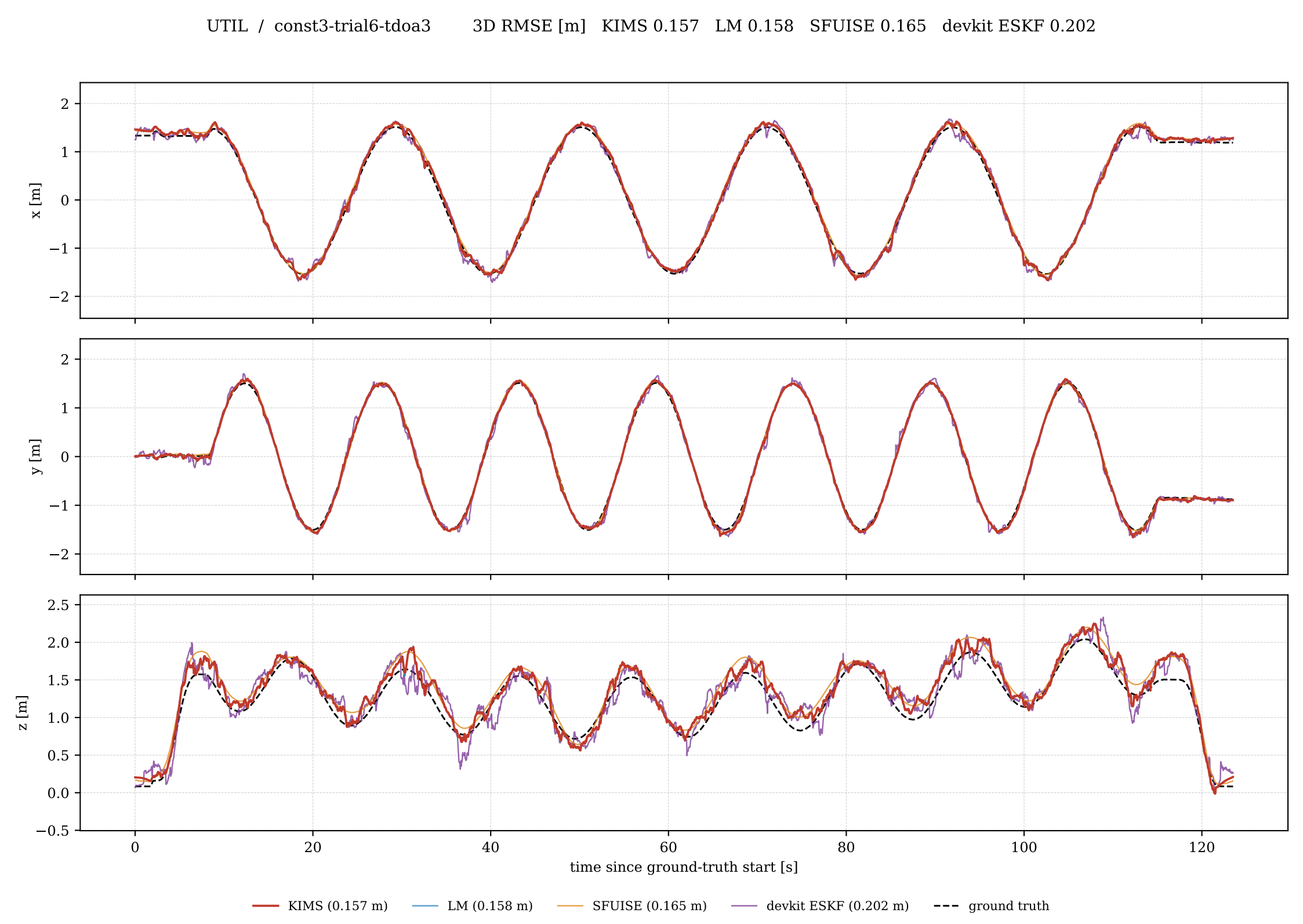

UTIL / const3-trial4-tdoa2 3D RMSE [m] KIMS 0.197 LM 0.201 SFUISE 0.183 devkit ESKF 0.217
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.197 m)
LM (0.201 m)
SFUISE (0.183 m)
devkit ESKF (0.217 m)
ground truth

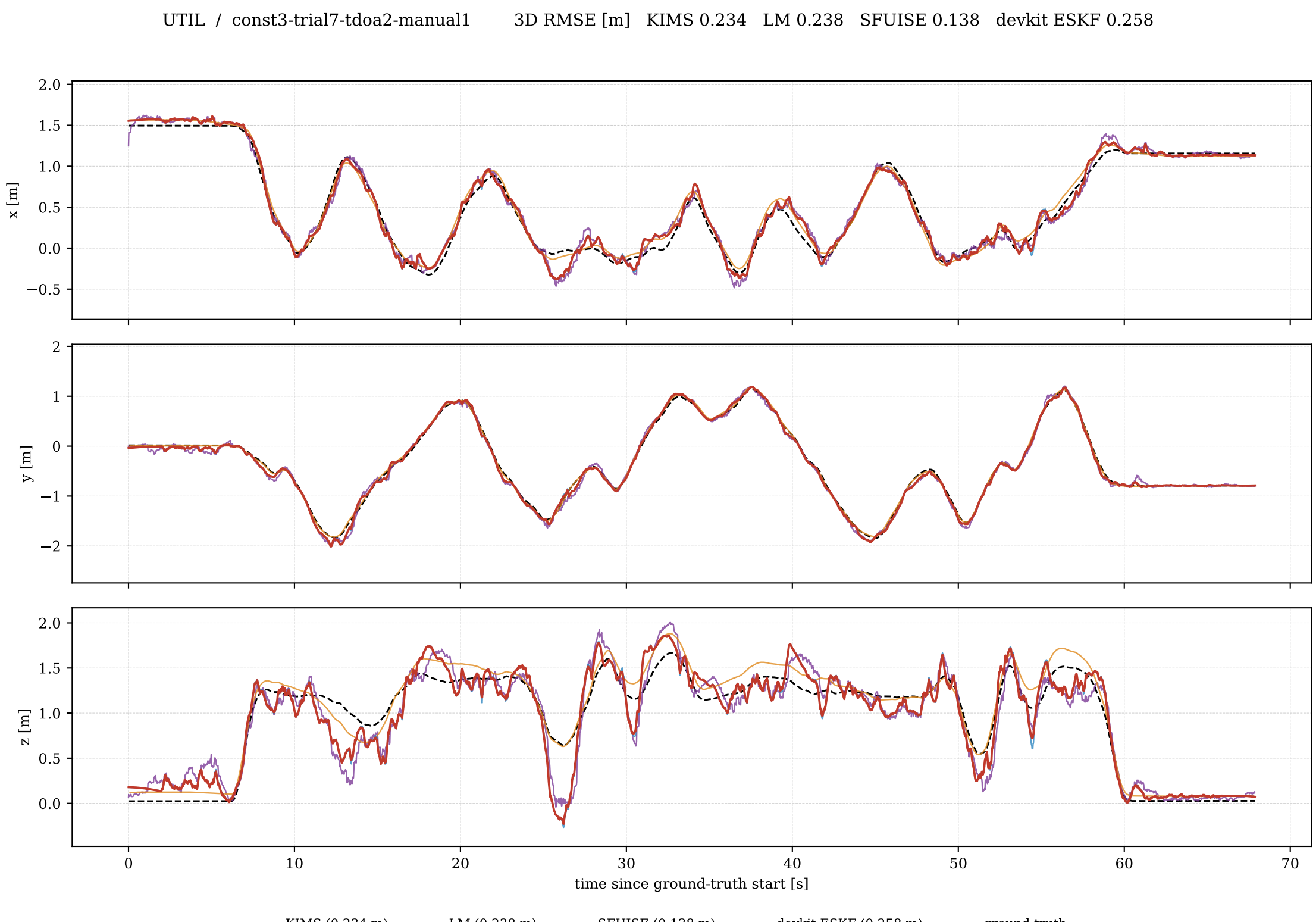

UTIL / const3-trial4-tdoa3 3D RMSE [m] KIMS 0.160 LM 0.160 SFUISE 0.174 devkit ESKF 0.199
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.160 m)
LM (0.160 m)
SFUISE (0.174 m)
devkit ESKF (0.199 m)
ground truth

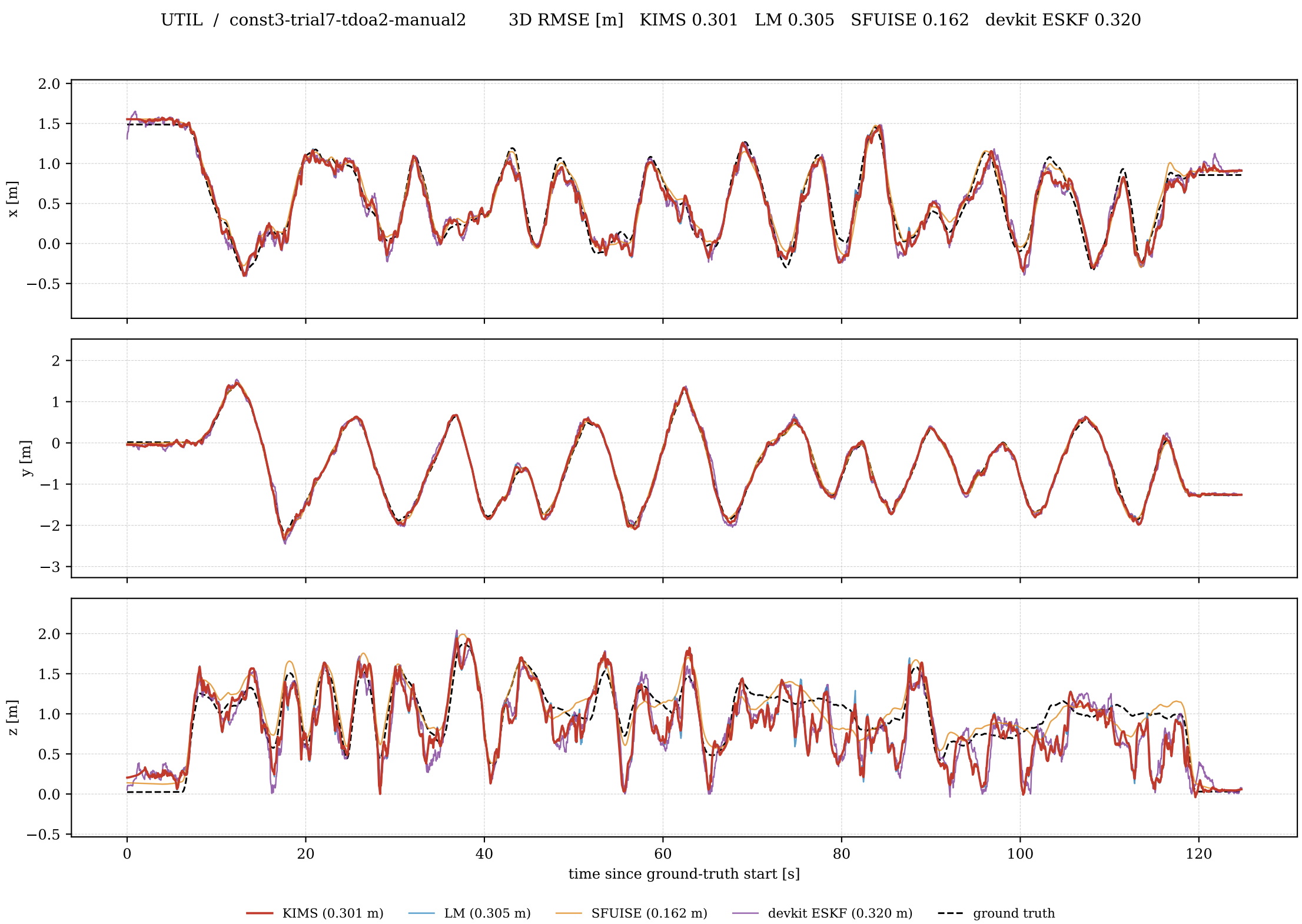

UTIL / const3-trial5-tdoa2 3D RMSE [m] KIMS 0.202 LM 0.206 SFUISE 0.186 devkit ESKF 0.221
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.202 m)
LM (0.206 m)
SFUISE (0.186 m)
devkit ESKF (0.221 m)
ground truth

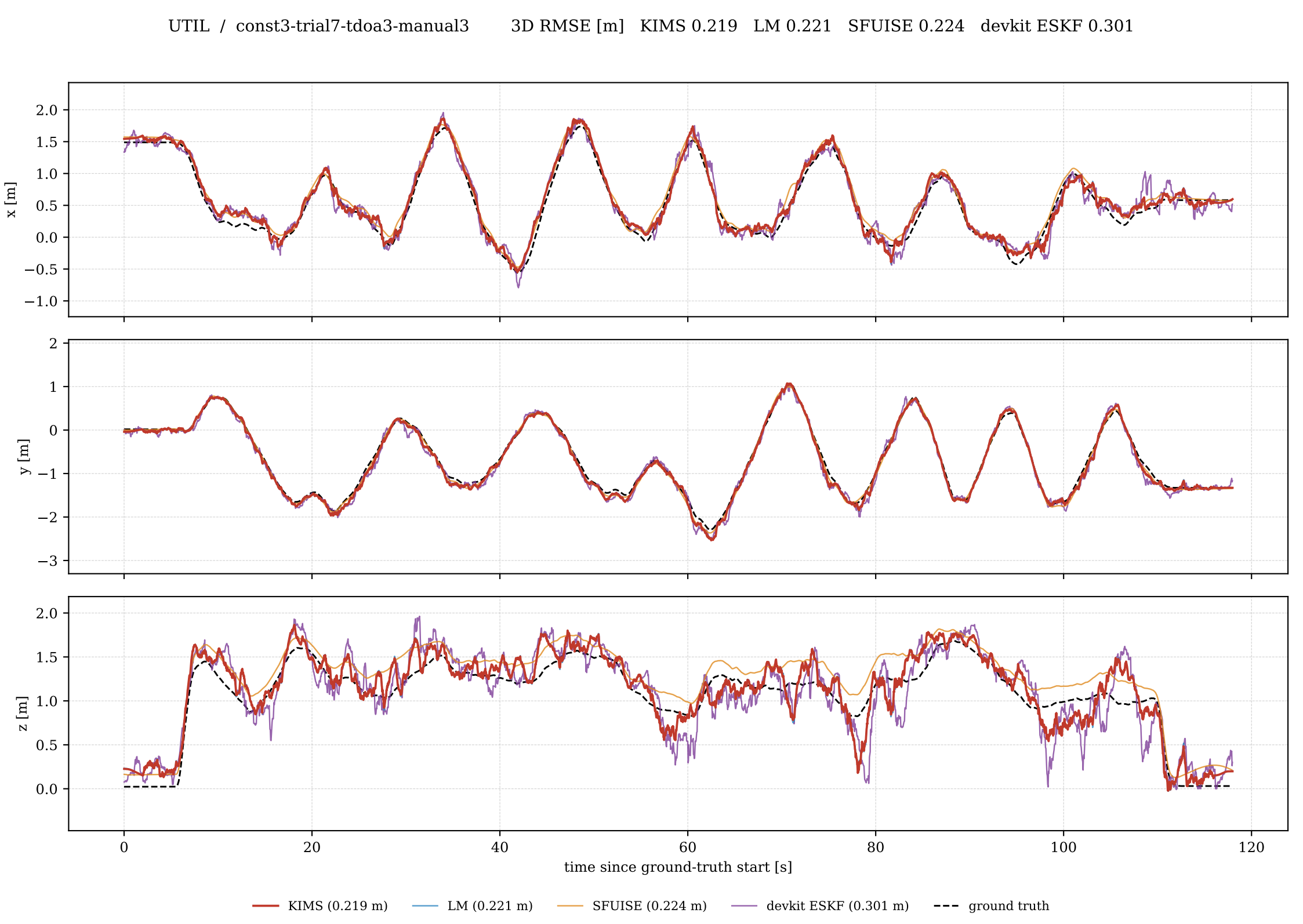

UTIL / const3-trial5-tdoa3 3D RMSE [m] KIMS 0.182 LM 0.184 SFUISE 0.177 devkit ESKF 0.230
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.182 m)
LM (0.184 m)
SFUISE (0.177 m)
devkit ESKF (0.230 m)
ground truth

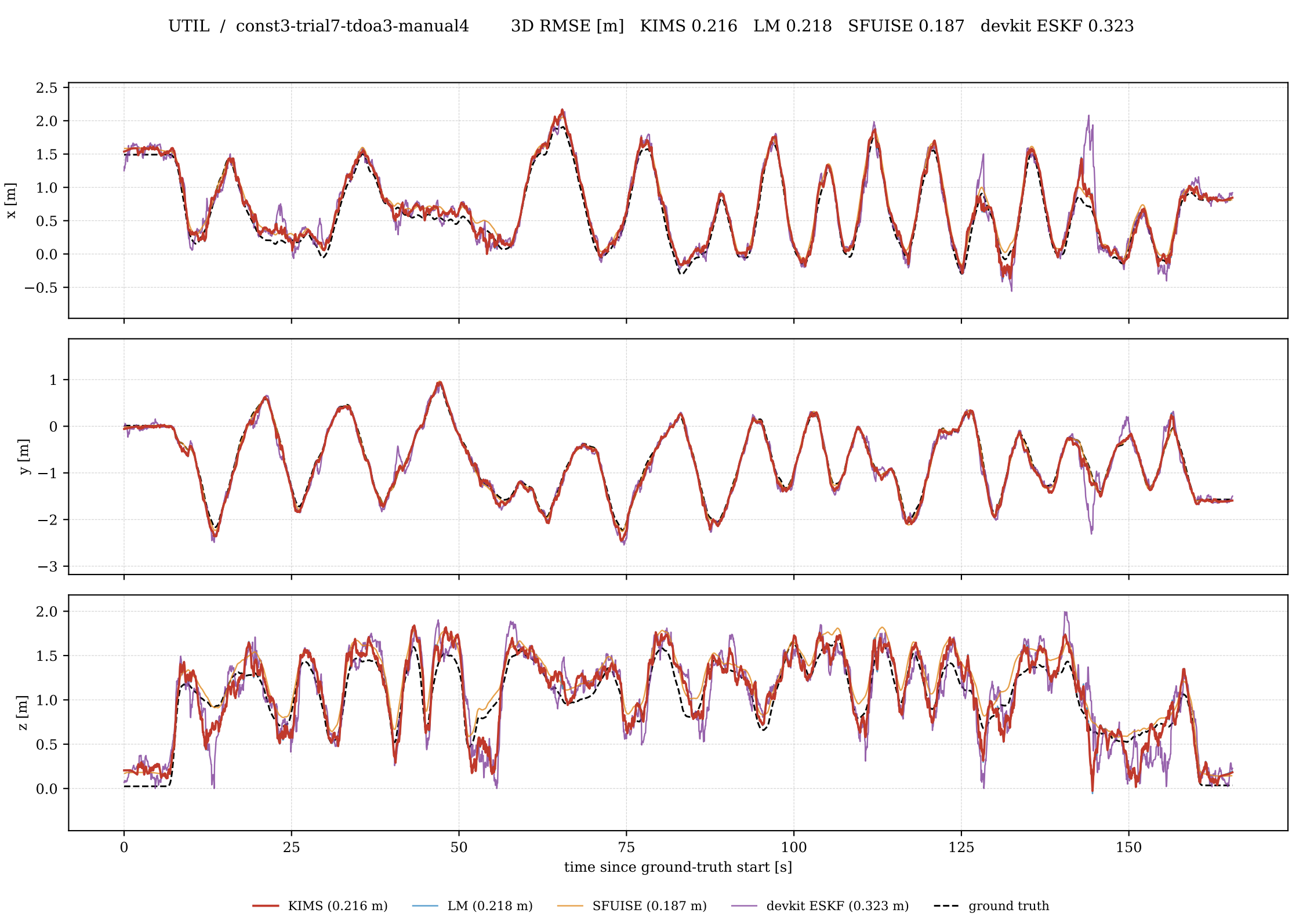

UTIL / const3-trial6-tdoa2 3D RMSE [m] KIMS 0.169 LM 0.174 SFUISE 0.171 devkit ESKF 0.195
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.169 m)
LM (0.174 m)
SFUISE (0.171 m)
devkit ESKF (0.195 m)
ground truth

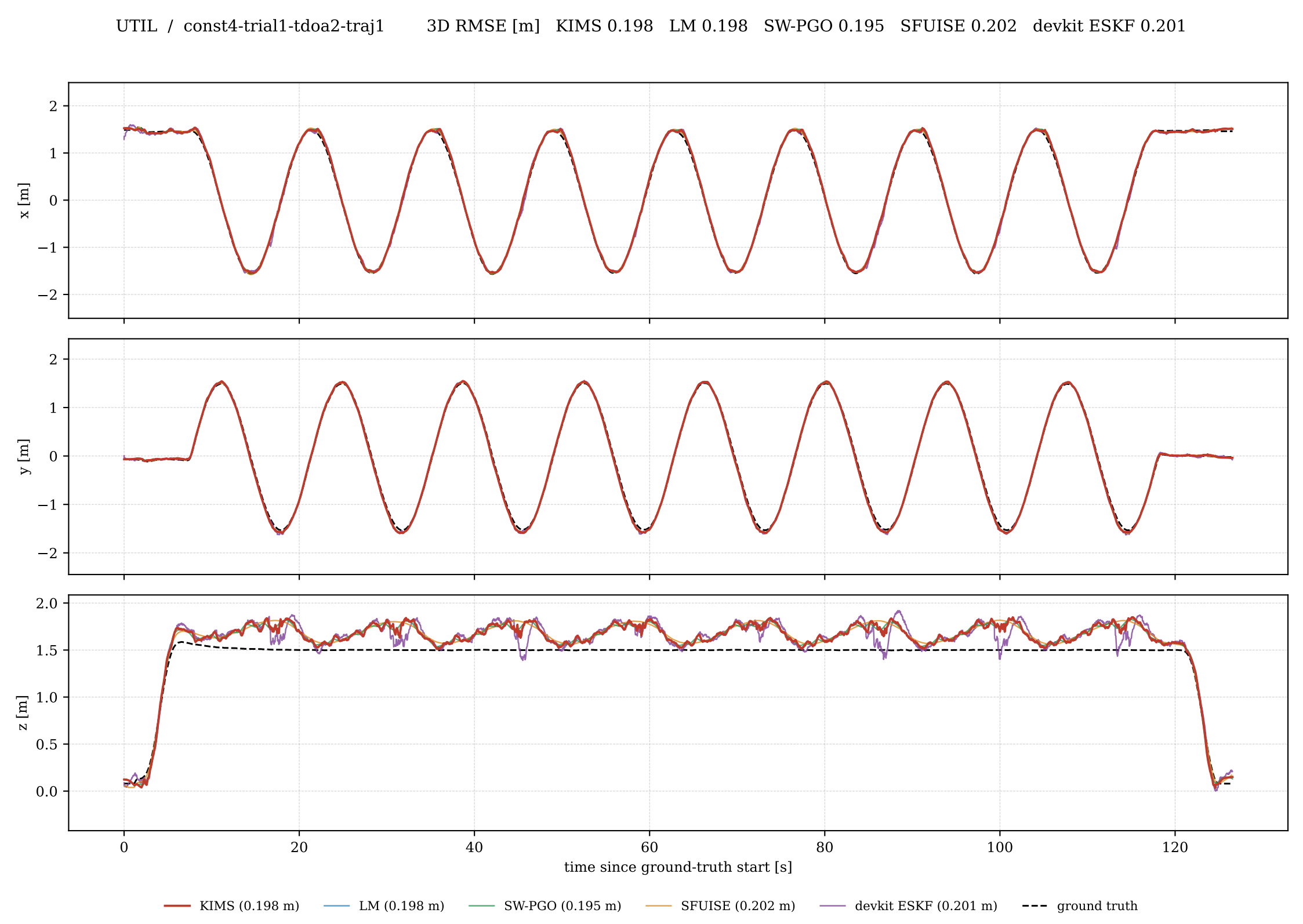

UTIL / const3-trial6-tdoa3 3D RMSE [m] KIMS 0.157 LM 0.158 SFUISE 0.165 devkit ESKF 0.202
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.157 m)
LM (0.158 m)
SFUISE (0.165 m)
devkit ESKF (0.202 m)
ground truth

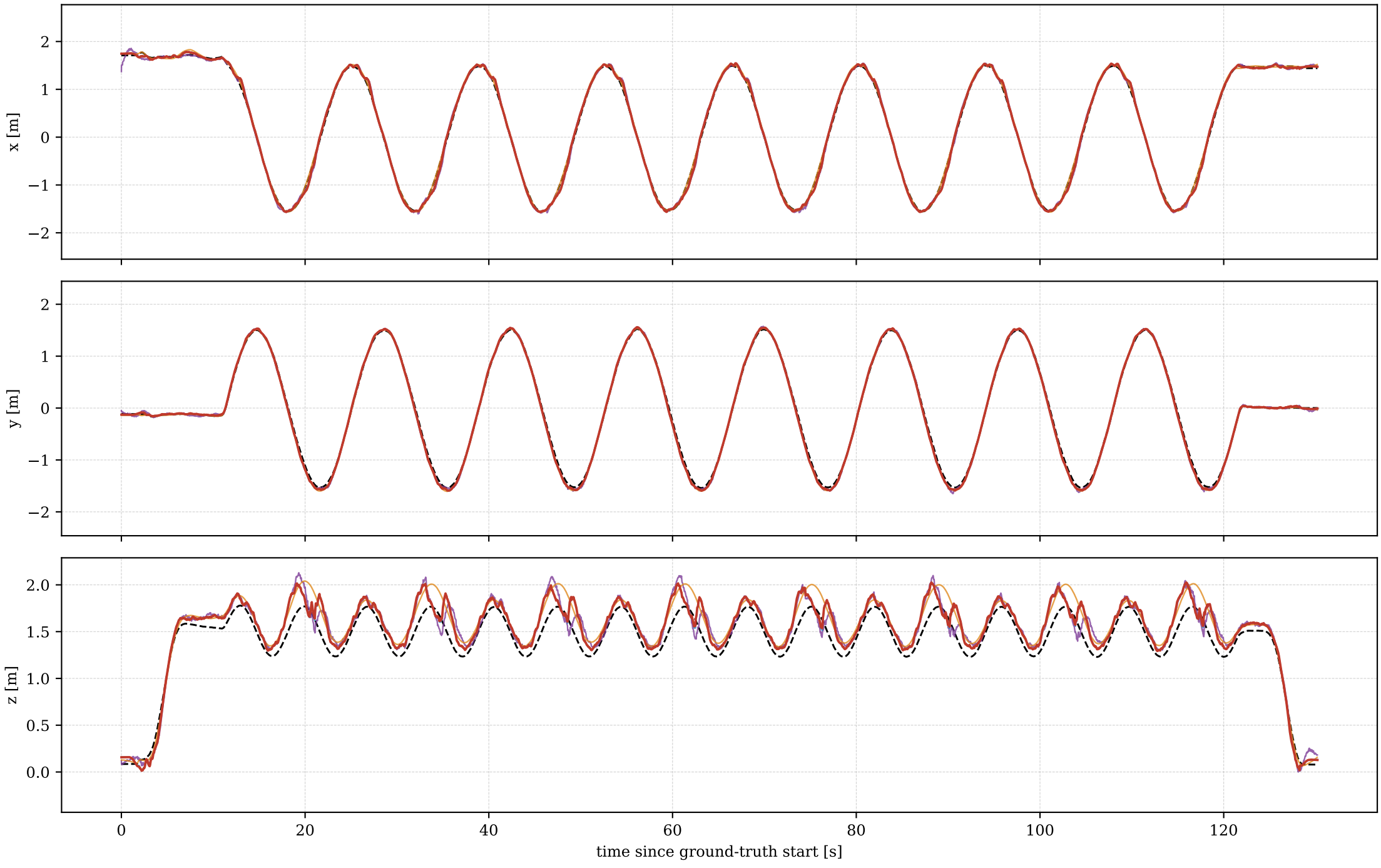

UTIL / const3-trial7-tdoa2-manual1 3D RMSE [m] KIMS 0.234 LM 0.238 SFUISE 0.138 devkit ESKF 0.258
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.234 m)
LM (0.238 m)
SFUISE (0.138 m)
devkit ESKF (0.258 m)
ground truth

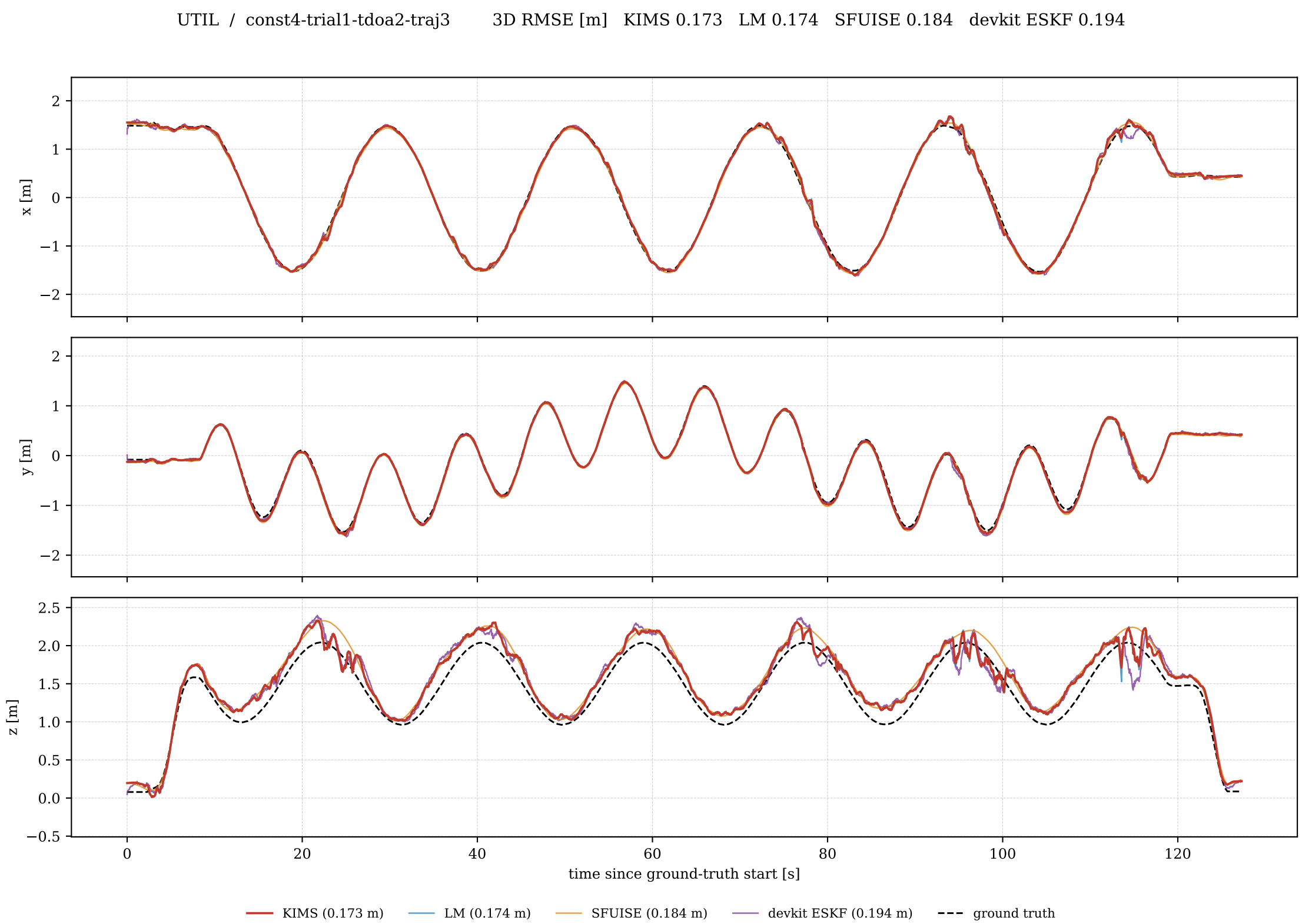

UTIL / const3-trial7-tdoa2-manual2 3D RMSE [m] KIMS 0.301 LM 0.305 SFUISE 0.162 devkit ESKF 0.320
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.301 m)
LM (0.305 m)
SFUISE (0.162 m)
devkit ESKF (0.320 m)
ground truth

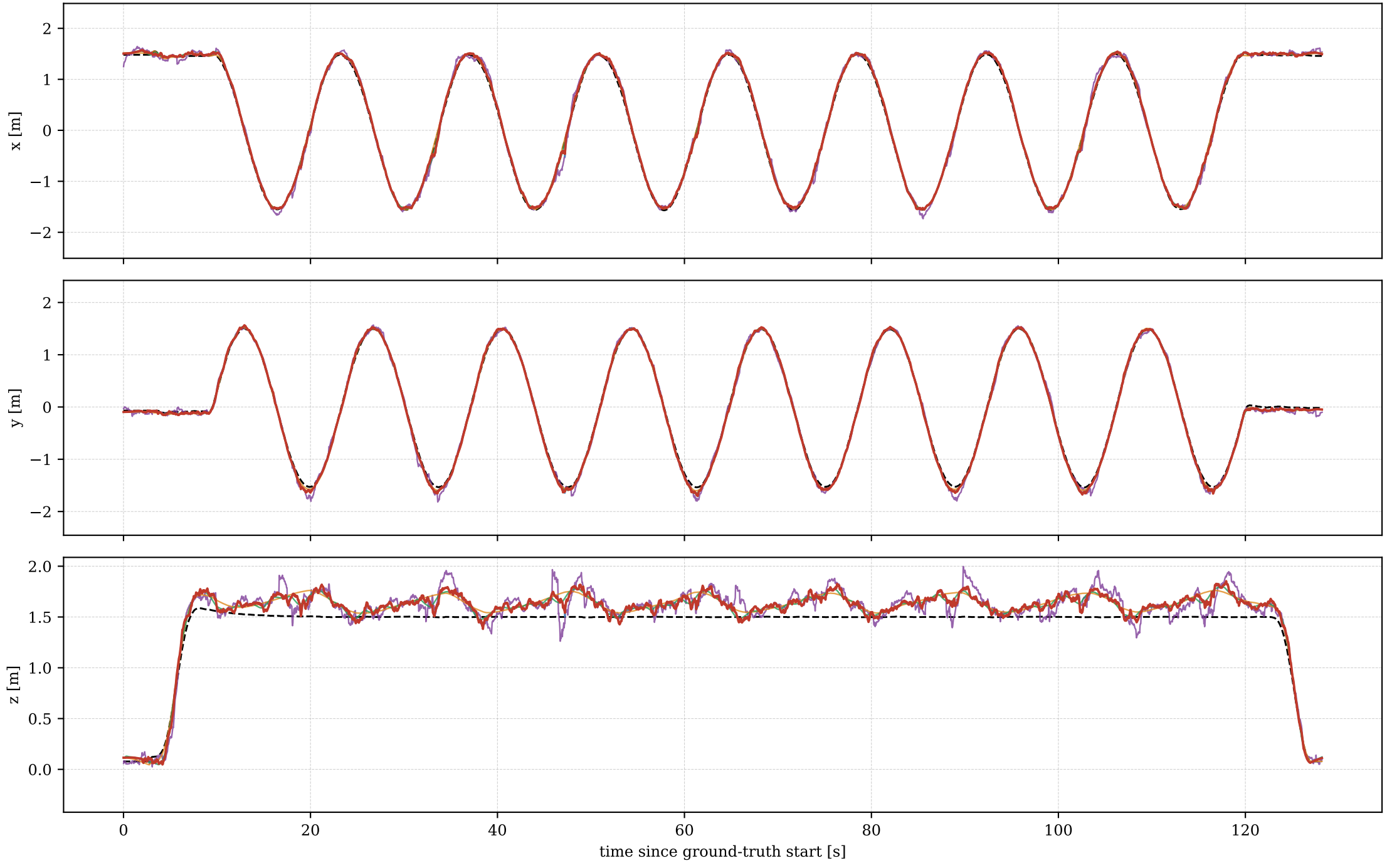

UTIL / const3-trial7-tdoa3-manual3 3D RMSE [m] KIMS 0.219 LM 0.221 SFUISE 0.224 devkit ESKF 0.301
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.219 m)
LM (0.221 m)
SFUISE (0.224 m)
devkit ESKF (0.301 m)
ground truth

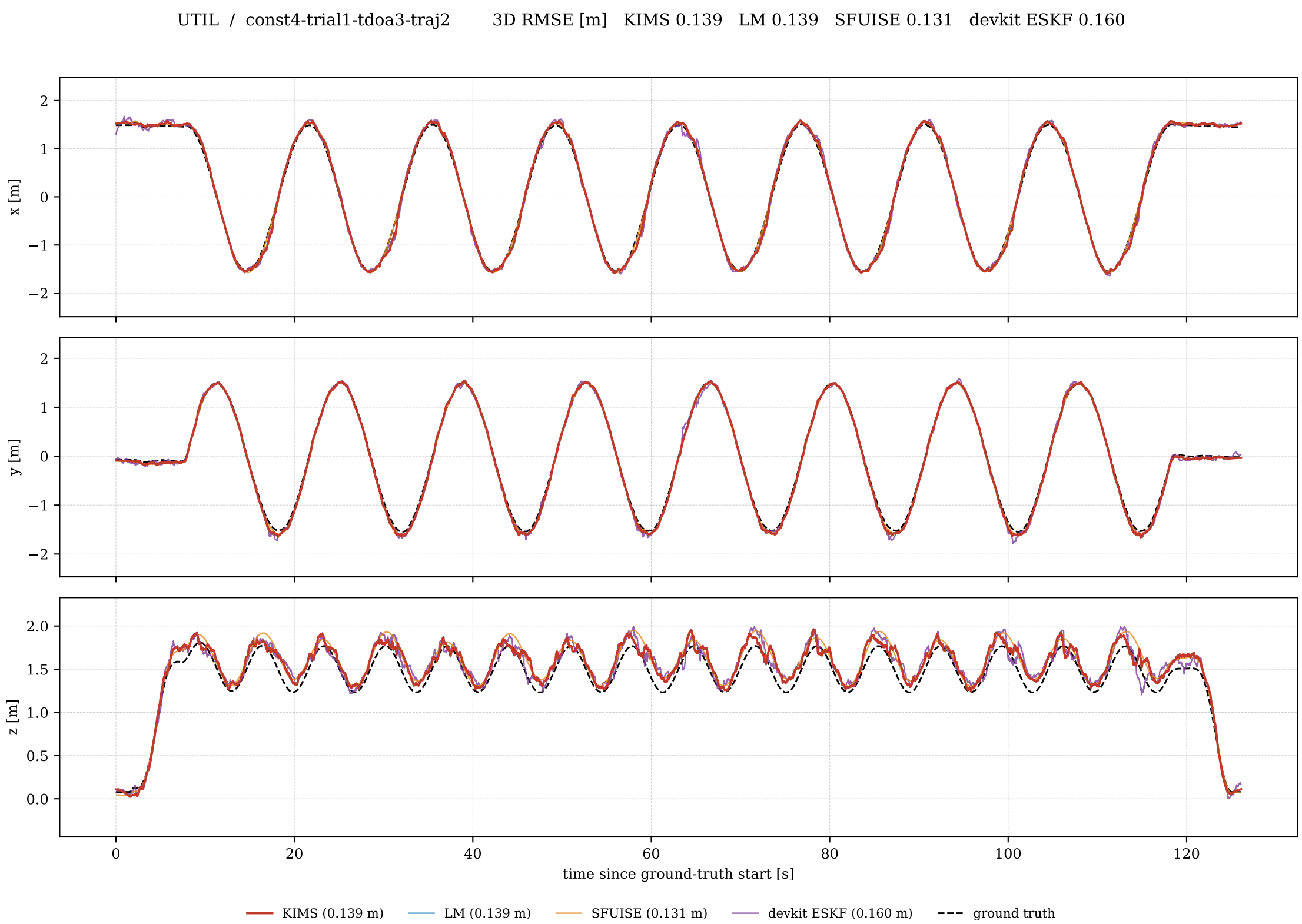

UTIL / const3-trial7-tdoa3-manual4 3D RMSE [m] KIMS 0.216 LM 0.218 SFUISE 0.187 devkit ESKF 0.323
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.216 m)
LM (0.218 m)
SFUISE (0.187 m)
devkit ESKF (0.323 m)
ground truth

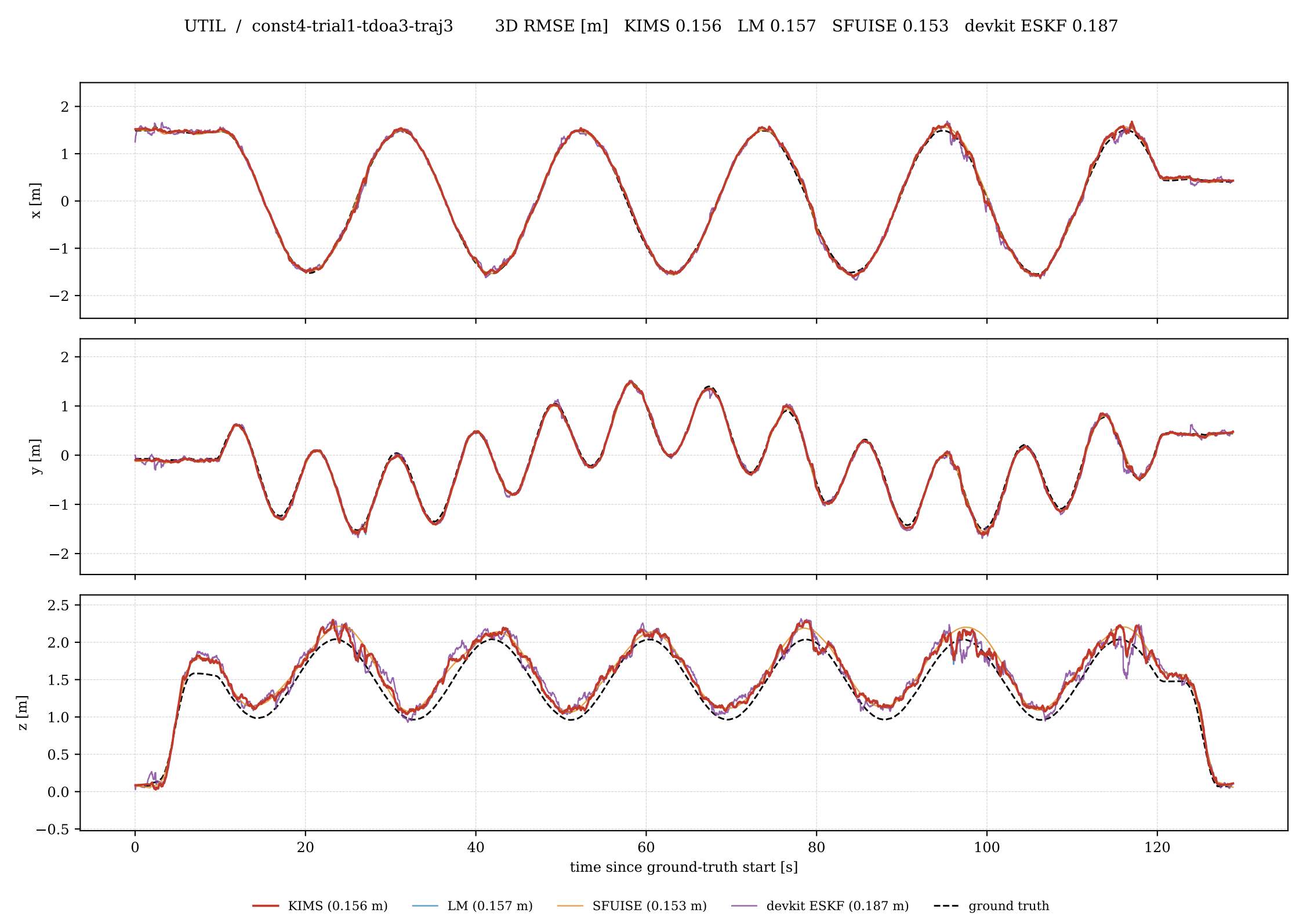

UTIL / const4-trial1-tdoa2-traj1 3D RMSE [m] KIMS 0.198 LM 0.198 SW-PGO 0.195 SFUISE 0.202 devkit ESKF 0.201
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.198 m)
LM (0.198 m)
SW-PGO (0.195 m)
SFUISE (0.202 m)
devkit ESKF (0.201 m)
ground truth

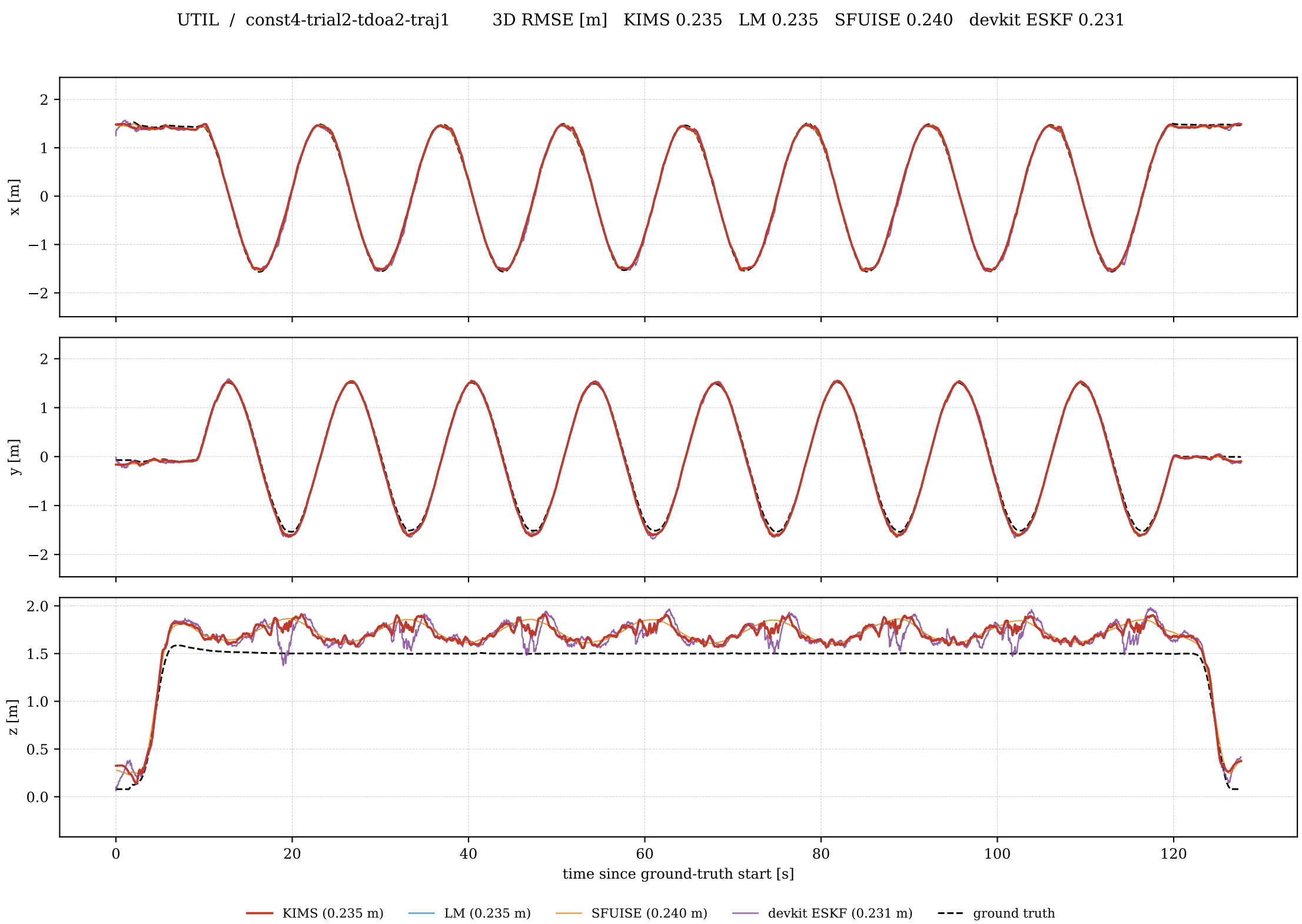

UTIL / const4-trial1-tdoa2-traj2 3D RMSE [m] KIMS 0.156 LM 0.156 SFUISE 0.159 devkit ESKF 0.168
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.156 m)
LM (0.156 m)
SFUISE (0.159 m)
devkit ESKF (0.168 m)
ground truth

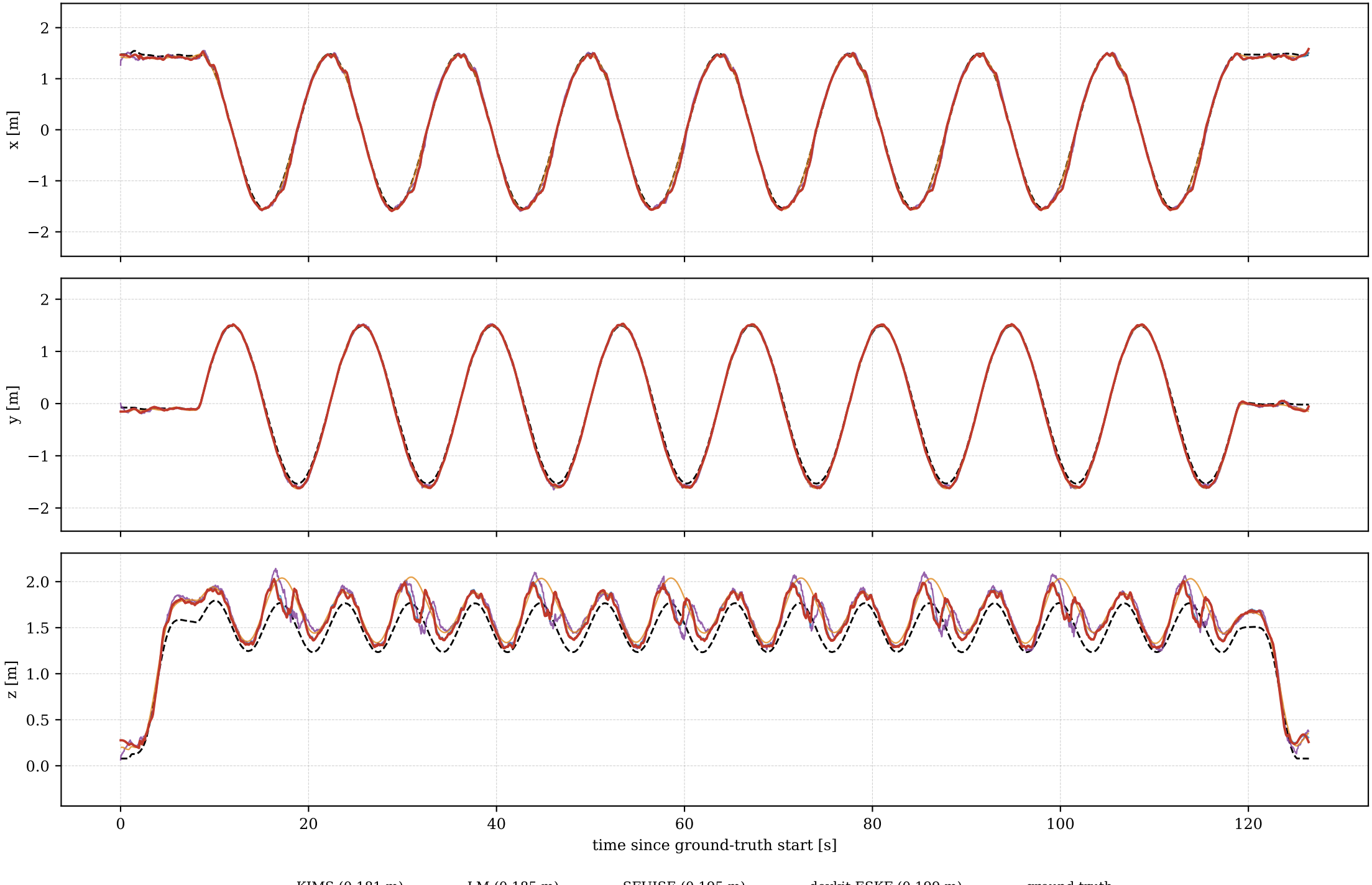

UTIL / const4-trial1-tdoa2-traj3 3D RMSE [m] KIMS 0.173 LM 0.174 SFUISE 0.184 devkit ESKF 0.194
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.173 m)
LM (0.174 m)
SFUISE (0.184 m)
devkit ESKF (0.194 m)
ground truth

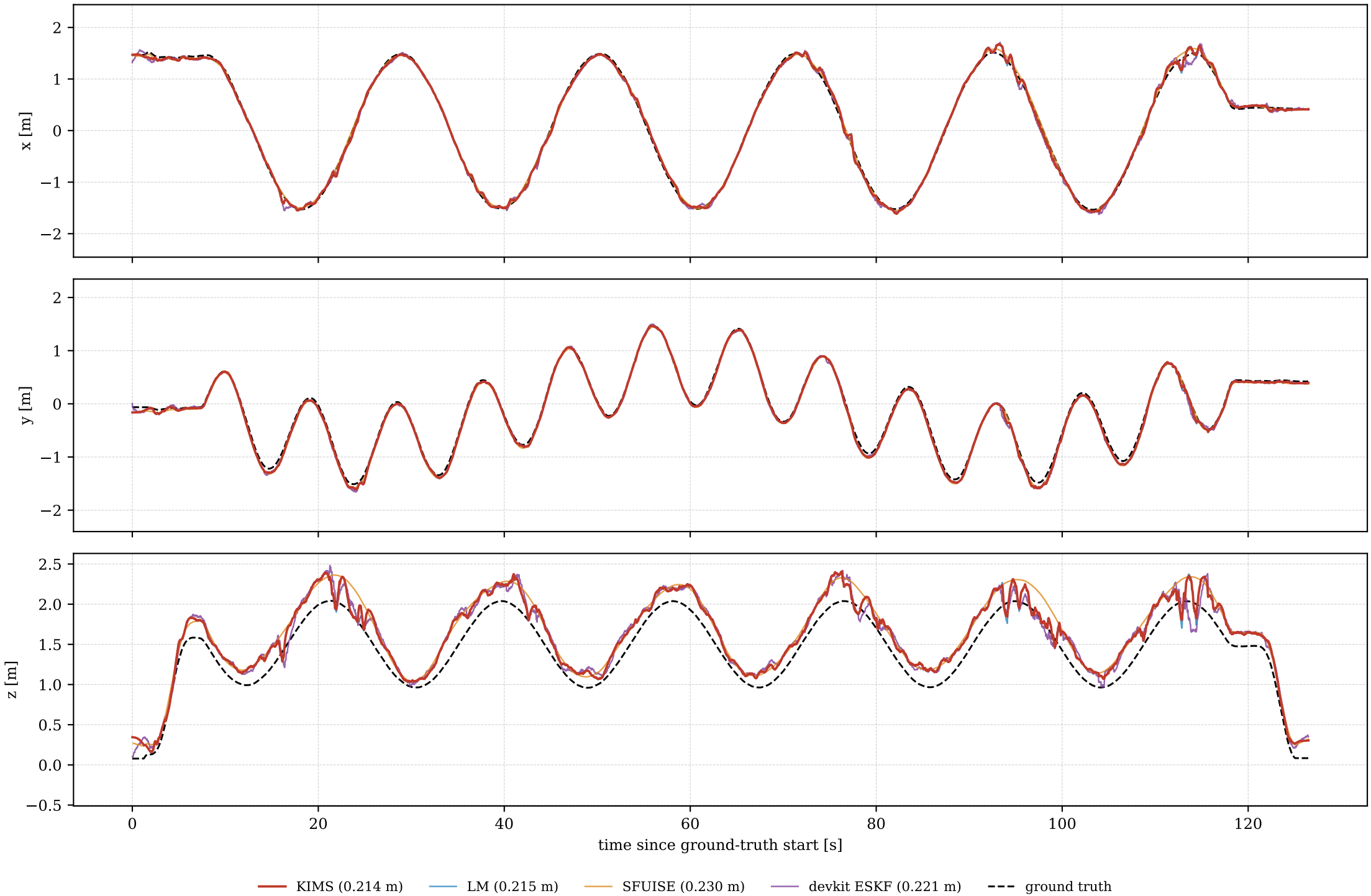

UTIL / const4-trial1-tdoa3-traj1 3D RMSE [m] KIMS 0.159 LM 0.159 SW-PGO 0.151 SFUISE 0.155 devkit ESKF 0.195
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.159 m)
LM (0.159 m)
SW-PGO (0.151 m)
SFUISE (0.155 m)
devkit ESKF (0.195 m)
ground truth

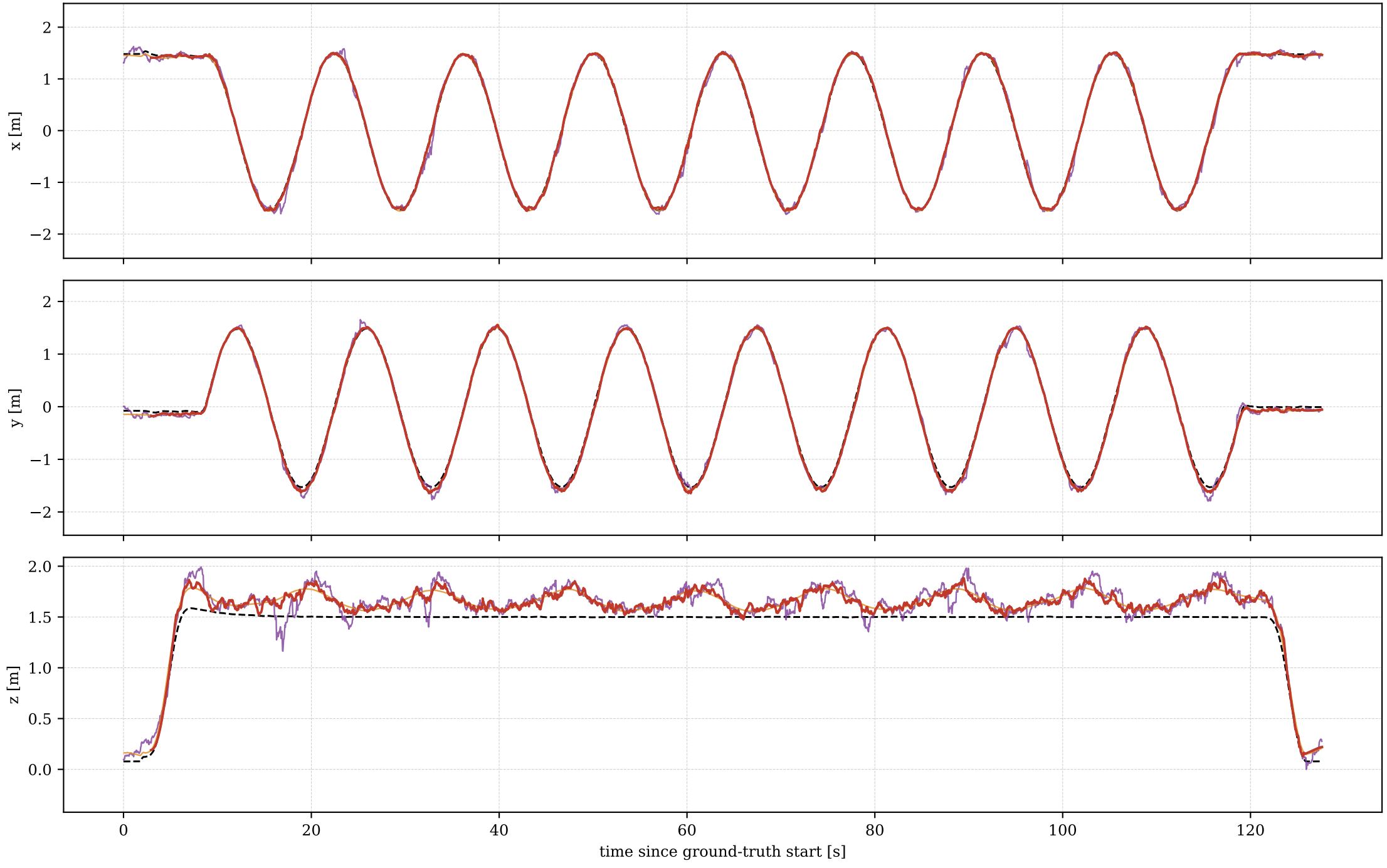

UTIL / const4-trial1-tdoa3-traj2 3D RMSE [m] KIMS 0.139 LM 0.139 SFUISE 0.131 devkit ESKF 0.160
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.139 m)
LM (0.139 m)
SFUISE (0.131 m)
devkit ESKF (0.160 m)
ground truth

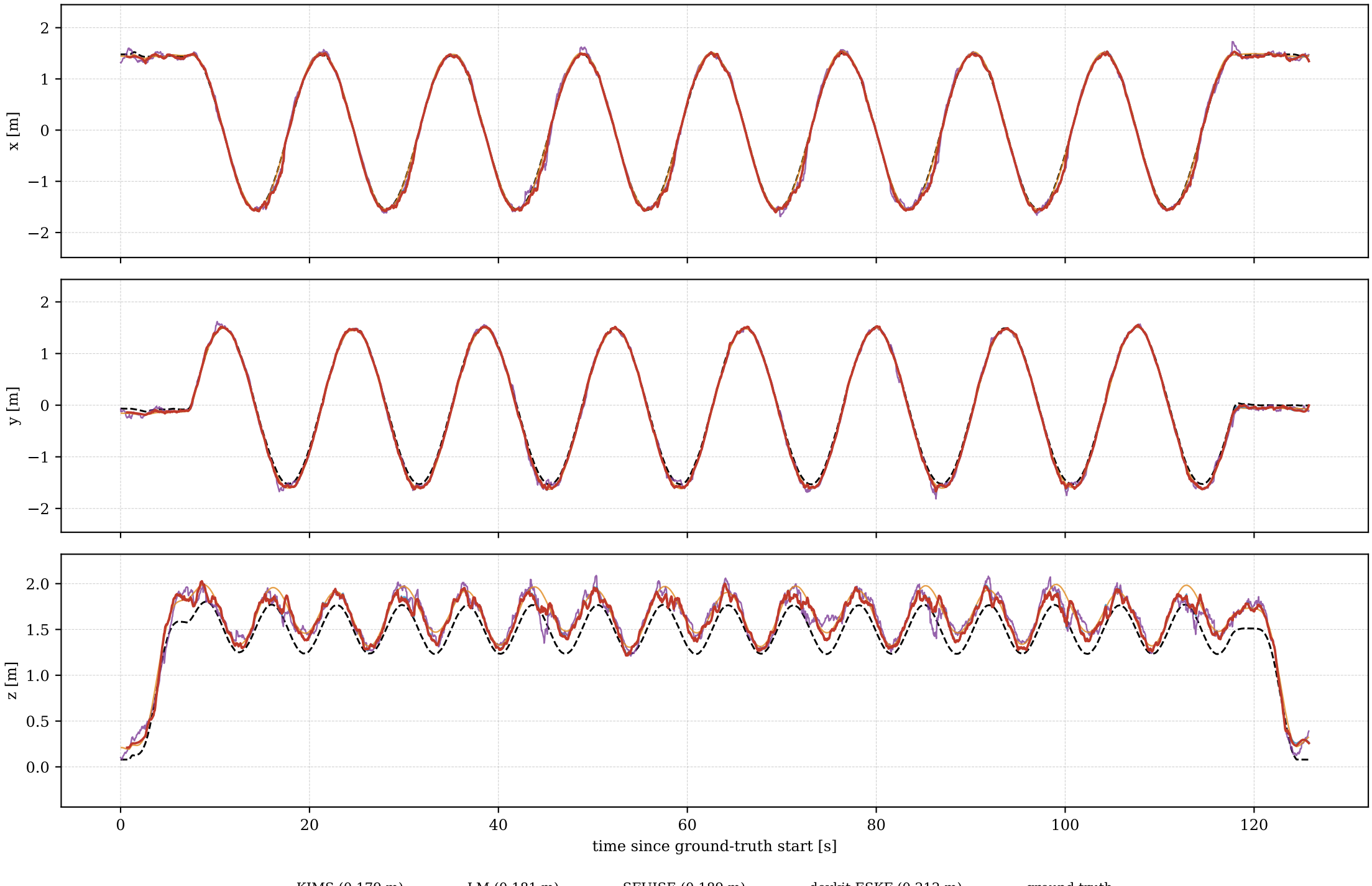

UTIL / const4-trial1-tdoa3-traj3 3D RMSE [m] KIMS 0.156 LM 0.157 SFUISE 0.153 devkit ESKF 0.187
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.156 m)
LM (0.157 m)
SFUISE (0.153 m)
devkit ESKF (0.187 m)
ground truth

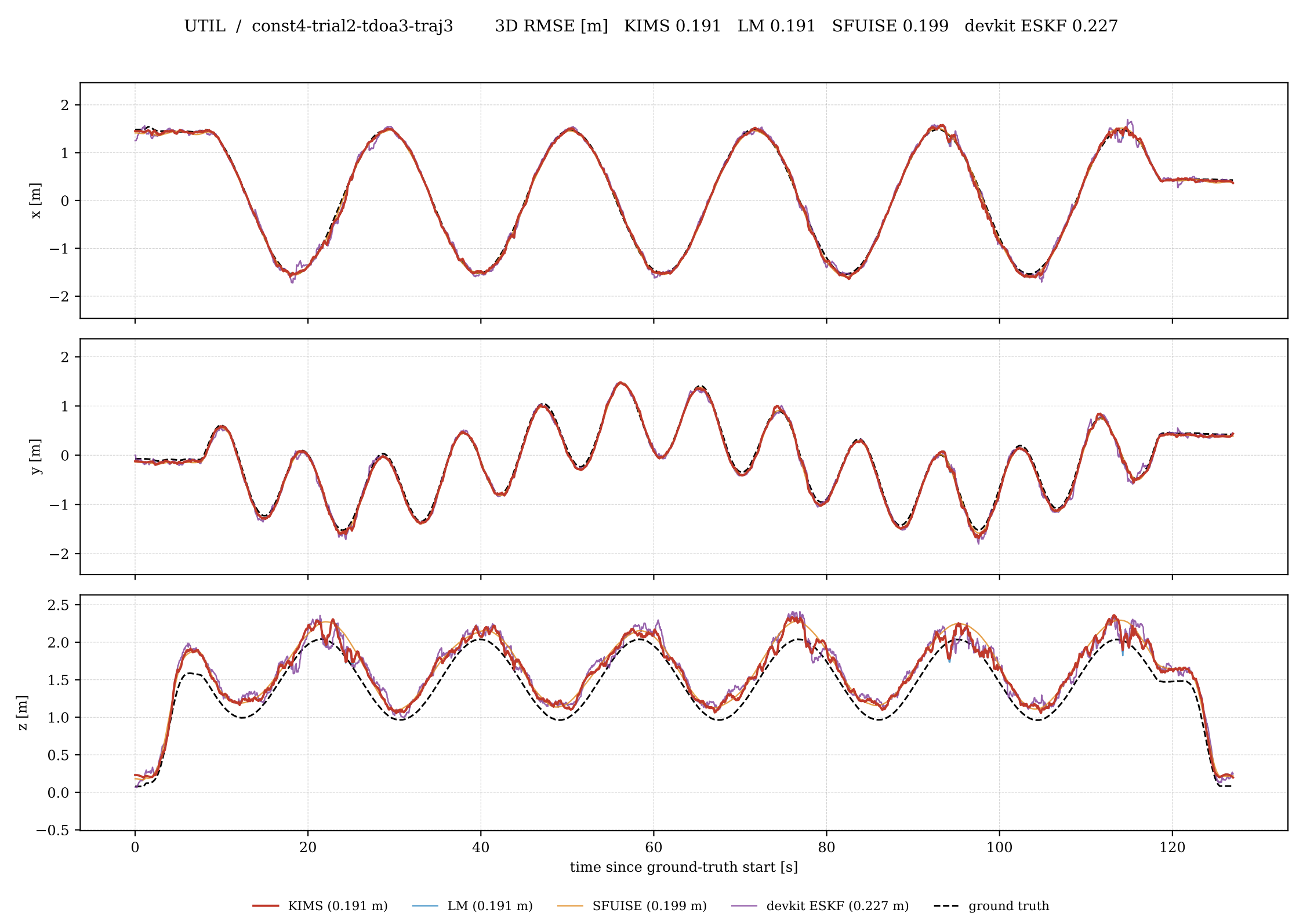

UTIL / const4-trial2-tdoa2-traj1 3D RMSE [m] KIMS 0.235 LM 0.235 SFUISE 0.240 devkit ESKF 0.231
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.235 m)
LM (0.235 m)
SFUISE (0.240 m)
devkit ESKF (0.231 m)
ground truth

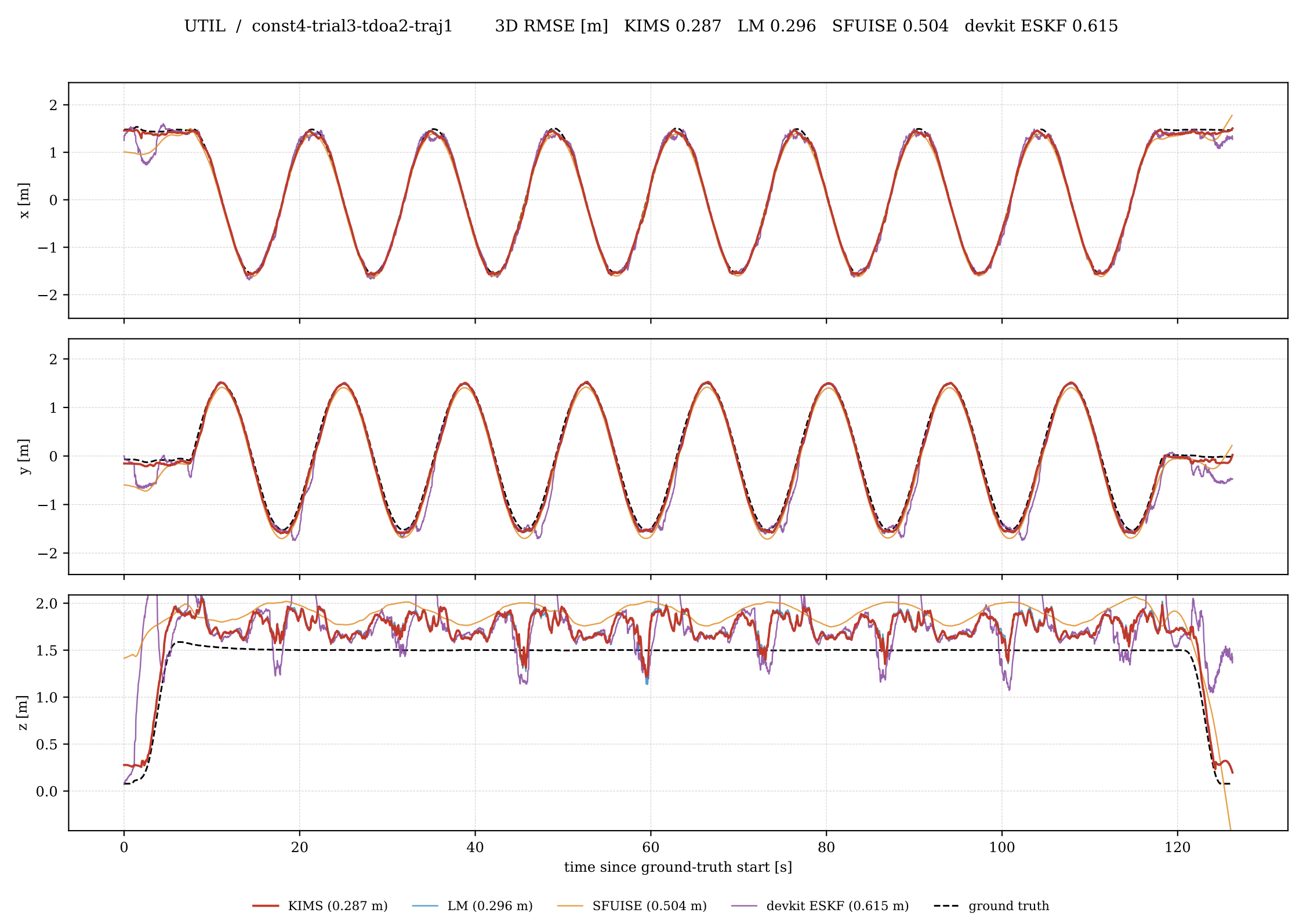

UTIL / const4-trial2-tdoa2-traj2 3D RMSE [m] KIMS 0.181 LM 0.185 SFUISE 0.195 devkit ESKF 0.199
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.181 m)
LM (0.185 m)
SFUISE (0.195 m)
devkit ESKF (0.199 m)
ground truth

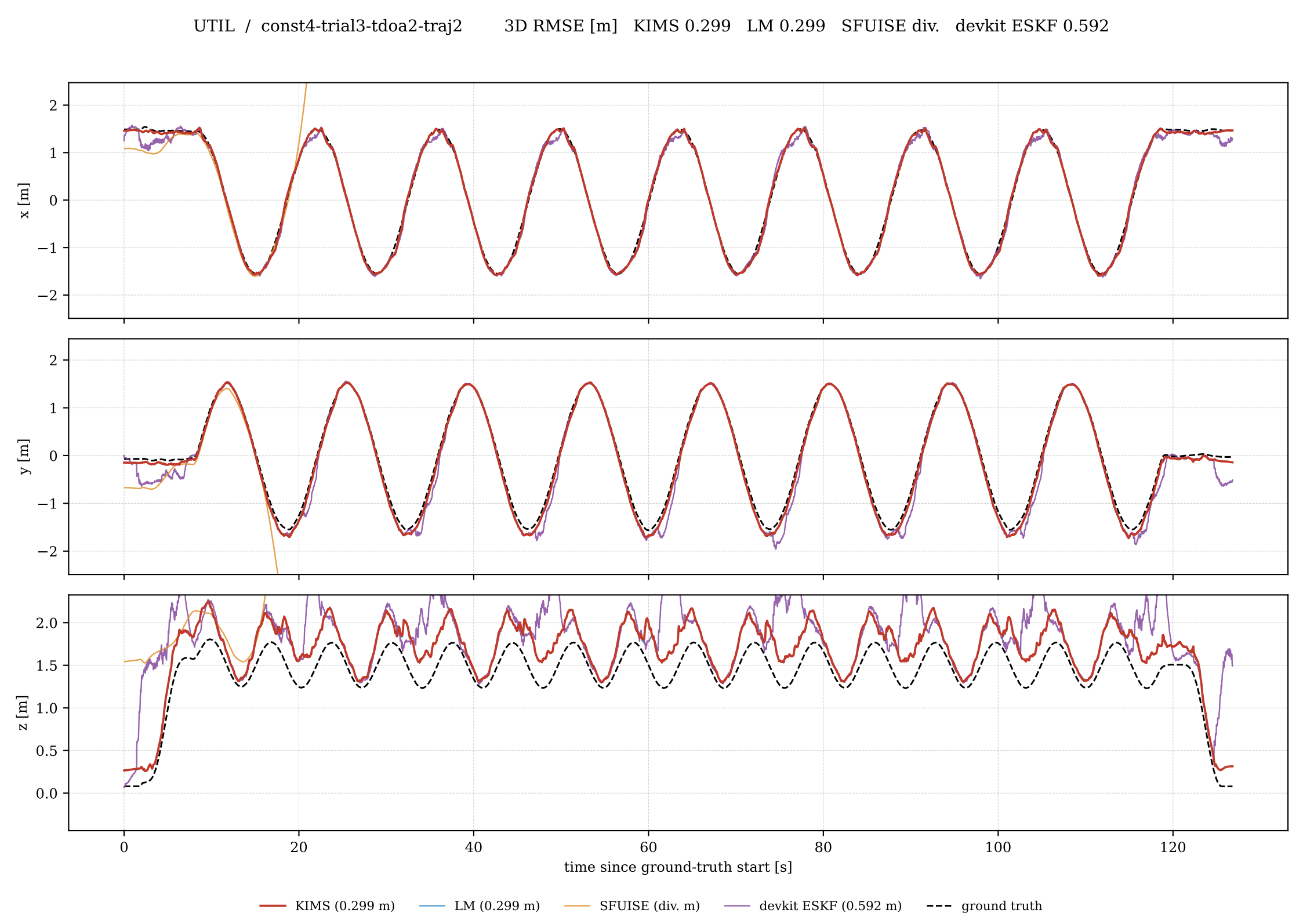

UTIL / const4-trial2-tdoa2-traj3 3D RMSE [m] KIMS 0.214 LM 0.215 SFUISE 0.230 devkit ESKF 0.221
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.214 m)
LM (0.215 m)
SFUISE (0.230 m)
devkit ESKF (0.221 m)
ground truth

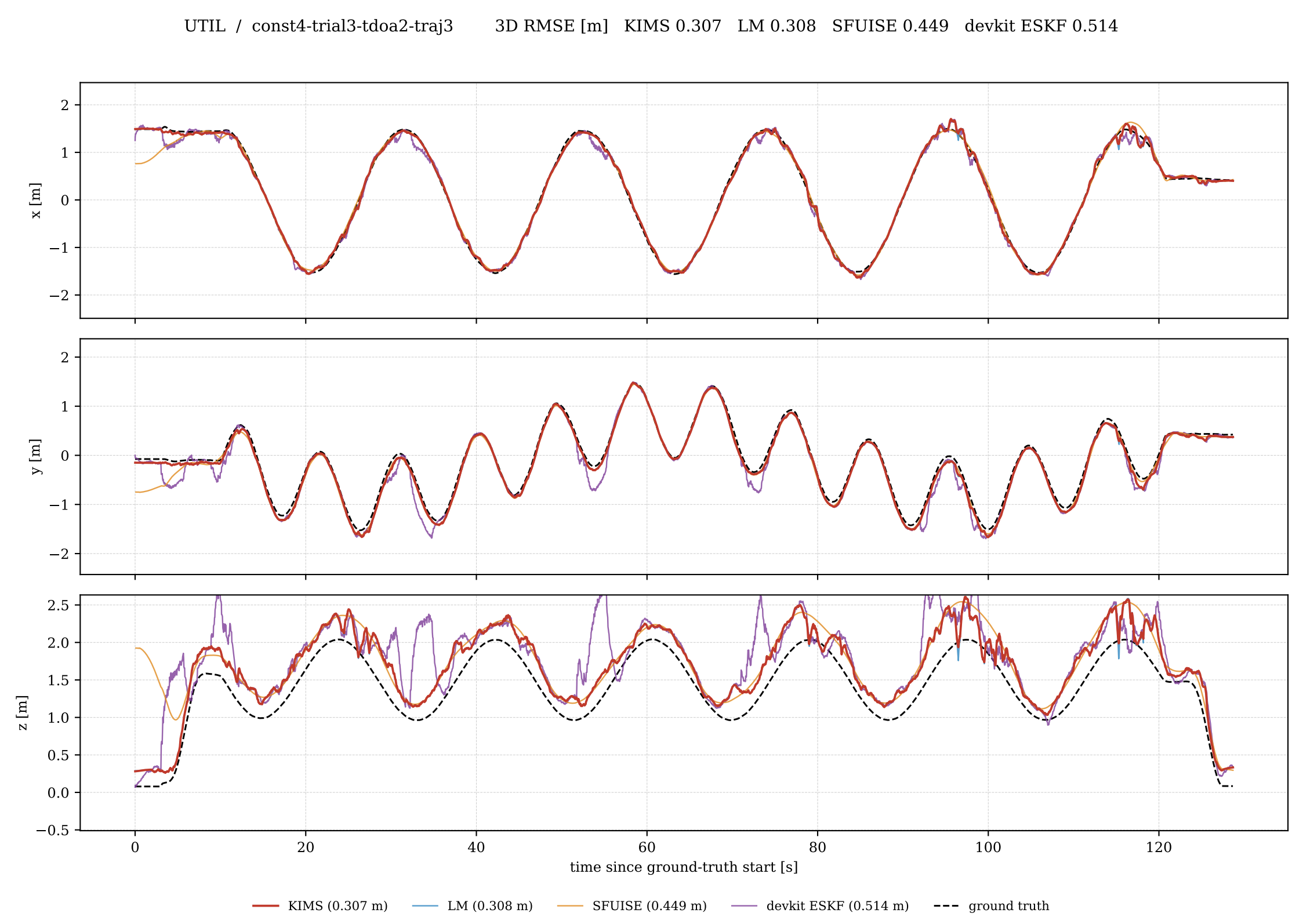

UTIL / const4-trial2-tdoa3-traj1 3D RMSE [m] KIMS 0.183 LM 0.183 SFUISE 0.177 devkit ESKF 0.217
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.183 m)
LM (0.183 m)
SFUISE (0.177 m)
devkit ESKF (0.217 m)
ground truth

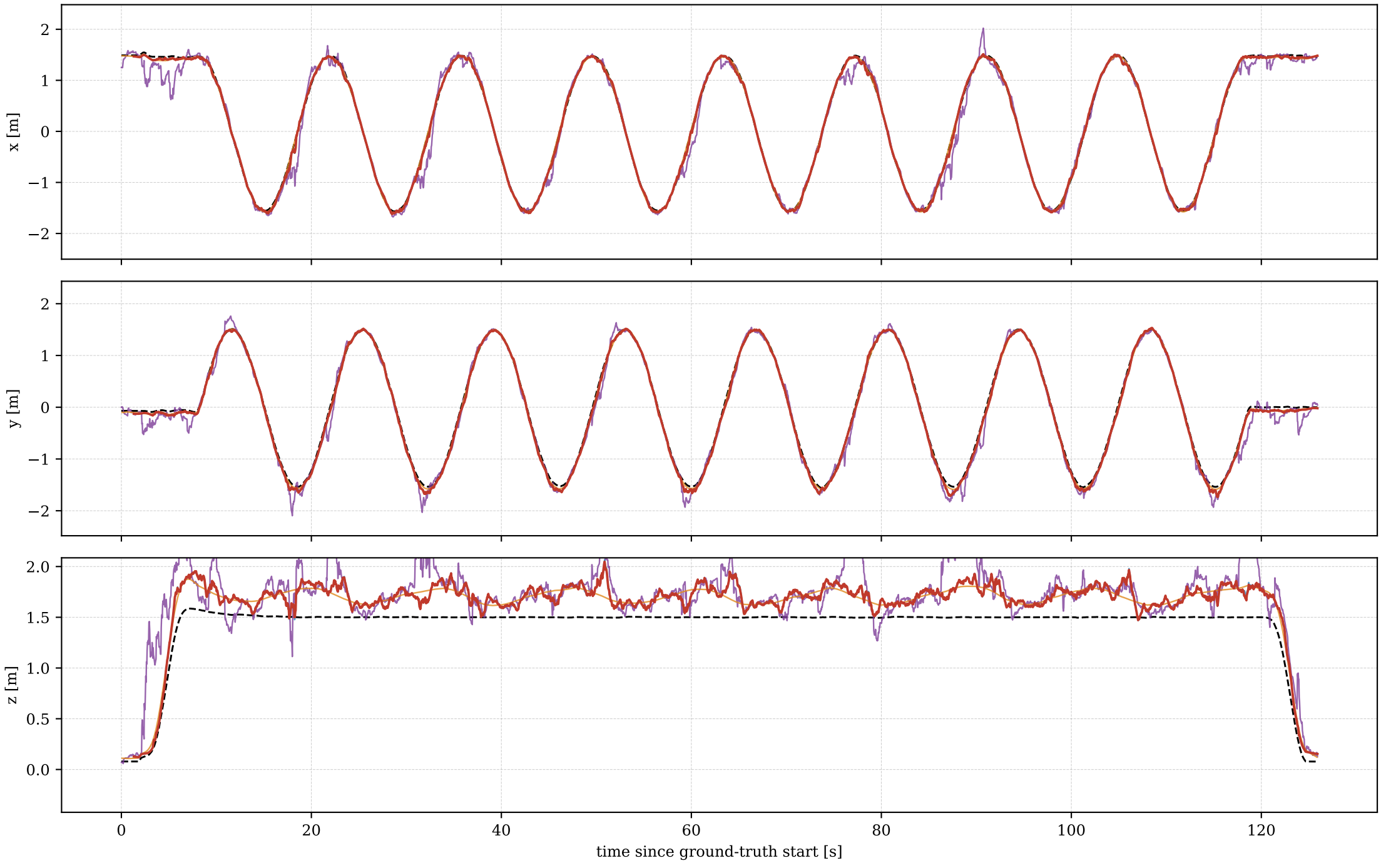

UTIL / const4-trial2-tdoa3-traj2 3D RMSE [m] KIMS 0.179 LM 0.181 SFUISE 0.189 devkit ESKF 0.212
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.179 m)
LM (0.181 m)
SFUISE (0.189 m)
devkit ESKF (0.212 m)
ground truth

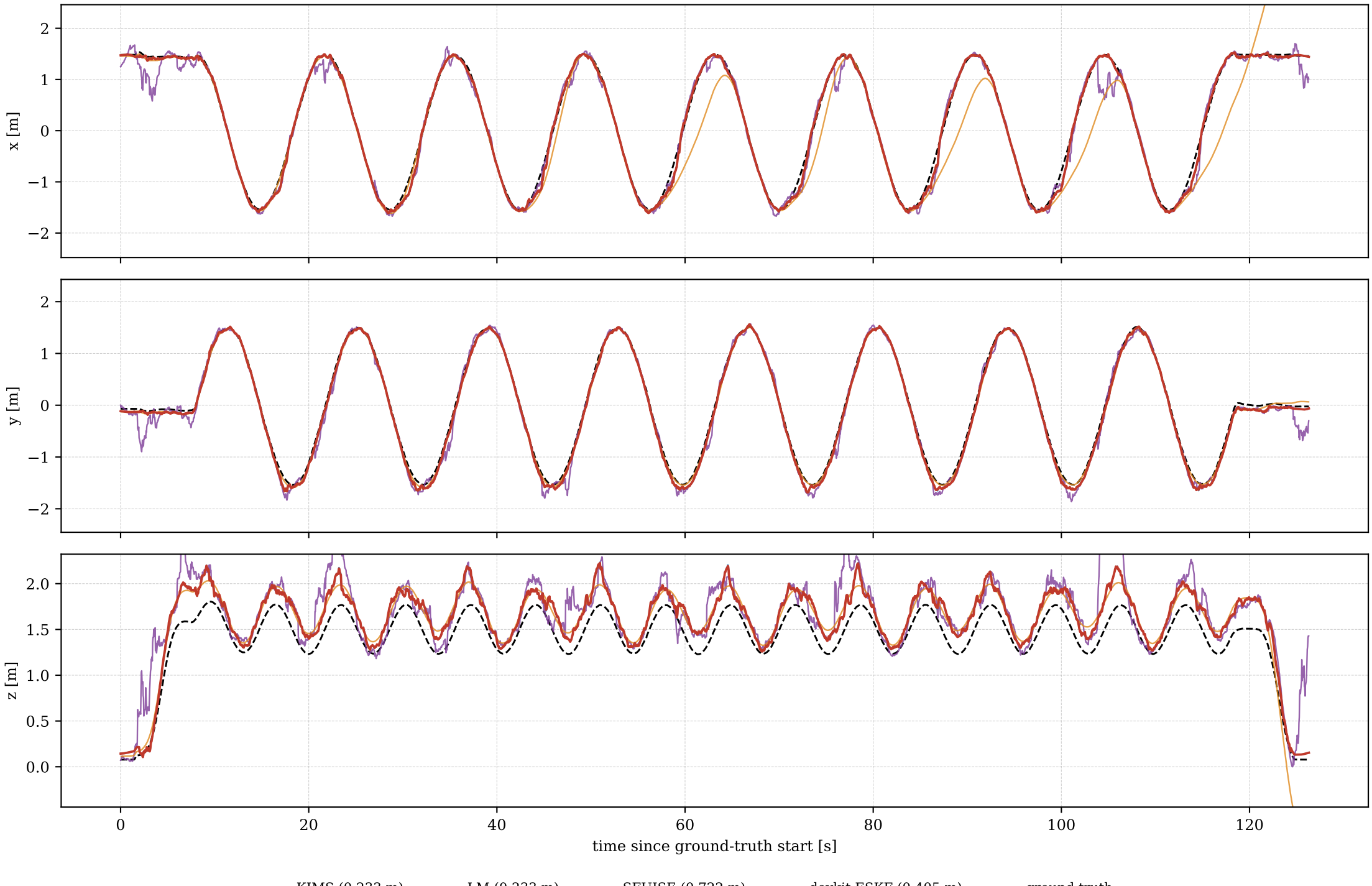

UTIL / const4-trial2-tdoa3-traj3 3D RMSE [m] KIMS 0.191 LM 0.191 SFUISE 0.199 devkit ESKF 0.227
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.191 m)
LM (0.191 m)
SFUISE (0.199 m)
devkit ESKF (0.227 m)
ground truth

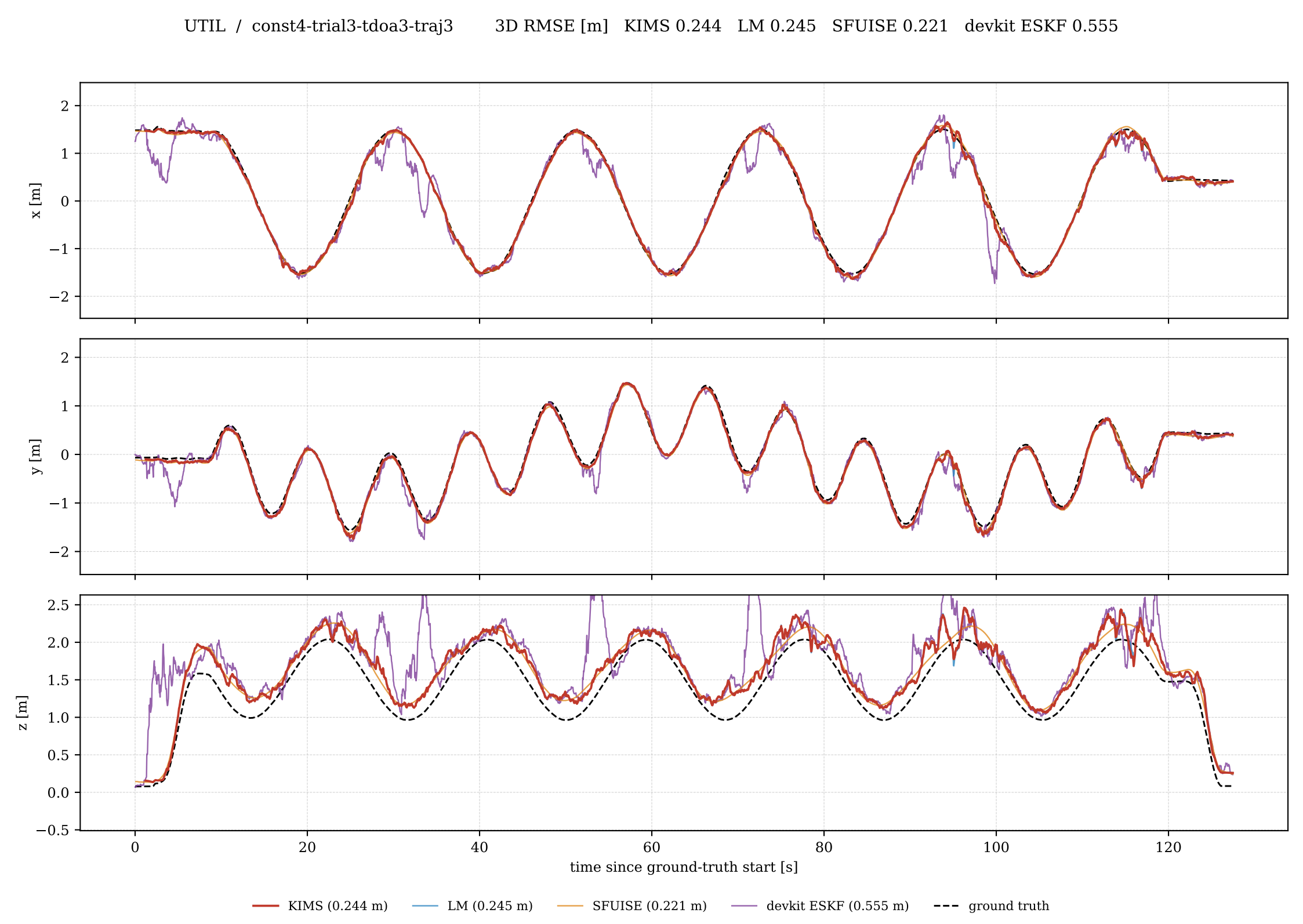

UTIL / const4-trial3-tdoa2-traj1 3D RMSE [m] KIMS 0.287 LM 0.296 SFUISE 0.504 devkit ESKF 0.615
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.287 m)
LM (0.296 m)
SFUISE (0.504 m)
devkit ESKF (0.615 m)
ground truth

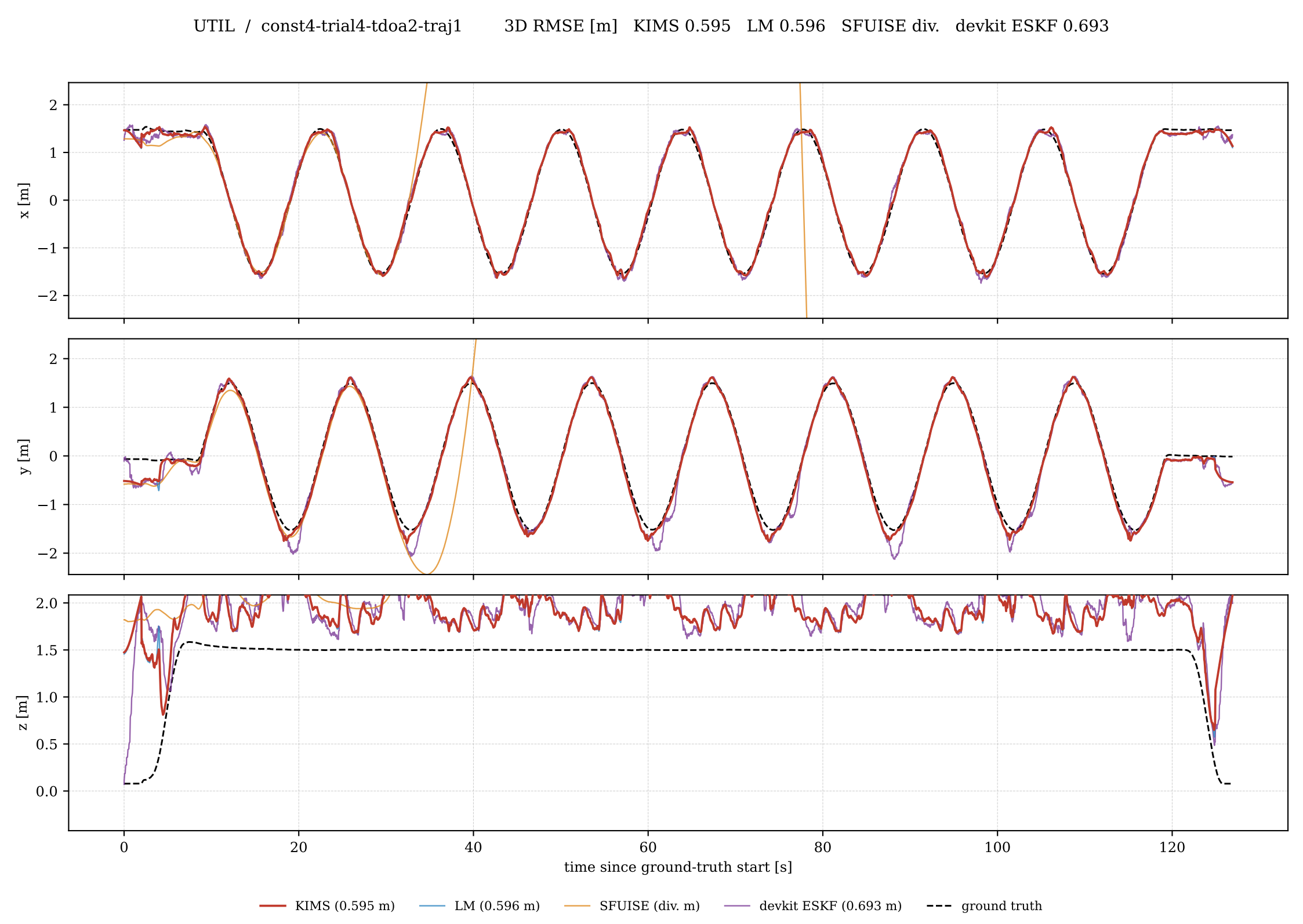

UTIL / const4-trial3-tdoa2-traj2 3D RMSE [m] KIMS 0.299 LM 0.299 SFUISE div. devkit ESKF 0.592
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.299 m)
LM (0.299 m)
SFUISE (div. m)
devkit ESKF (0.592 m)
ground truth

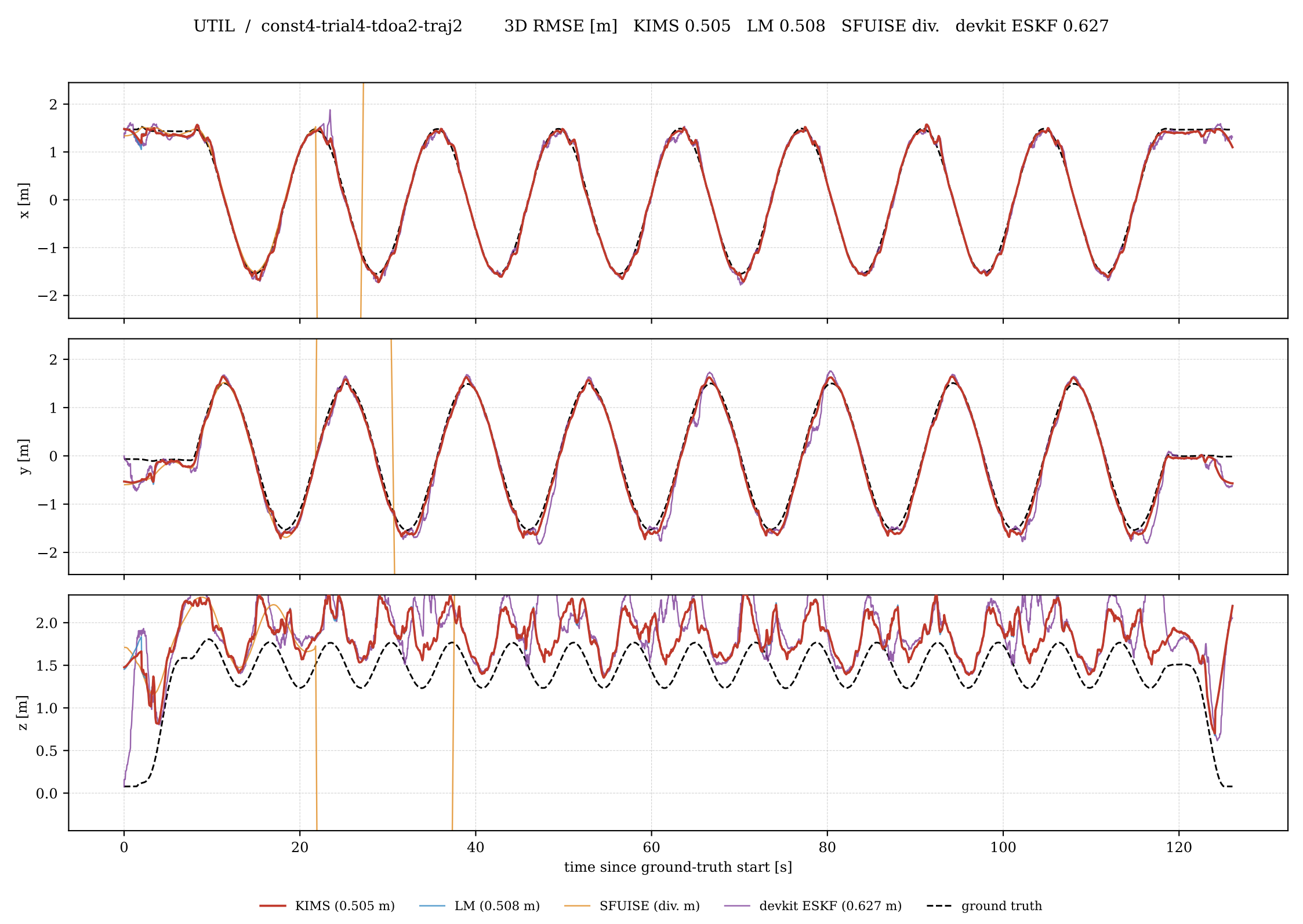

UTIL / const4-trial3-tdoa2-traj3 3D RMSE [m] KIMS 0.307 LM 0.308 SFUISE 0.449 devkit ESKF 0.514
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.307 m)
LM (0.308 m)
SFUISE (0.449 m)
devkit ESKF (0.514 m)
ground truth

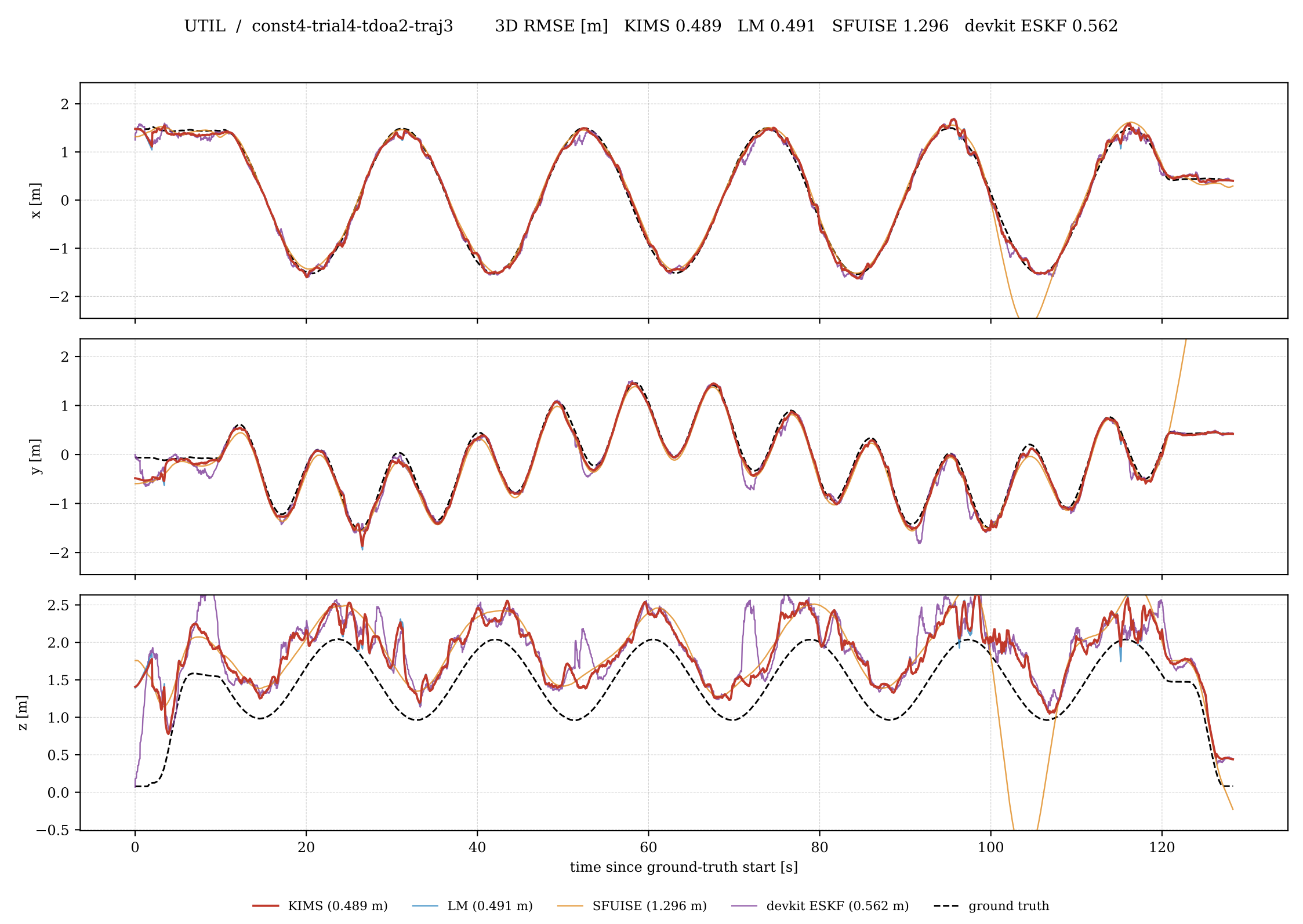

UTIL / const4-trial3-tdoa3-traj1 3D RMSE [m] KIMS 0.242 LM 0.242 SFUISE 0.216 devkit ESKF 0.431
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.242 m)
LM (0.242 m)
SFUISE (0.216 m)
devkit ESKF (0.431 m)
ground truth

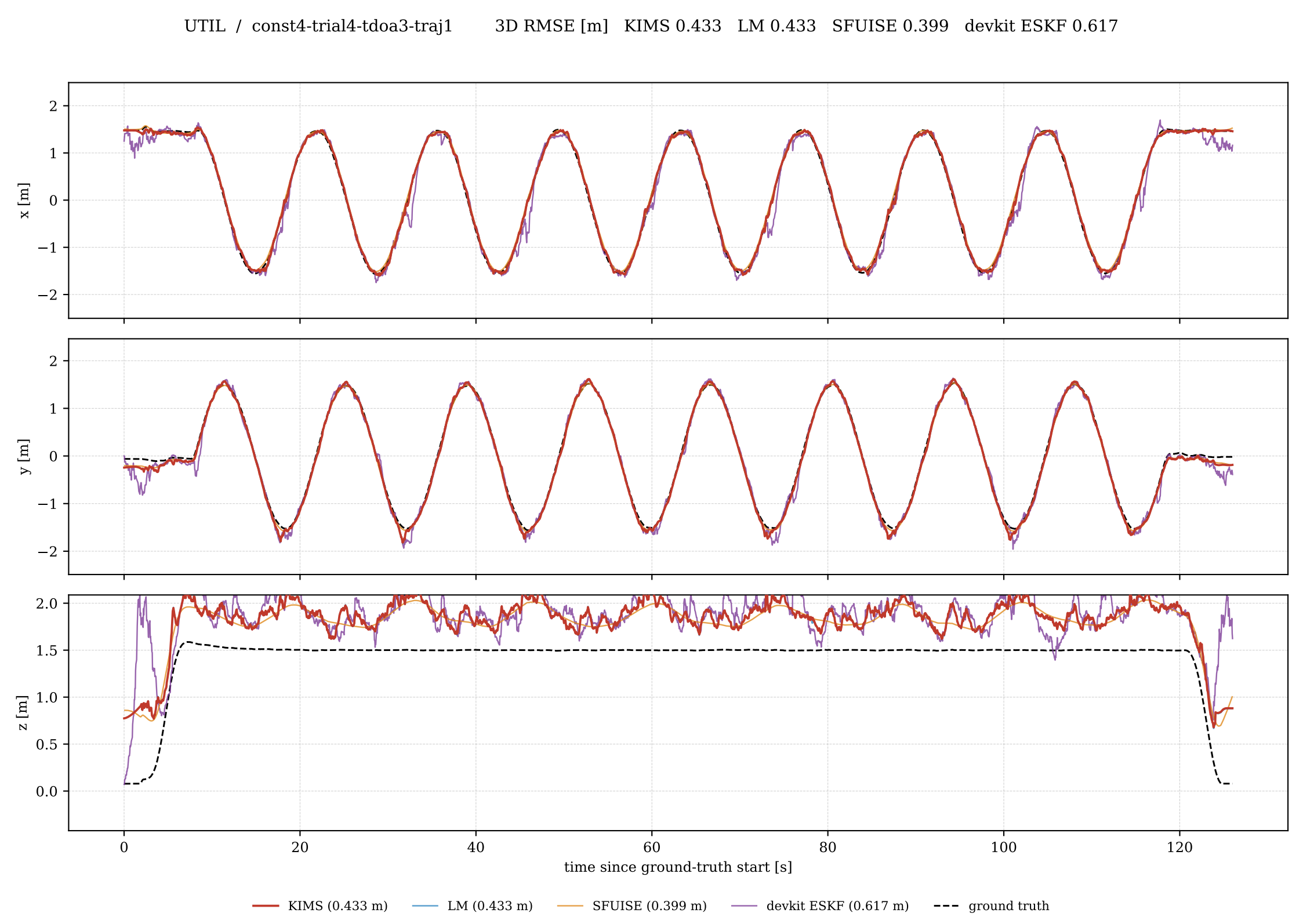

UTIL / const4-trial3-tdoa3-traj2 3D RMSE [m] KIMS 0.233 LM 0.233 SFUISE 0.722 devkit ESKF 0.405
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.233 m)
LM (0.233 m)
SFUISE (0.722 m)
devkit ESKF (0.405 m)
ground truth

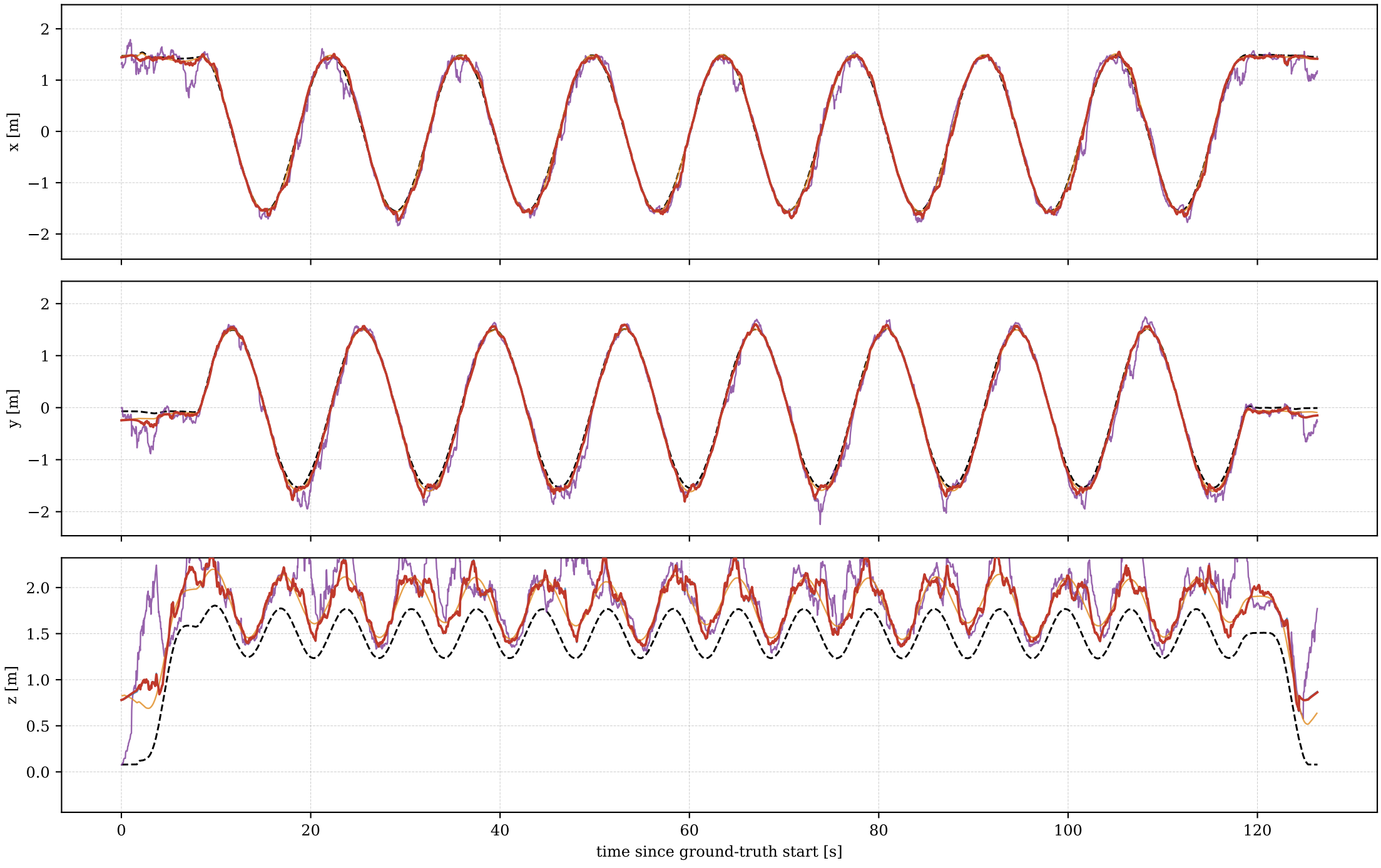

UTIL / const4-trial3-tdoa3-traj3 3D RMSE [m] KIMS 0.244 LM 0.245 SFUISE 0.221 devkit ESKF 0.555
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.244 m)
LM (0.245 m)
SFUISE (0.221 m)
devkit ESKF (0.555 m)
ground truth

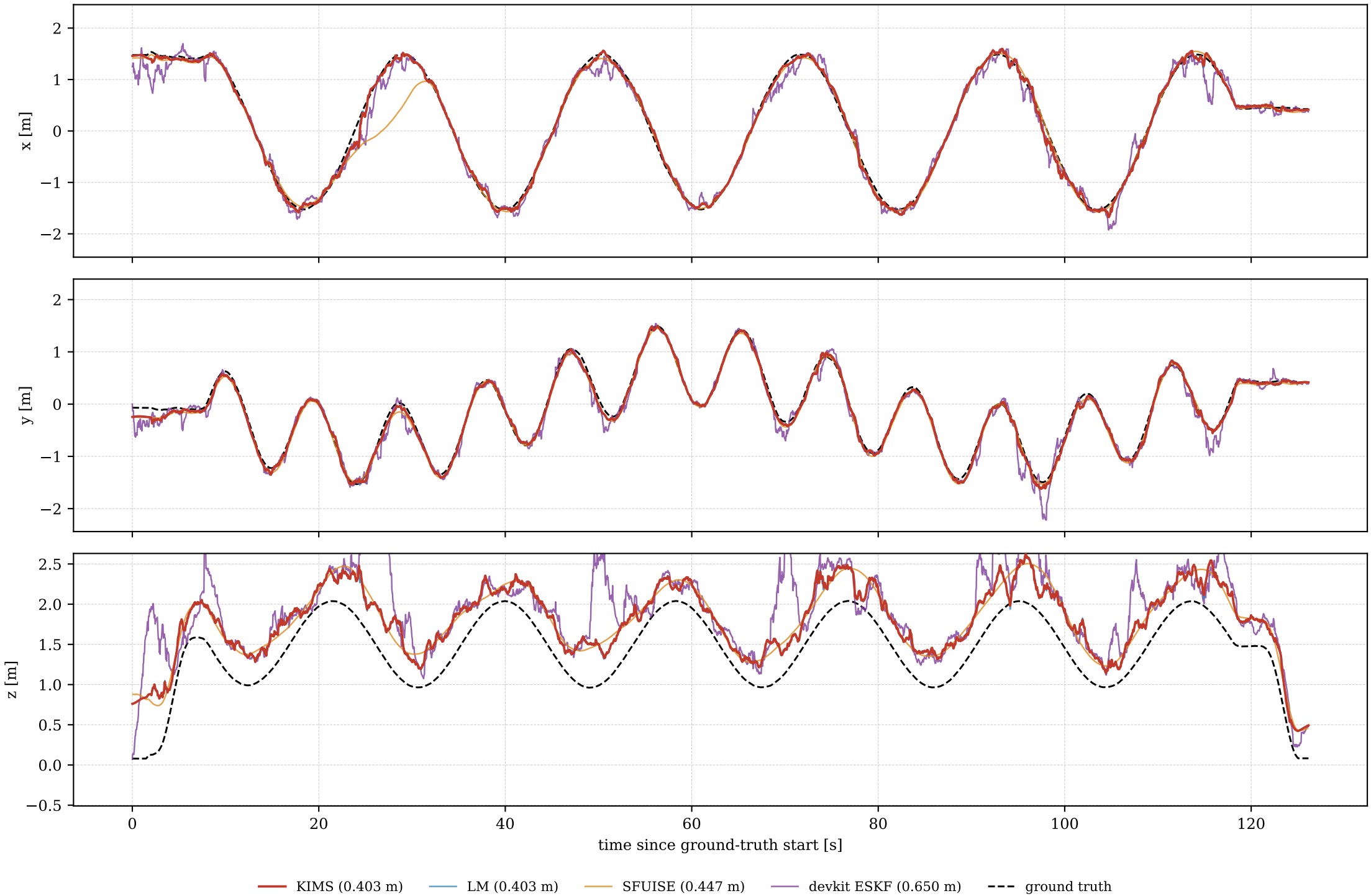

UTIL / const4-trial4-tdoa2-traj1 3D RMSE [m] KIMS 0.595 LM 0.596 SFUISE div. devkit ESKF 0.693
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.595 m)
LM (0.596 m)
SFUISE (div. m)
devkit ESKF (0.693 m)
ground truth

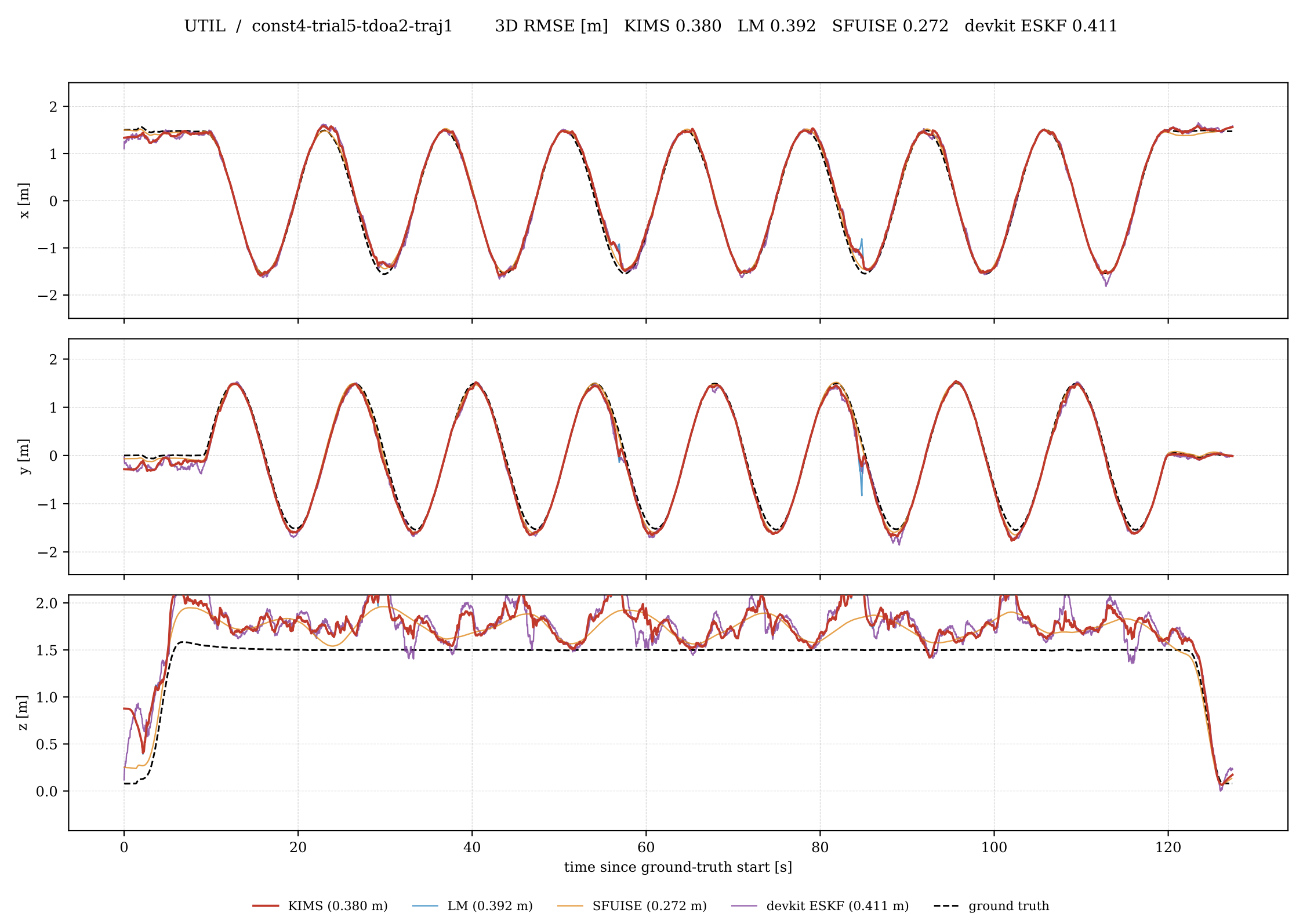

UTIL / const4-trial4-tdoa2-traj2 3D RMSE [m] KIMS 0.505 LM 0.508 SFUISE div. devkit ESKF 0.627
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.505 m)
LM (0.508 m)
SFUISE (div. m)
devkit ESKF (0.627 m)
ground truth

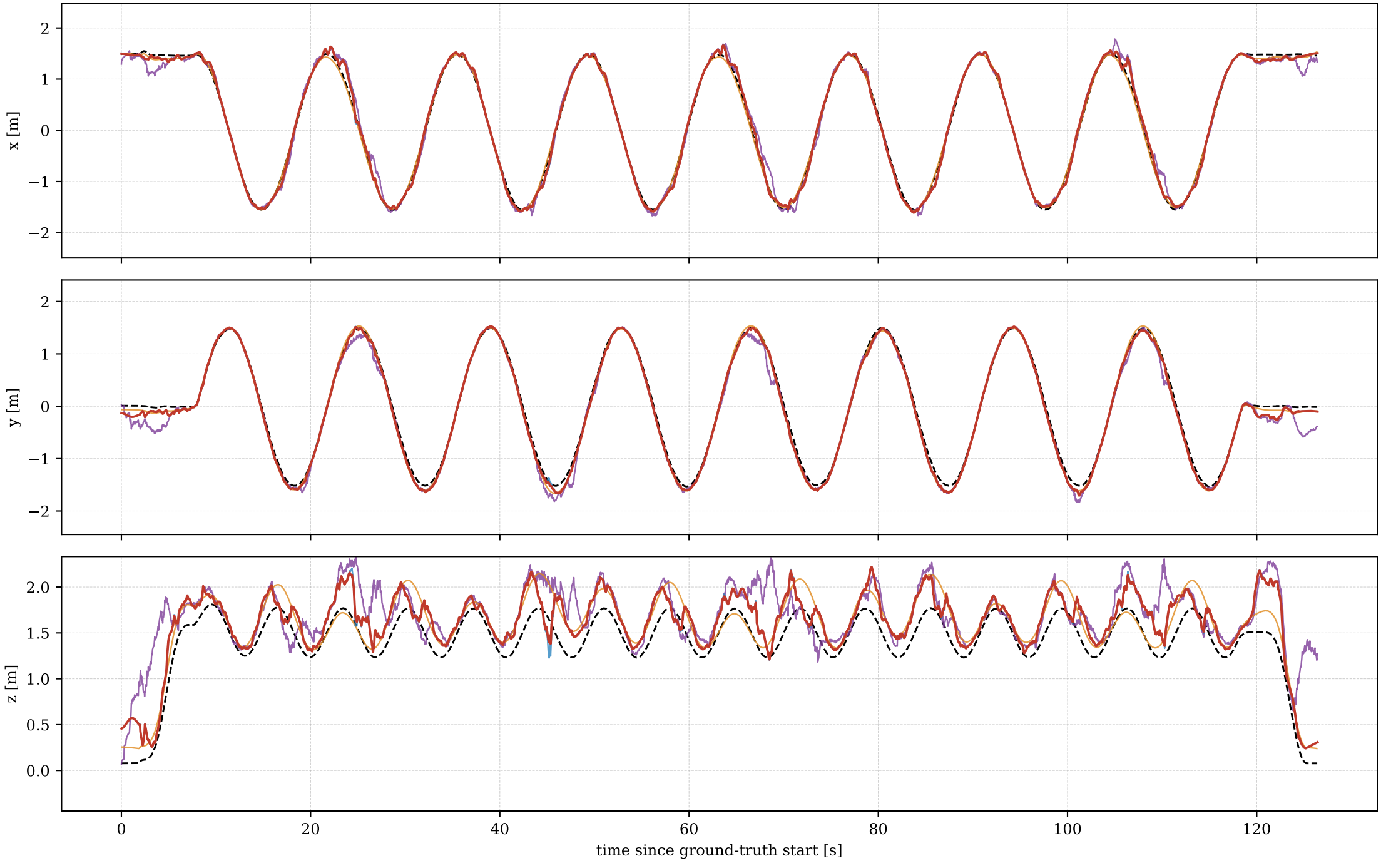

UTIL / const4-trial4-tdoa2-traj3 3D RMSE [m] KIMS 0.489 LM 0.491 SFUISE 1.296 devkit ESKF 0.562
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.489 m)
LM (0.491 m)
SFUISE (1.296 m)
devkit ESKF (0.562 m)
ground truth

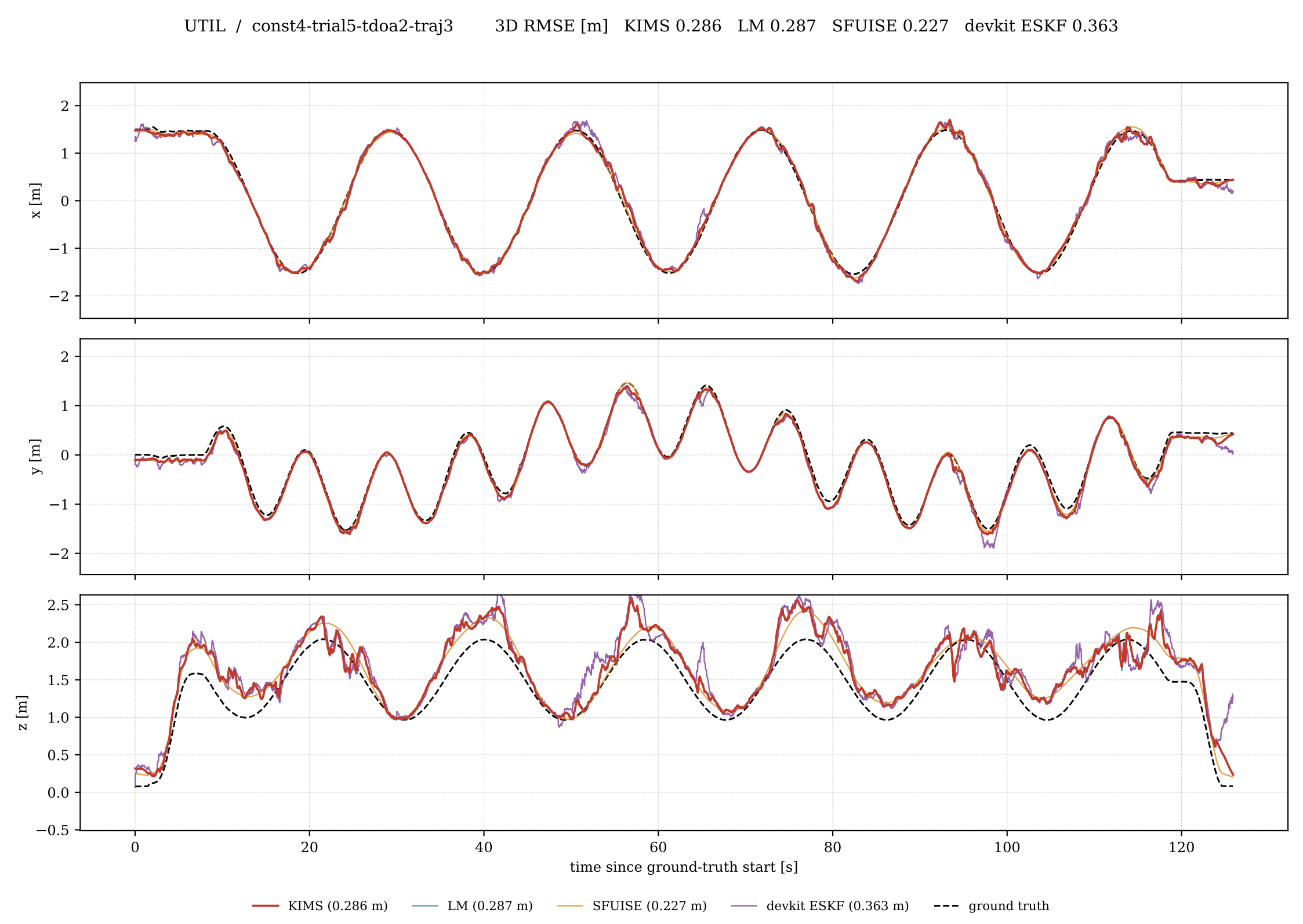

UTIL / const4-trial4-tdoa3-traj1 3D RMSE [m] KIMS 0.433 LM 0.433 SFUISE 0.399 devkit ESKF 0.617
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.433 m)
LM (0.433 m)
SFUISE (0.399 m)
devkit ESKF (0.617 m)
ground truth

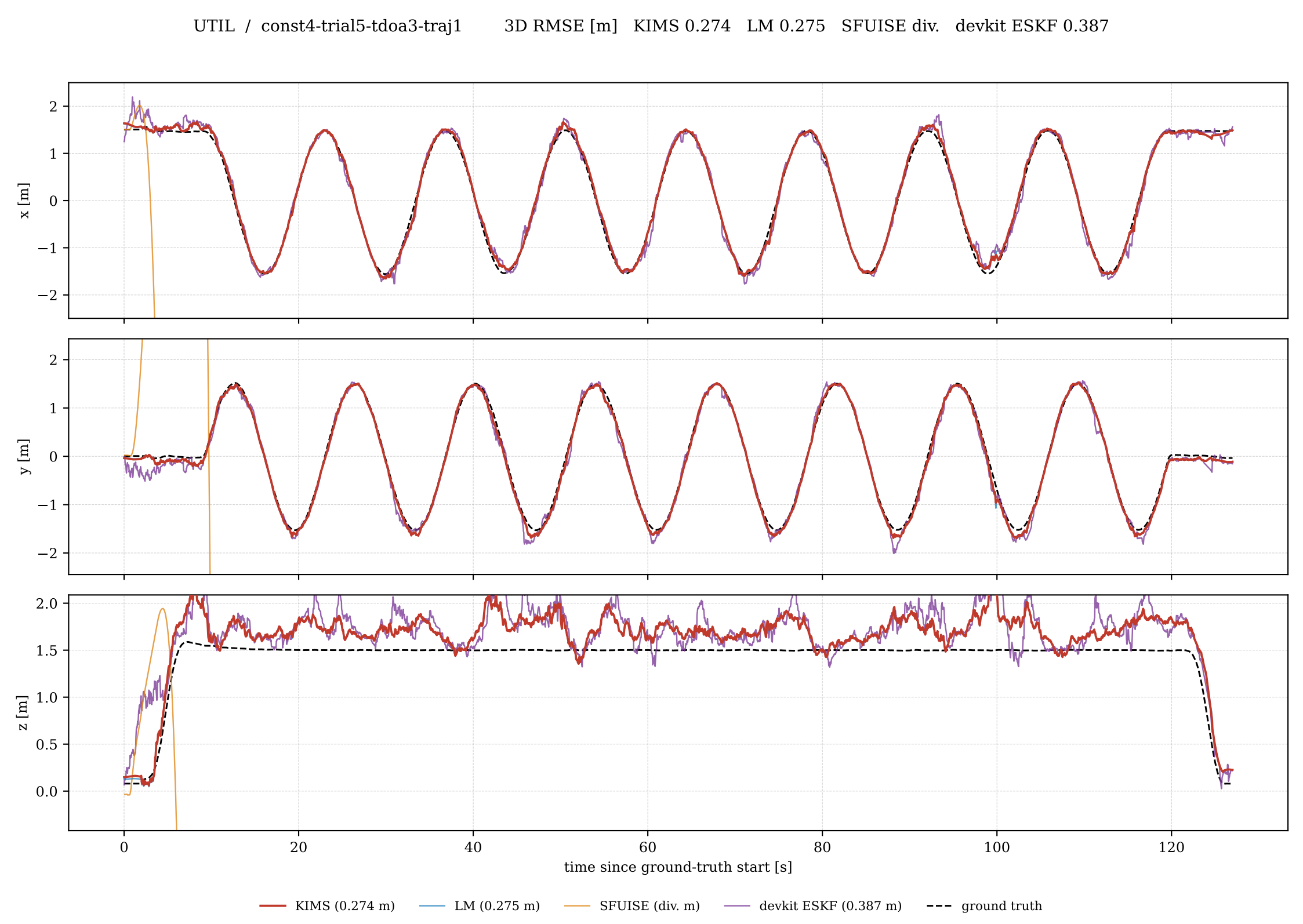

UTIL / const4-trial4-tdoa3-traj2 3D RMSE [m] KIMS 0.396 LM 0.396 SFUISE 0.349 devkit ESKF 0.631
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.396 m)
LM (0.396 m)
SFUISE (0.349 m)
devkit ESKF (0.631 m)
ground truth

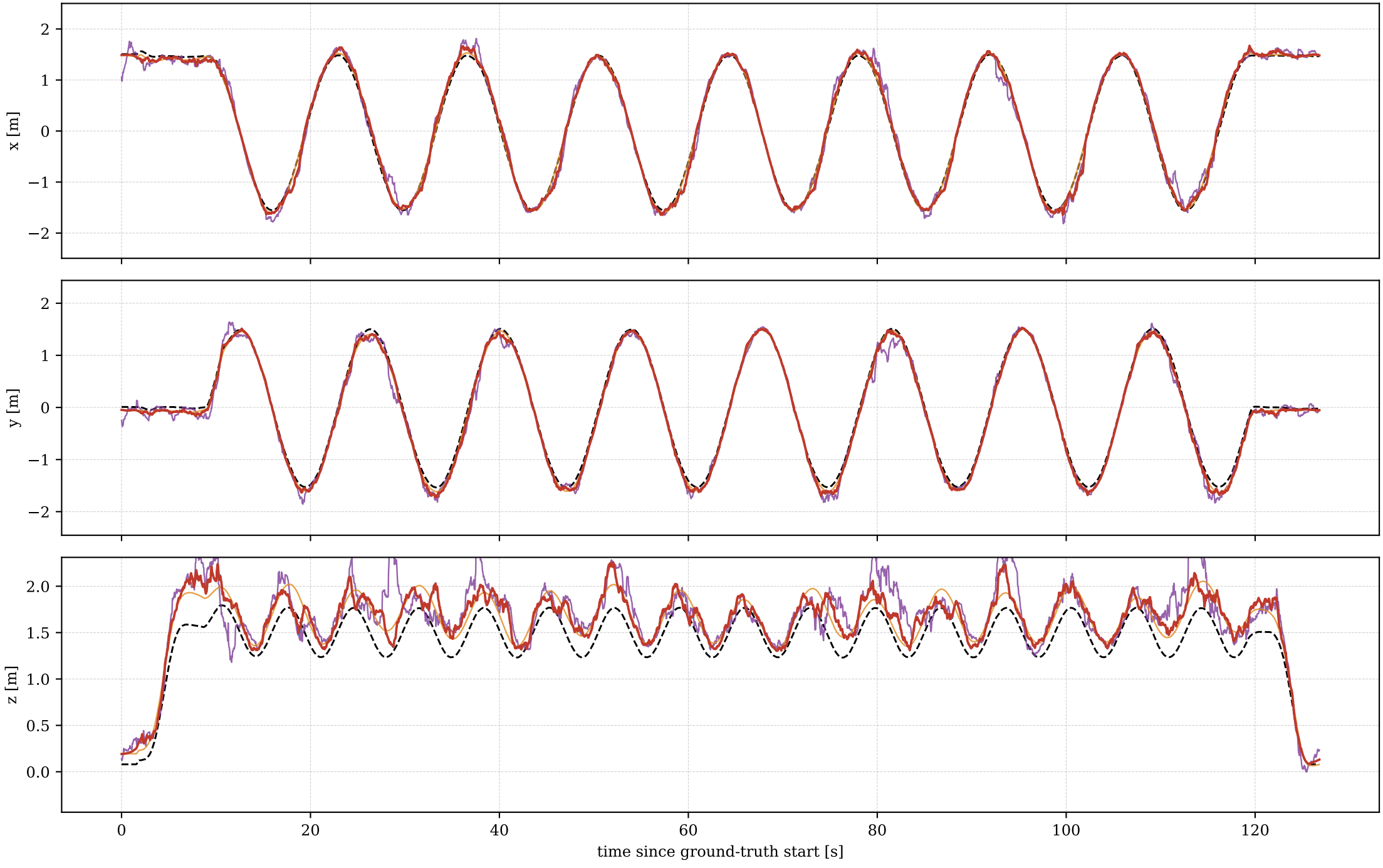

UTIL / const4-trial4-tdoa3-traj3 3D RMSE [m] KIMS 0.403 LM 0.403 SFUISE 0.447 devkit ESKF 0.650
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.403 m)
LM (0.403 m)
SFUISE (0.447 m)
devkit ESKF (0.650 m)
ground truth

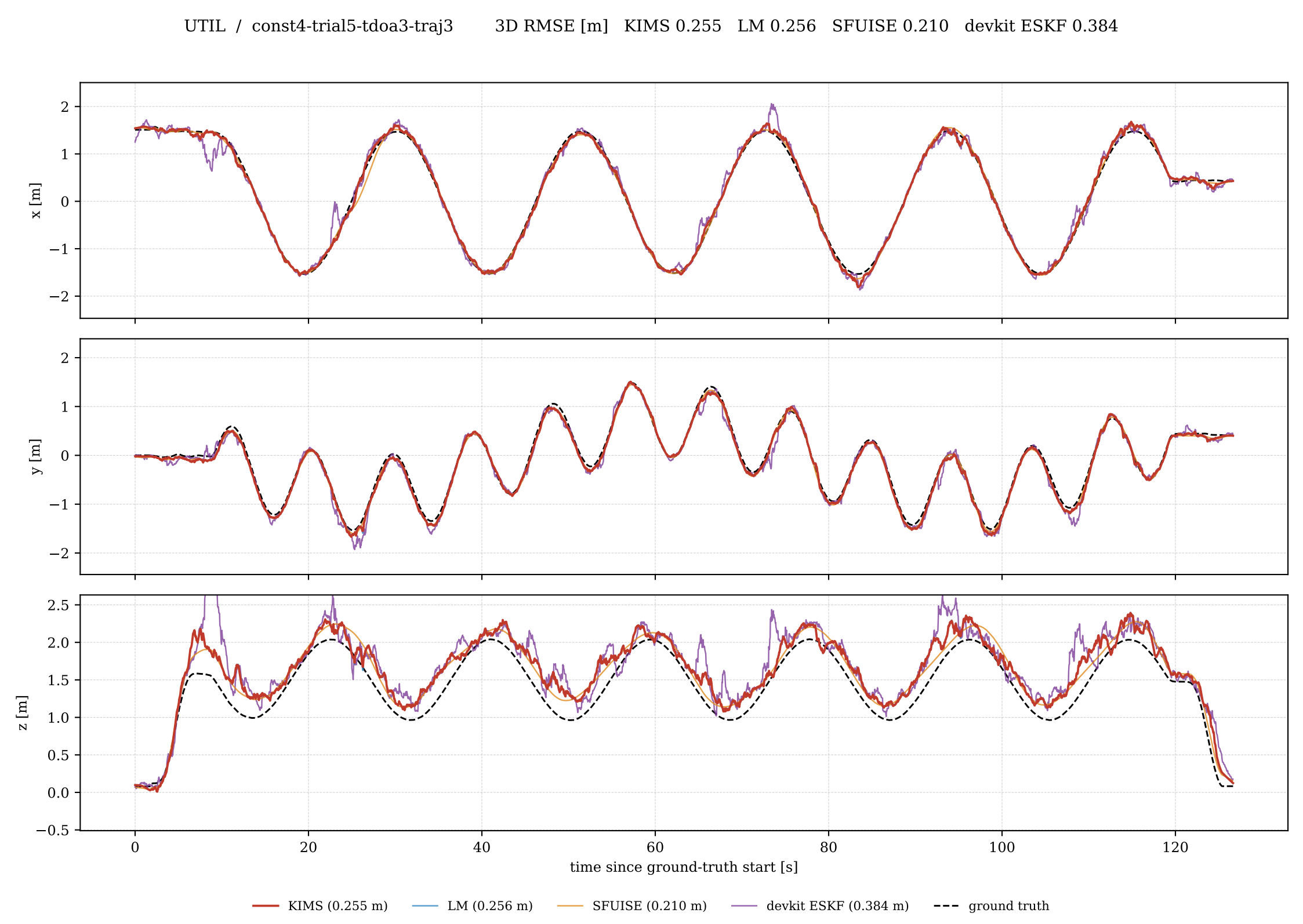

UTIL / const4-trial5-tdoa2-traj1 3D RMSE [m] KIMS 0.380 LM 0.392 SFUISE 0.272 devkit ESKF 0.411
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.380 m)
LM (0.392 m)
SFUISE (0.272 m)
devkit ESKF (0.411 m)
ground truth

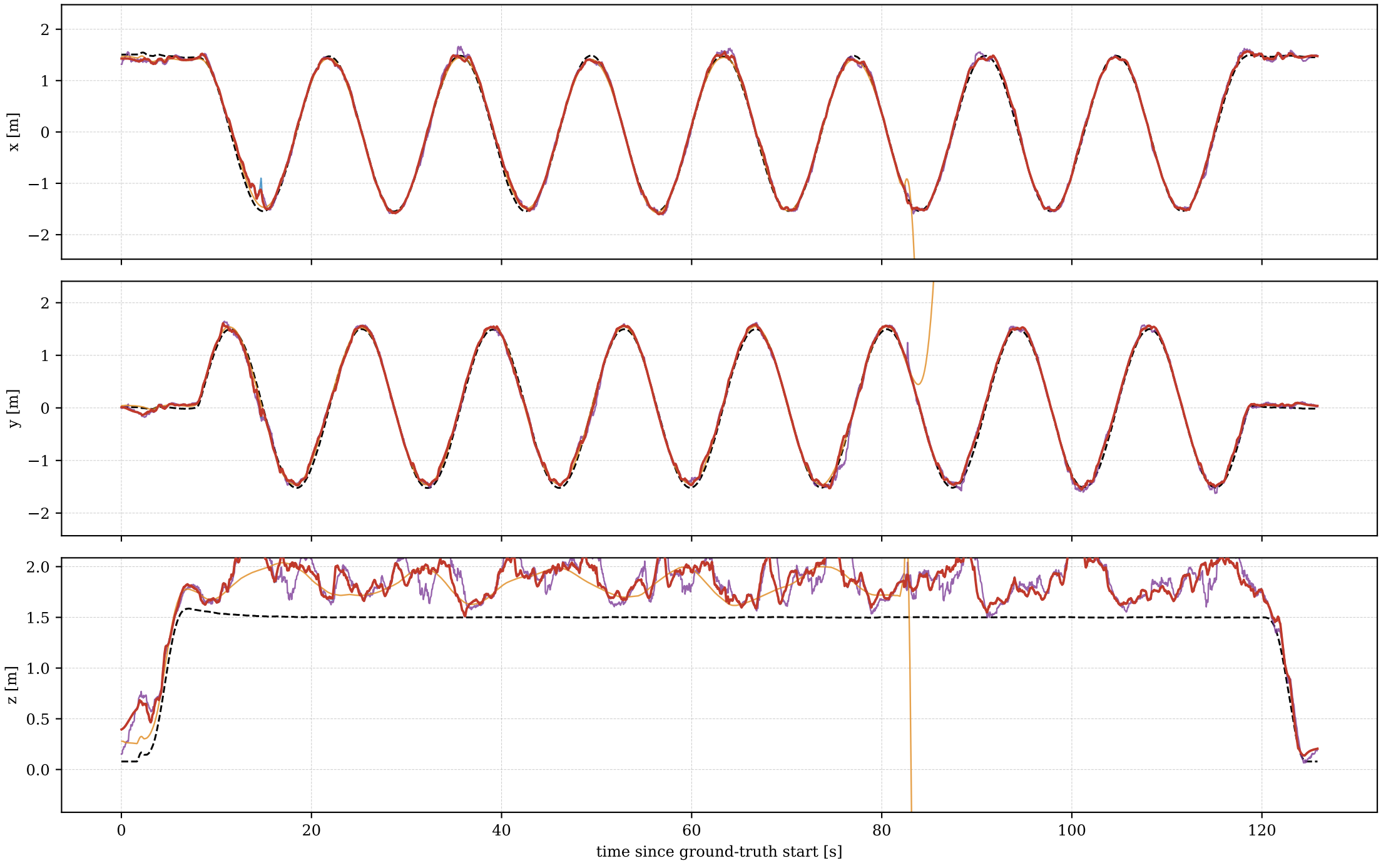

UTIL / const4-trial5-tdoa2-traj2 3D RMSE [m] KIMS 0.262 LM 0.264 SFUISE 0.218 devkit ESKF 0.426
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.262 m)
LM (0.264 m)
SFUISE (0.218 m)
devkit ESKF (0.426 m)
ground truth

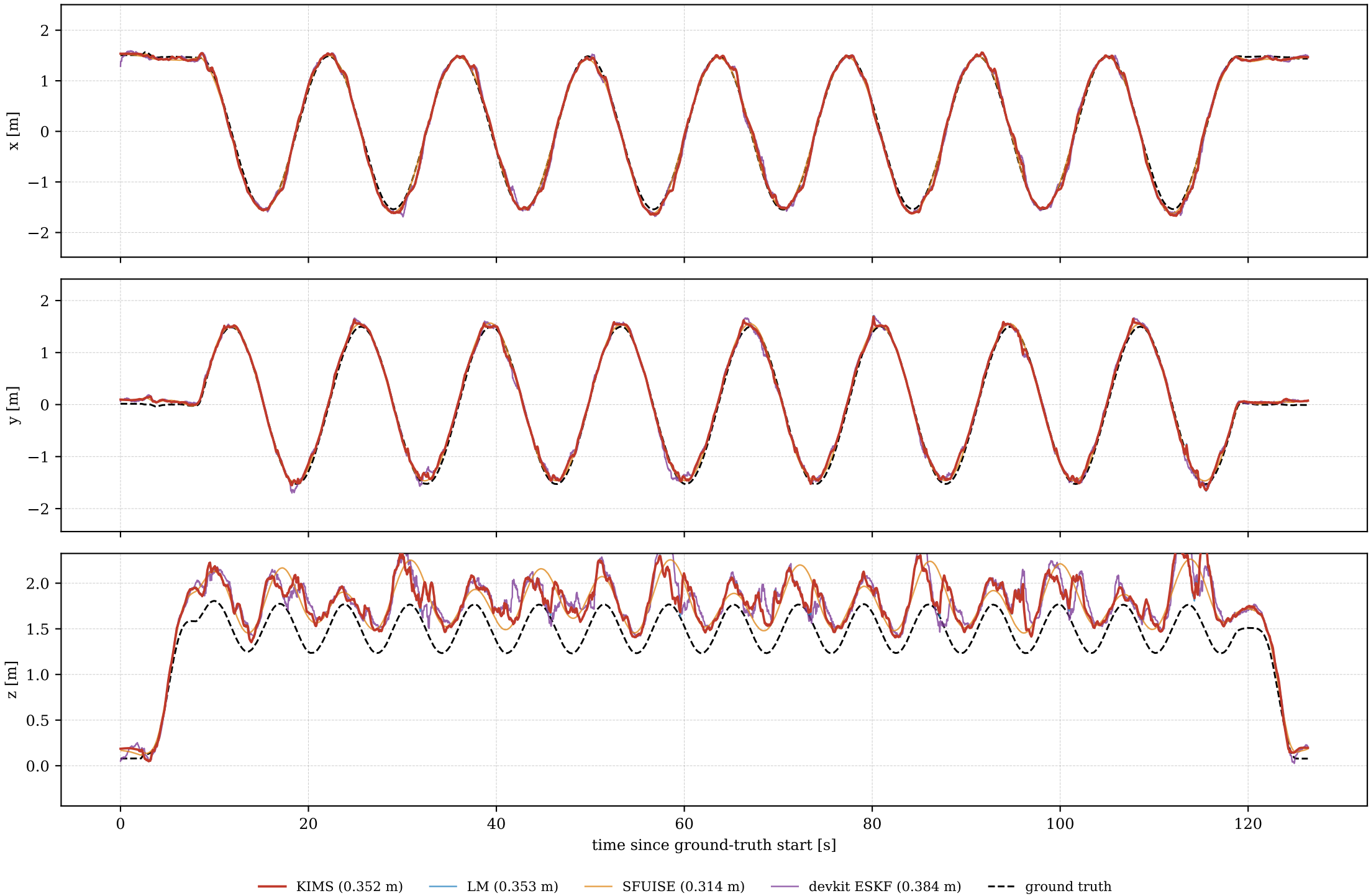

UTIL / const4-trial5-tdoa2-traj3 3D RMSE [m] KIMS 0.286 LM 0.287 SFUISE 0.227 devkit ESKF 0.363
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.286 m)
LM (0.287 m)
SFUISE (0.227 m)
devkit ESKF (0.363 m)
ground truth

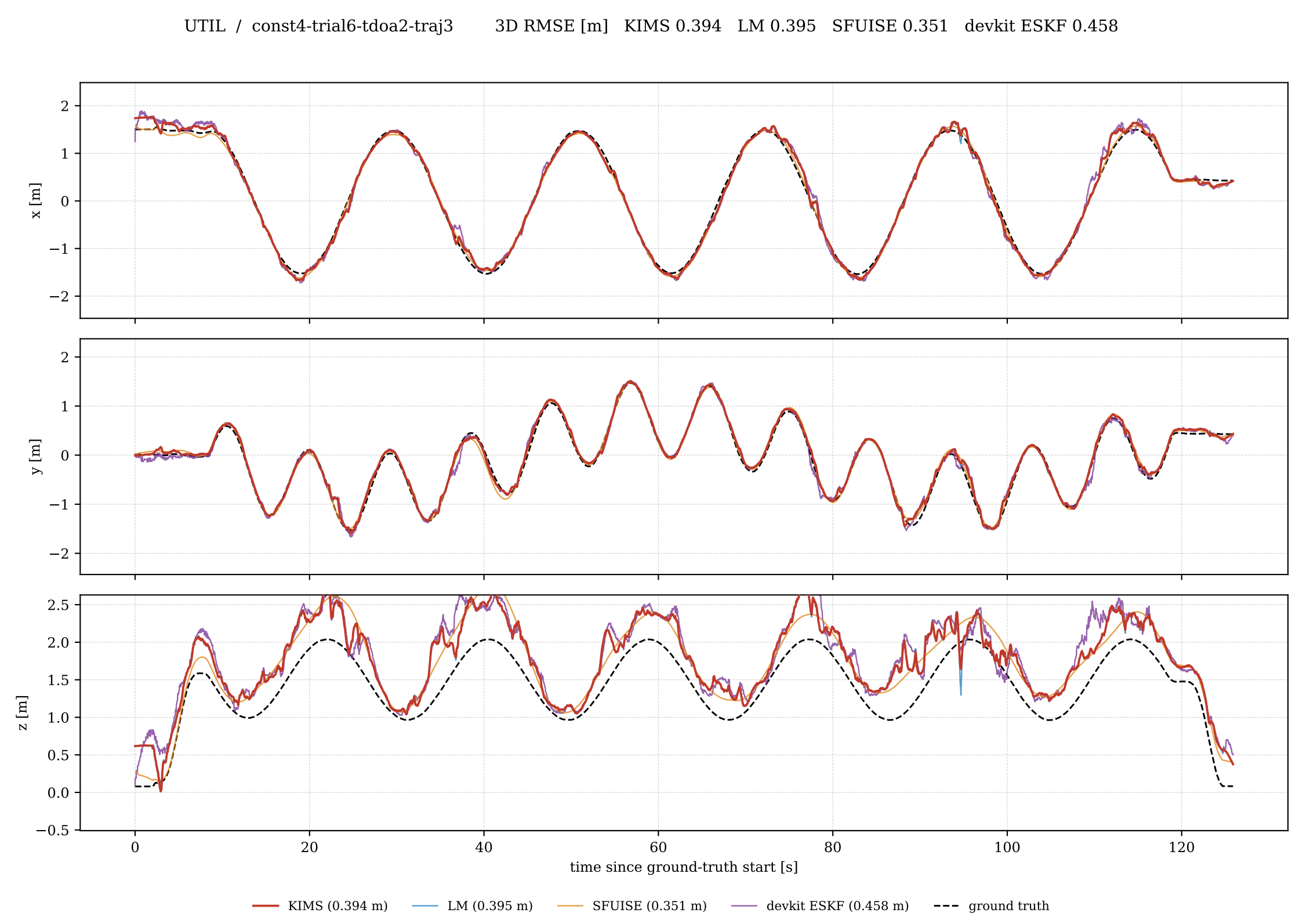

UTIL / const4-trial5-tdoa3-traj1 3D RMSE [m] KIMS 0.274 LM 0.275 SFUISE div. devkit ESKF 0.387
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.274 m)
LM (0.275 m)
SFUISE (div. m)
devkit ESKF (0.387 m)
ground truth

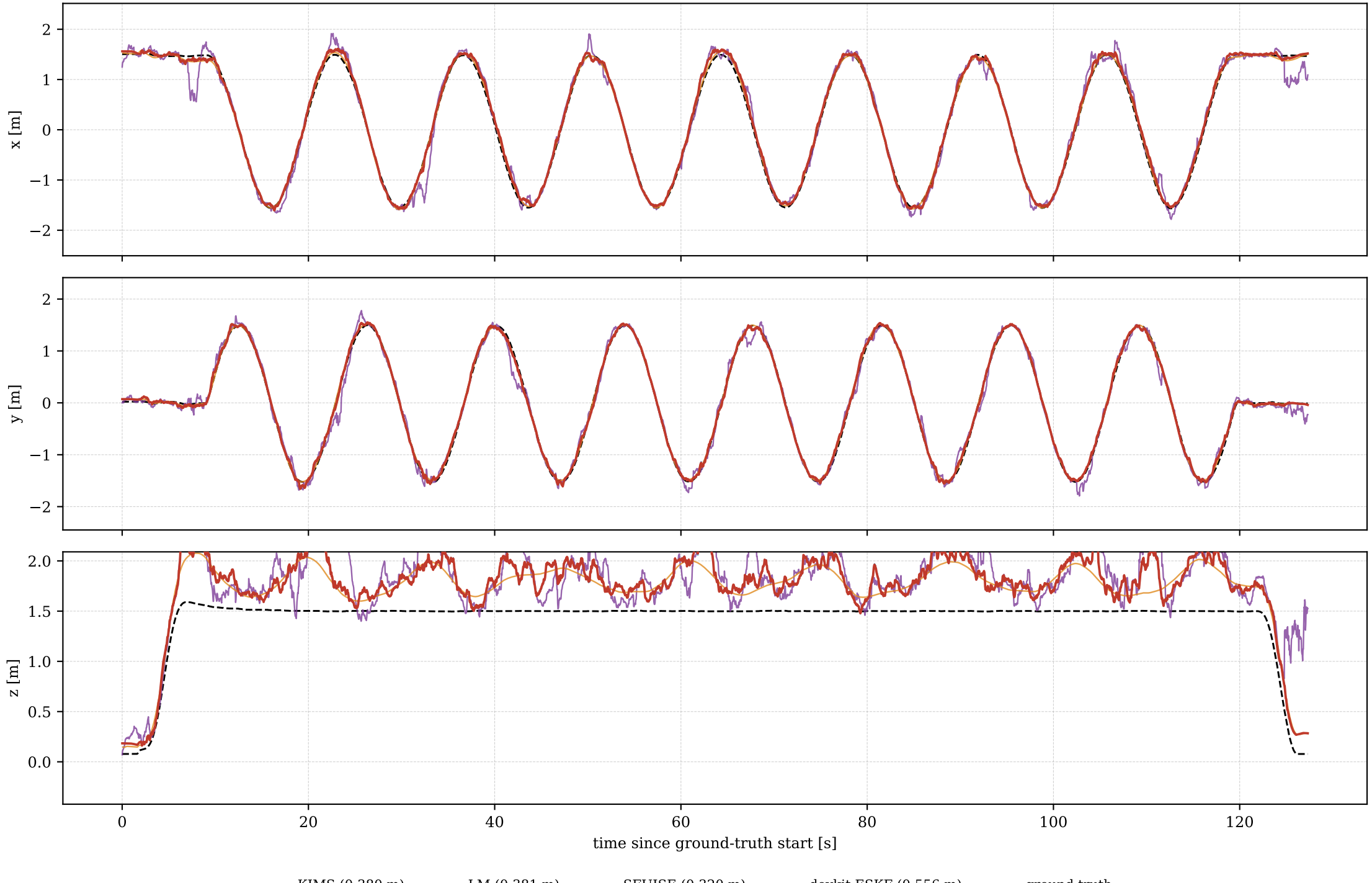

UTIL / const4-trial5-tdoa3-traj2 3D RMSE [m] KIMS 0.258 LM 0.259 SFUISE 0.206 devkit ESKF 0.366
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.258 m)
LM (0.259 m)
SFUISE (0.206 m)
devkit ESKF (0.366 m)
ground truth

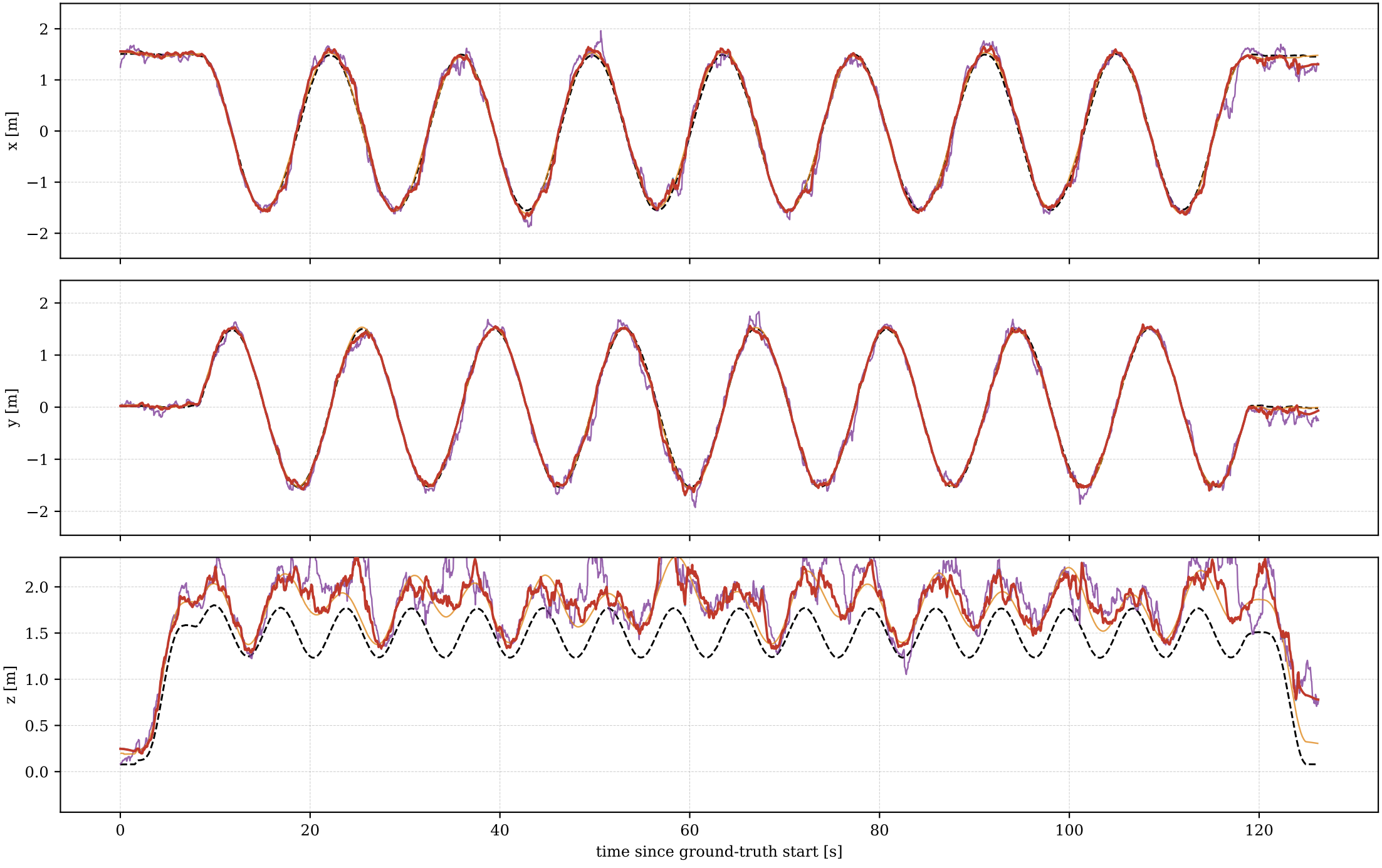

UTIL / const4-trial5-tdoa3-traj3 3D RMSE [m] KIMS 0.255 LM 0.256 SFUISE 0.210 devkit ESKF 0.384
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.255 m)
LM (0.256 m)
SFUISE (0.210 m)
devkit ESKF (0.384 m)
ground truth

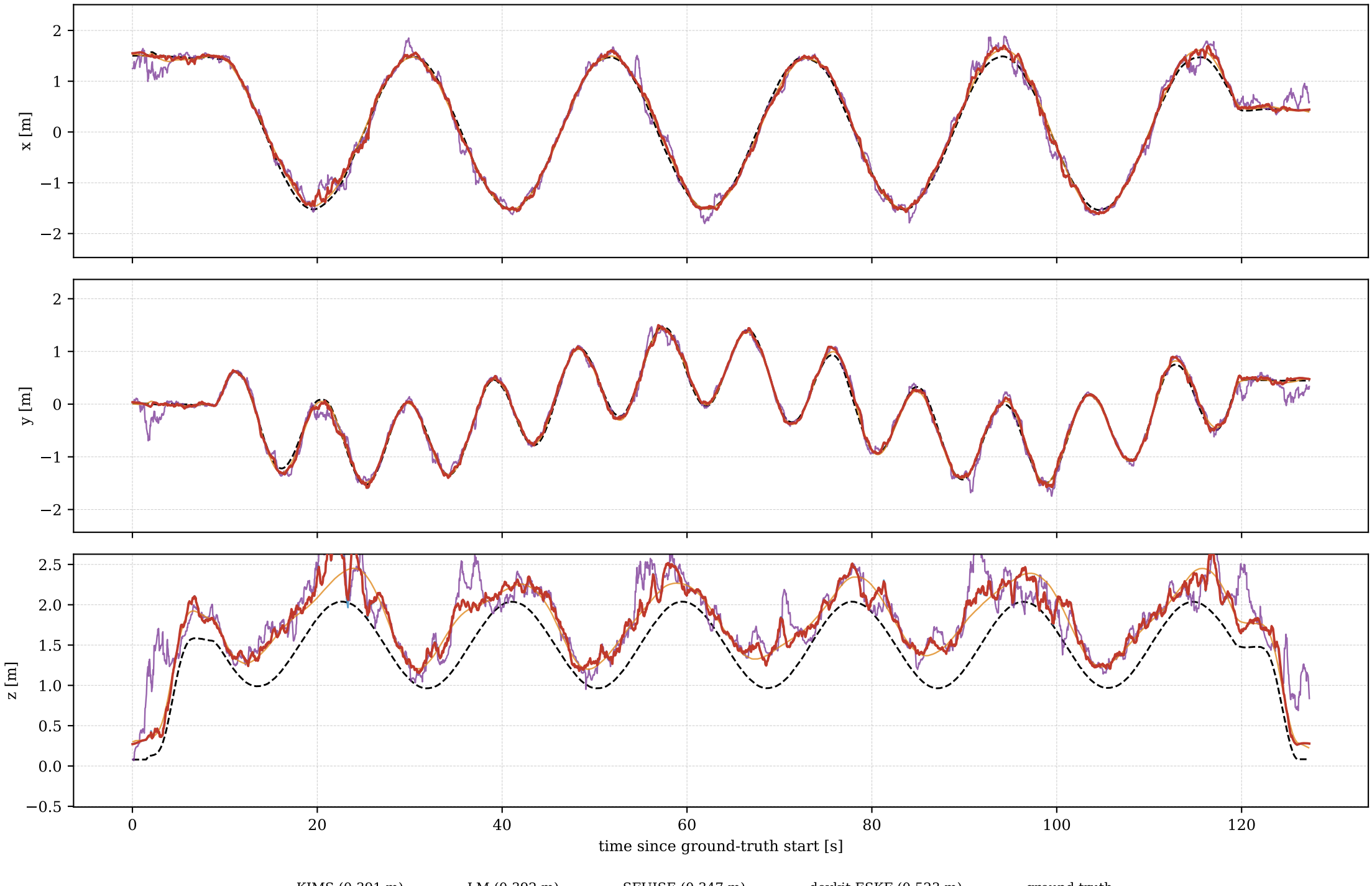

UTIL / const4-trial6-tdoa2-traj1 3D RMSE [m] KIMS 0.412 LM 0.413 SFUISE div. devkit ESKF 0.454
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.412 m)
LM (0.413 m)
SFUISE (div. m)
devkit ESKF (0.454 m)
ground truth

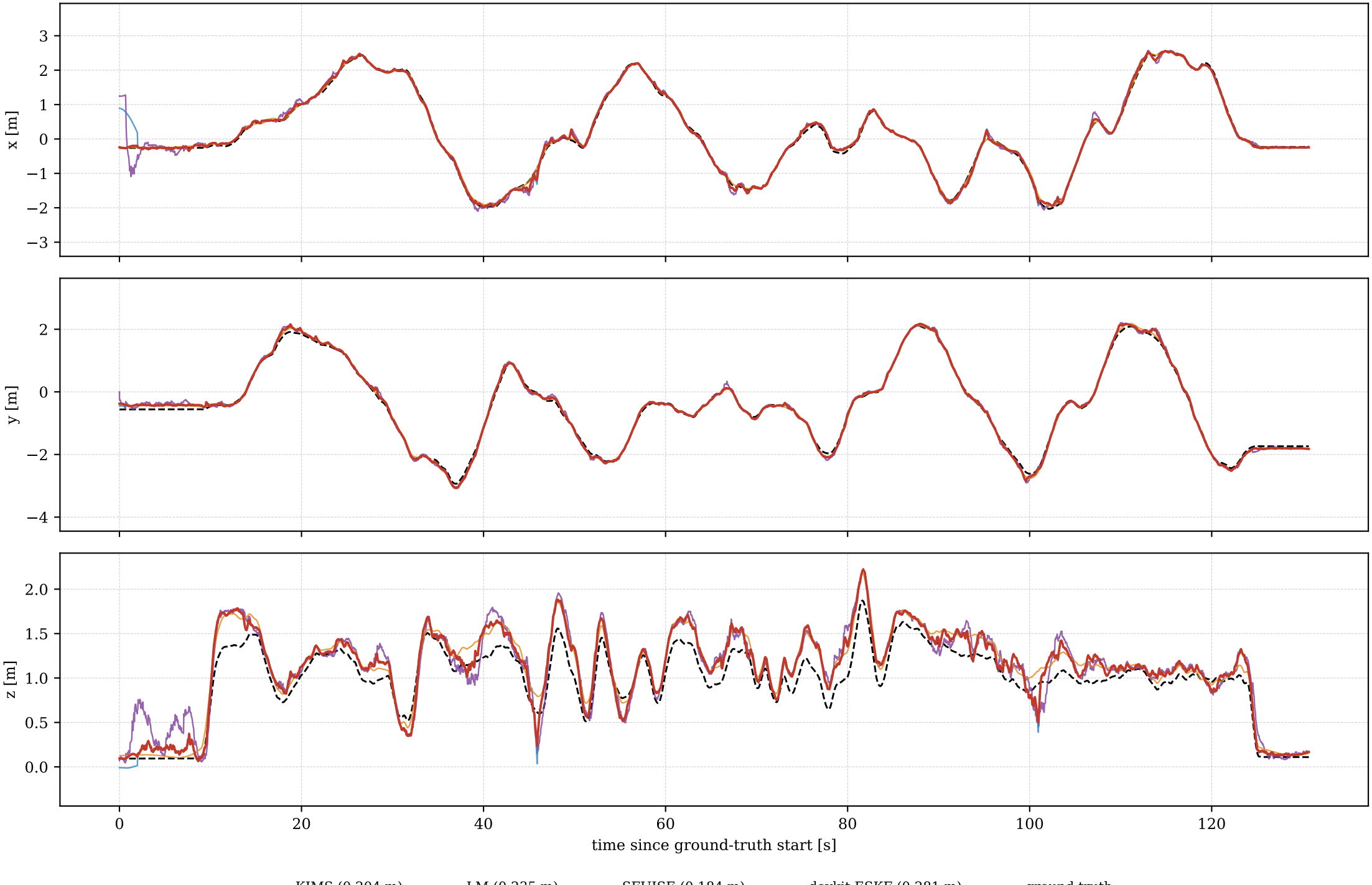

UTIL / const4-trial6-tdoa2-traj2 3D RMSE [m] KIMS 0.352 LM 0.353 SFUISE 0.314 devkit ESKF 0.384
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.352 m)
LM (0.353 m)
SFUISE (0.314 m)
devkit ESKF (0.384 m)
ground truth

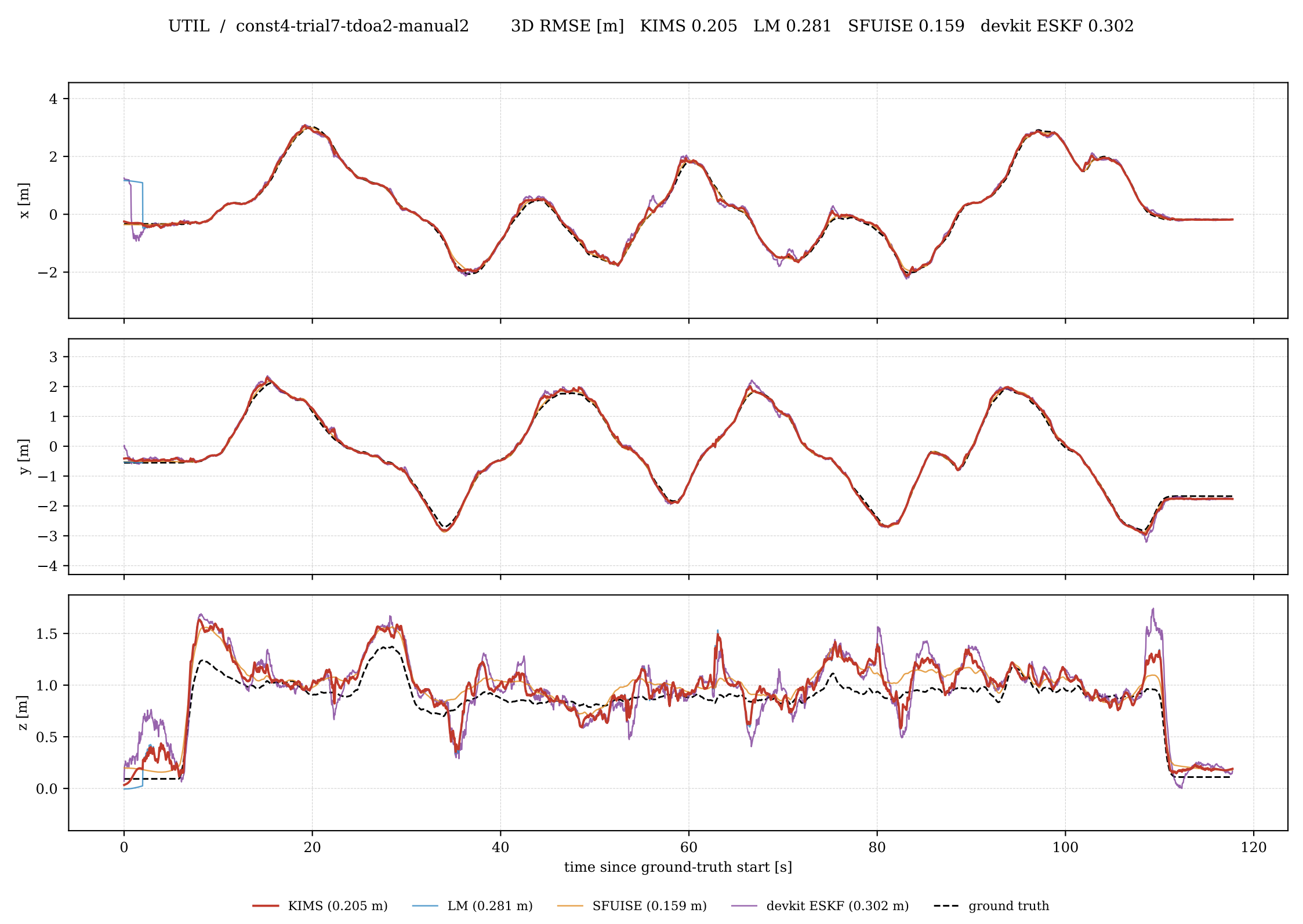

UTIL / const4-trial6-tdoa2-traj3 3D RMSE [m] KIMS 0.394 LM 0.395 SFUISE 0.351 devkit ESKF 0.458
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.394 m)
LM (0.395 m)
SFUISE (0.351 m)
devkit ESKF (0.458 m)
ground truth

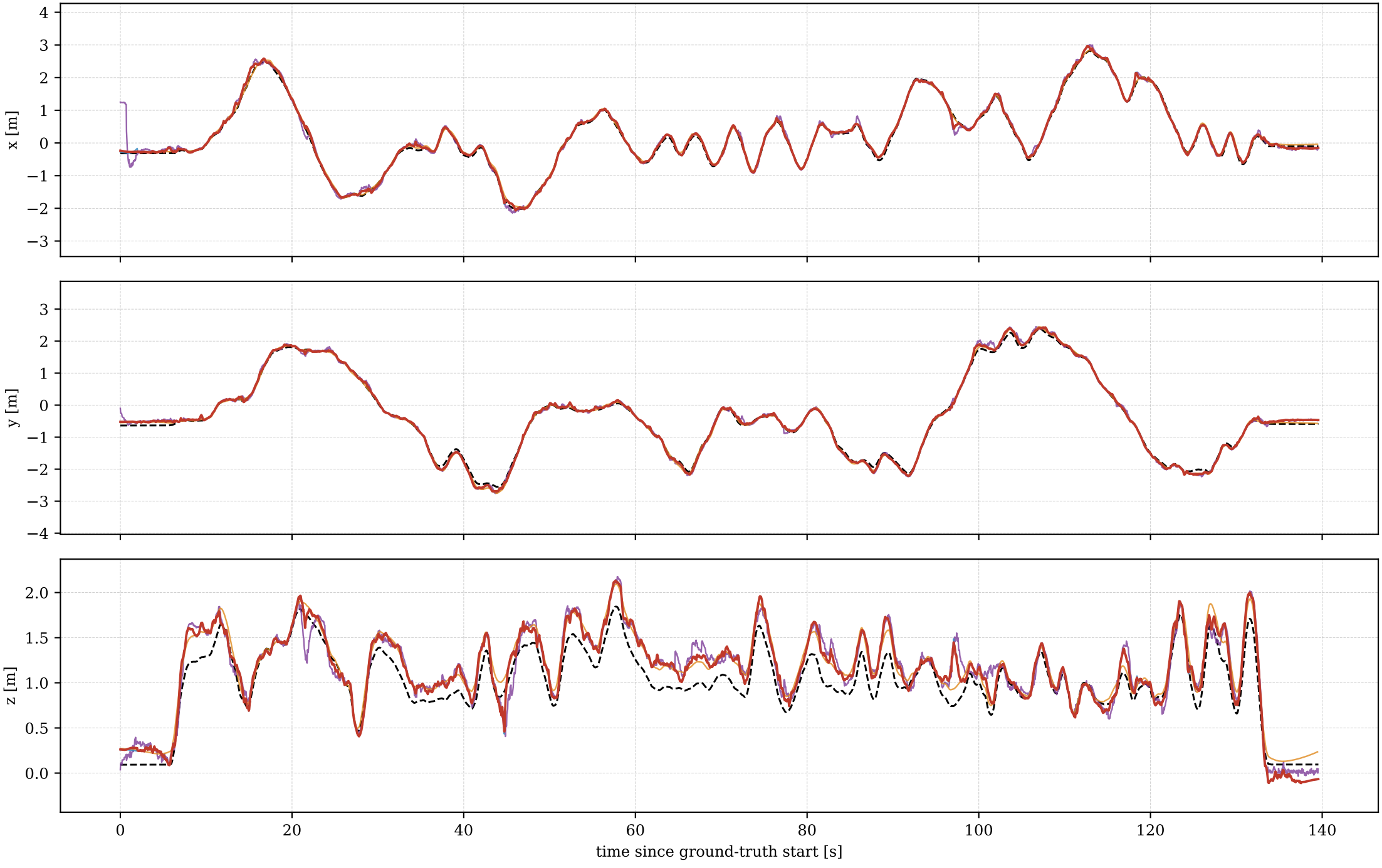

UTIL / const4-trial6-tdoa3-traj1 3D RMSE [m] KIMS 0.380 LM 0.381 SFUISE 0.320 devkit ESKF 0.556
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.380 m)
LM (0.381 m)
SFUISE (0.320 m)
devkit ESKF (0.556 m)
ground truth


UTIL / const4-trial6-tdoa3-traj2 3D RMSE [m] KIMS 0.401 LM 0.402 SFUISE 0.320 devkit ESKF 0.540
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.401 m)
LM (0.402 m)
SFUISE (0.320 m)
devkit ESKF (0.540 m)
ground truth


UTIL / const4-trial6-tdoa3-traj3 3D RMSE [m] KIMS 0.391 LM 0.392 SFUISE 0.347 devkit ESKF 0.523
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.391 m)
LM (0.392 m)
SFUISE (0.347 m)
devkit ESKF (0.523 m)
ground truth


UTIL / const4-trial7-tdoa2-manual1 3D RMSE [m] KIMS 0.204 LM 0.235 SFUISE 0.184 devkit ESKF 0.281
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.204 m)
LM (0.235 m)
SFUISE (0.184 m)
devkit ESKF (0.281 m)
ground truth


UTIL / const4-trial7-tdoa2-manual2 3D RMSE [m] KIMS 0.205 LM 0.281 SFUISE 0.159 devkit ESKF 0.302
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.205 m)
LM (0.281 m)
SFUISE (0.159 m)
devkit ESKF (0.302 m)
ground truth


UTIL / const4-trial7-tdoa2-manual3 3D RMSE [m] KIMS 0.211 LM 0.212 SFUISE 0.190 devkit ESKF 0.270
x [m]
y [m]
z [m]
time since ground-truth start [s]
KIMS (0.211 m)
LM (0.212 m)
SFUISE (0.190 m)
devkit ESKF (0.270 m)
ground truth